

# Package ‘cTMed’

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**Title** Continuous-Time Mediation

**Version** 1.0.10

**Description** Computes effect sizes, standard errors, and confidence intervals for total, direct, and indirect effects in continuous-time mediation models as described in Pesigan, Russell, and Chow (2025) <[doi:10.1037/met0000779](https://doi.org/10.1037/met0000779)>.

**URL** <https://github.com/jeksterslab/cTMed>,  
<https://jeksterslab.github.io/cTMed/>

**BugReports** <https://github.com/jeksterslab/cTMed/issues>

**License** GPL (>= 3)

**Encoding** UTF-8

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**Depends** R (>= 3.5.0)

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bootStateSpace

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BootBeta

*Bootstrap Sampling Distribution for the Elements of the Matrix of Lagged Coefficients Over a Specific Time Interval or a Range of Time Intervals*

**Description**

This function generates a bootstrap method sampling distribution for the elements of the matrix of lagged coefficients  $\beta$  over a specific time interval  $\Delta t$  or a range of time intervals using the first-order stochastic differential equation model drift matrix  $\Phi$ .

**Usage**

```
BootBeta(phi, phi_hat, delta_t, ncores = NULL, tol = 0.001)
```

**Arguments**

|         |                                                                                                                                                                             |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phi     | List of numeric matrices. Each element of the list is a bootstrap estimate of the drift matrix ( $\Phi$ ).                                                                  |
| phi_hat | Numeric matrix. The estimated drift matrix ( $\hat{\Phi}$ ) from the original data set. phi_hat should have row and column names pertaining to the variables in the system. |
| delta_t | Numeric. Time interval ( $\Delta t$ ).                                                                                                                                      |
| ncores  | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of replications R is a large value.                |
| tol     | Numeric. Smallest possible time interval to allow.                                                                                                                          |

**Details**

See [Total\(\)](#).

**Value**

Returns an object of class `ctmedboot` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("BootBeta").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**est** Estimated elements of the matrix of lagged coefficients.

**thetahatstar** A matrix of bootstrap elements of the matrix of lagged coefficients.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**References**

- Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. [doi:10.2307/271028](#)
- Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. [doi:10.1080/10705511.2014.973960](#)
- Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. [doi:10.1037/met0000779](#)
- Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. [doi:10.1007/s11336021097670](#)

**See Also**

Other Continuous-Time Mediation Functions: [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

**Examples**

```
## Not run:
library(bootStateSpace)
# prepare parameters
## number of individuals
n <- 50
## time points
time <- 100
delta_t <- 0.10
## dynamic structure
p <- 3
mu0 <- rep(x = 0, times = p)
sigma0 <- matrix(
  data = c(
    1.0,
    0.2,
    0.2,
    0.2,
    1.0,
    0.2,
    0.2,
    0.2,
    1.0
  ),
  nrow = p
)
sigma0_l <- t(chol(sigma0))
mu <- rep(x = 0, times = p)
phi <- matrix(
  data = c(
    -0.357,
    0.771,
    -0.450,
    0.0,
    -0.511,
    0.729,
  )
)
```

```

      0,
      0,
      -0.693
    ),
    nrow = p
  )
  sigma <- matrix(
    data = c(
      0.24455556,
      0.02201587,
      -0.05004762,
      0.02201587,
      0.07067800,
      0.01539456,
      -0.05004762,
      0.01539456,
      0.07553061
    ),
    nrow = p
  )
  sigma_l <- t(chol(sigma))
  ## measurement model
  k <- 3
  nu <- rep(x = 0, times = k)
  lambda <- diag(k)
  theta <- 0.2 * diag(k)
  theta_l <- t(chol(theta))

  boot <- PBSSMOUFixed(
    R = 10L, # use at least 1000 in actual research
    path = getwd(),
    prefix = "ou",
    n = n,
    time = time,
    delta_t = delta_t,
    mu0 = mu0,
    sigma0_l = sigma0_l,
    mu = mu,
    phi = phi,
    sigma_l = sigma_l,
    nu = nu,
    lambda = lambda,
    theta_l = theta_l,
    ncores = NULL, # consider using multiple cores
    seed = 42
  )
  phi_hat <- phi
  colnames(phi_hat) <- rownames(phi_hat) <- c("x", "m", "y")
  phi <- extract(object = boot, what = "phi")

  # Specific time interval -----
  BootBeta(
    phi = phi,

```

```

    phi_hat = phi_hat,
    delta_t = 1
)

# Range of time intervals -----
boot <- BootBeta(
  phi = phi,
  phi_hat = phi_hat,
  delta_t = 1:5
)
plot(boot)
plot(boot, type = "bc") # bias-corrected

# Methods -----
# BootBeta has a number of methods including
# print, summary, confint, and plot
print(boot)
summary(boot)
confint(boot, level = 0.95)
print(boot, type = "bc") # bias-corrected
summary(boot, type = "bc")
confint(boot, level = 0.95, type = "bc")

## End(Not run)

```

---

BootBetaStd

*Bootstrap Sampling Distribution for the Elements of the Standardized Matrix of Lagged Coefficients Over a Specific Time Interval or a Range of Time Intervals*

---

## Description

This function generates a bootstrap method sampling distribution for the elements of the standardized matrix of lagged coefficients  $\beta$  over a specific time interval  $\Delta t$  or a range of time intervals using the first-order stochastic differential equation model drift matrix  $\Phi$  and process noise covariance matrix  $\Sigma$ .

## Usage

```

BootBetaStd(
  phi,
  sigma,
  phi_hat,
  sigma_hat,
  delta_t,
  ncores = NULL,
  tol = 0.001
)

```

**Arguments**

|           |                                                                                                                                                                             |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phi       | List of numeric matrices. Each element of the list is a bootstrap estimate of the drift matrix ( $\Phi$ ).                                                                  |
| sigma     | List of numeric matrices. Each element of the list is a bootstrap estimate of the process noise covariance matrix ( $\Sigma$ ).                                             |
| phi_hat   | Numeric matrix. The estimated drift matrix ( $\hat{\Phi}$ ) from the original data set. phi_hat should have row and column names pertaining to the variables in the system. |
| sigma_hat | Numeric matrix. The estimated process noise covariance matrix ( $\hat{\Sigma}$ ) from the original data set.                                                                |
| delta_t   | Numeric. Time interval ( $\Delta t$ ).                                                                                                                                      |
| ncores    | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of replications R is a large value.                |
| tol       | Numeric. Smallest possible time interval to allow.                                                                                                                          |

**Details**

See [TotalStd\(\)](#).

**Value**

Returns an object of class `ctmedboot` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("BootBetaStd").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**est** Estimated elements of the standardized matrix of lagged coefficients.

**thetahatstar** A matrix of bootstrap elements of the standardized matrix of lagged coefficients.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**References**

Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. [doi:10.2307/271028](#)

Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. [doi:10.1080/10705511.2014.973960](#)

Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. [doi:10.1037/met0000779](#)



Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. doi:[10.1007/s11336021097670](https://doi.org/10.1007/s11336021097670)

### See Also

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

### Examples

```
## Not run:
library(bootStateSpace)
# prepare parameters
## number of individuals
n <- 50
## time points
time <- 100
delta_t <- 0.10
## dynamic structure
p <- 3
mu0 <- rep(x = 0, times = p)
sigma0 <- matrix(
  data = c(
    1.0,
    0.2,
    0.2,
    0.2,
    1.0,
    0.2,
    0.2,
    0.2,
    1.0
  ),
  nrow = p
)
sigma0_l <- t(chol(sigma0))
mu <- rep(x = 0, times = p)
phi <- matrix(
  data = c(
    -0.357,
    0.771,
    -0.450,
```

```

      0.0,
      -0.511,
      0.729,
      0,
      0,
      -0.693
    ),
    nrow = p
  )
  sigma <- matrix(
    data = c(
      0.24455556,
      0.02201587,
      -0.05004762,
      0.02201587,
      0.07067800,
      0.01539456,
      -0.05004762,
      0.01539456,
      0.07553061
    ),
    nrow = p
  )
  sigma_l <- t(chol(sigma))
  ## measurement model
  k <- 3
  nu <- rep(x = 0, times = k)
  lambda <- diag(k)
  theta <- 0.2 * diag(k)
  theta_l <- t(chol(theta))

  boot <- PBSSMOUFixed(
    R = 10L, # use at least 1000 in actual research
    path = getwd(),
    prefix = "ou",
    n = n,
    time = time,
    delta_t = delta_t,
    mu0 = mu0,
    sigma0_l = sigma0_l,
    mu = mu,
    phi = phi,
    sigma_l = sigma_l,
    nu = nu,
    lambda = lambda,
    theta_l = theta_l,
    ncores = NULL, # consider using multiple cores
    seed = 42
  )
  phi_hat <- phi
  colnames(phi_hat) <- rownames(phi_hat) <- c("x", "m", "y")
  sigma_hat <- sigma
  phi <- extract(object = boot, what = "phi")

```

```

sigma <- extract(object = boot, what = "sigma")

# Specific time interval -----
BootBetaStd(
  phi = phi,
  sigma = sigma,
  phi_hat = phi_hat,
  sigma_hat = sigma_hat,
  delta_t = 1
)

# Range of time intervals -----
boot <- BootBetaStd(
  phi = phi,
  sigma = sigma,
  phi_hat = phi_hat,
  sigma_hat = sigma_hat,
  delta_t = 1:5
)
plot(boot)
plot(boot, type = "bc") # bias-corrected

# Methods -----
# BootBetaStd has a number of methods including
# print, summary, confint, and plot
print(boot)
summary(boot)
confint(boot, level = 0.95)
print(boot, type = "bc") # bias-corrected
summary(boot, type = "bc")
confint(boot, level = 0.95, type = "bc")

## End(Not run)

```

---

BootDirectCentral

*Bootstrap Sampling Distribution for the Direct Effect Centrality Over  
a Specific Time Interval or a Range of Time Intervals*


---

## Description

This function generates a bootstrap method sampling distribution for the direct effect centrality over a specific time interval  $\Delta t$  or a range of time intervals using the first-order stochastic differential equation model drift matrix  $\Phi$ .

## Usage

```
BootDirectCentral(phi, phi_hat, delta_t, ncores = NULL, tol = 0.001)
```

**Arguments**

|         |                                                                                                                                                                             |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phi     | List of numeric matrices. Each element of the list is a bootstrap estimate of the drift matrix ( $\Phi$ ).                                                                  |
| phi_hat | Numeric matrix. The estimated drift matrix ( $\hat{\Phi}$ ) from the original data set. phi_hat should have row and column names pertaining to the variables in the system. |
| delta_t | Numeric. Time interval ( $\Delta t$ ).                                                                                                                                      |
| ncores  | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of replications R is a large value.                |
| tol     | Numeric. Smallest possible time interval to allow.                                                                                                                          |

**Details**

See [DirectCentral\(\)](#) for more details.

**Value**

Returns an object of class `ctmedboot` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("BootDirectCentral").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**est** A vector of direct effect centrality.

**thetahatstar** A matrix of bootstrap direct effect centrality.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**References**

- Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. [doi:10.2307/271028](#)
- Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. [doi:10.1080/10705511.2014.973960](#)
- Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. [doi:10.1037/met0000779](#)
- Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. [doi:10.1007/s11336021097670](#)

**See Also**

Other Continuous-Time Mediation Functions: `BootBeta()`, `BootBetaStd()`, `BootDirectCentralStd()`, `BootIndirectCentral()`, `BootIndirectCentralStd()`, `BootMed()`, `BootMedStd()`, `BootTotalCentral()`, `BootTotalCentralStd()`, `DeltaBeta()`, `DeltaBetaStd()`, `DeltaDirectCentral()`, `DeltaDirectCentralStd()`, `DeltaIndirectCentral()`, `DeltaMed()`, `DeltaMedStd()`, `DeltaTotalCentral()`, `DeltaTotalCentralStd()`, `Direct()`, `DirectCentral()`, `DirectCentralStd()`, `DirectStd()`, `Indirect()`, `IndirectCentral()`, `IndirectCentralStd()`, `IndirectStd()`, `MCBeta()`, `MCBetaStd()`, `MCDirectCentral()`, `MCDirectCentralStd()`, `MCIndirectCentral()`, `MCIndirectCentralStd()`, `MCMed()`, `MCMedStd()`, `MCPHi()`, `MCPHiSigma()`, `MCTotalCentral()`, `MCTotalCentralStd()`, `Med()`, `MedStd()`, `PosteriorBeta()`, `PosteriorBetaStd()`, `PosteriorDirectCentral()`, `PosteriorDirectCentralStd()`, `PosteriorIndirectCentral()`, `PosteriorIndirectCentralStd()`, `PosteriorMed()`, `PosteriorMedStd()`, `PosteriorTotalCentral()`, `PosteriorTotalCentralStd()`, `Total()`, `TotalCentral()`, `TotalCentralStd()`, `TotalStd()`, `Trajectory()`

**Examples**

```
## Not run:
library(bootStateSpace)
# prepare parameters
## number of individuals
n <- 50
## time points
time <- 100
delta_t <- 0.10
## dynamic structure
p <- 3
mu0 <- rep(x = 0, times = p)
sigma0 <- matrix(
  data = c(
    1.0,
    0.2,
    0.2,
    0.2,
    1.0,
    0.2,
    0.2,
    0.2,
    1.0
  ),
  nrow = p
)
sigma0_l <- t(chol(sigma0))
mu <- rep(x = 0, times = p)
phi <- matrix(
  data = c(
    -0.357,
    0.771,
    -0.450,
    0.0,
    -0.511,
    0.729,

```

```

      0,
      0,
      -0.693
    ),
    nrow = p
  )
sigma <- matrix(
  data = c(
    0.24455556,
    0.02201587,
    -0.05004762,
    0.02201587,
    0.07067800,
    0.01539456,
    -0.05004762,
    0.01539456,
    0.07553061
  ),
  nrow = p
)
sigma_l <- t(chol(sigma))
## measurement model
k <- 3
nu <- rep(x = 0, times = k)
lambda <- diag(k)
theta <- 0.2 * diag(k)
theta_l <- t(chol(theta))

boot <- PBSSMOUFixed(
  R = 10L, # use at least 1000 in actual research
  path = getwd(),
  prefix = "ou",
  n = n,
  time = time,
  delta_t = delta_t,
  mu0 = mu0,
  sigma0_l = sigma0_l,
  mu = mu,
  phi = phi,
  sigma_l = sigma_l,
  nu = nu,
  lambda = lambda,
  theta_l = theta_l,
  ncores = NULL, # consider using multiple cores
  seed = 42
)
phi_hat <- phi
colnames(phi_hat) <- rownames(phi_hat) <- c("x", "m", "y")
phi <- extract(object = boot, what = "phi")

# Specific time interval -----
BootDirectCentral(
  phi = phi,

```

```

    phi_hat = phi_hat,
    delta_t = 1
)

# Range of time intervals -----
boot <- BootDirectCentral(
  phi = phi,
  phi_hat = phi_hat,
  delta_t = 1:5
)
plot(boot)
plot(boot, type = "bc") # bias-corrected

# Methods -----
# BootDirectCentral has a number of methods including
# print, summary, confint, and plot
print(boot)
summary(boot)
confint(boot, level = 0.95)
print(boot, type = "bc") # bias-corrected
summary(boot, type = "bc")
confint(boot, level = 0.95, type = "bc")

## End(Not run)

```

---

|                      |                                                                                                                                                 |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| BootDirectCentralStd | <i>Bootstrap Sampling Distribution for the Standardized Direct Effect Centrality Over a Specific Time Interval or a Range of Time Intervals</i> |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|

---

## Description

This function generates a bootstrap method sampling distribution for the standardized direct effect centrality over a specific time interval  $\Delta t$  or a range of time intervals using the first-order stochastic differential equation model drift matrix  $\Phi$  and process noise covariance matrix  $\Sigma$ .

## Usage

```

BootDirectCentralStd(
  phi,
  sigma,
  phi_hat,
  sigma_hat,
  delta_t,
  ncores = NULL,
  tol = 0.001
)

```

**Arguments**

|           |                                                                                                                                                                             |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phi       | List of numeric matrices. Each element of the list is a bootstrap estimate of the drift matrix ( $\Phi$ ).                                                                  |
| sigma     | List of numeric matrices. Each element of the list is a bootstrap estimate of the process noise covariance matrix ( $\Sigma$ ).                                             |
| phi_hat   | Numeric matrix. The estimated drift matrix ( $\hat{\Phi}$ ) from the original data set. phi_hat should have row and column names pertaining to the variables in the system. |
| sigma_hat | Numeric matrix. The estimated process noise covariance matrix ( $\hat{\Sigma}$ ) from the original data set.                                                                |
| delta_t   | Numeric. Time interval ( $\Delta t$ ).                                                                                                                                      |
| ncores    | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of replications R is a large value.                |
| tol       | Numeric. Smallest possible time interval to allow.                                                                                                                          |

**Details**

See [DirectCentralStd\(\)](#) for more details.

**Value**

Returns an object of class `ctmedboot` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("BootDirectCentralStd").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**est** A vector of standardized direct effect centrality.

**thetahatstar** A matrix of bootstrap standardized direct effect centrality.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**See Also**

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#),



```

PosteriorDirectCentral(), PosteriorDirectCentralStd(), PosteriorIndirectCentral(),
PosteriorIndirectCentralStd(), PosteriorMed(), PosteriorMedStd(), PosteriorTotalCentral(),
PosteriorTotalCentralStd(), Total(), TotalCentral(), TotalCentralStd(), TotalStd(),
Trajectory()

```

## Examples

```

## Not run:
library(bootStateSpace)
# prepare parameters
## number of individuals
n <- 50
## time points
time <- 100
delta_t <- 0.10
## dynamic structure
p <- 3
mu0 <- rep(x = 0, times = p)
sigma0 <- matrix(
  data = c(
    1.0,
    0.2,
    0.2,
    0.2,
    1.0,
    0.2,
    0.2,
    0.2,
    1.0
  ),
  nrow = p
)
sigma0_l <- t(chol(sigma0))
mu <- rep(x = 0, times = p)
phi <- matrix(
  data = c(
    -0.357,
    0.771,
    -0.450,
    0.0,
    -0.511,
    0.729,
    0,
    0,
    -0.693
  ),
  nrow = p
)
sigma <- matrix(
  data = c(
    0.24455556,
    0.02201587,

```

```

      -0.05004762,
      0.02201587,
      0.07067800,
      0.01539456,
      -0.05004762,
      0.01539456,
      0.07553061
    ),
    nrow = p
  )
  sigma_l <- t(chol(sigma))
  ## measurement model
  k <- 3
  nu <- rep(x = 0, times = k)
  lambda <- diag(k)
  theta <- 0.2 * diag(k)
  theta_l <- t(chol(theta))

  boot <- PBSSMOUFixed(
    R = 10L, # use at least 1000 in actual research
    path = getwd(),
    prefix = "ou",
    n = n,
    time = time,
    delta_t = delta_t,
    mu0 = mu0,
    sigma0_l = sigma0_l,
    mu = mu,
    phi = phi,
    sigma_l = sigma_l,
    nu = nu,
    lambda = lambda,
    theta_l = theta_l,
    ncores = NULL, # consider using multiple cores
    seed = 42
  )
  phi_hat <- phi
  sigma_hat <- sigma
  colnames(phi_hat) <- rownames(phi_hat) <- c("x", "m", "y")
  phi <- extract(object = boot, what = "phi")
  sigma <- extract(object = boot, what = "sigma")

  # Specific time interval -----
  BootDirectCentralStd(
    phi = phi,
    sigma = sigma,
    phi_hat = phi_hat,
    sigma_hat = sigma_hat,
    delta_t = 1
  )

  # Range of time intervals -----
  boot <- BootDirectCentralStd(

```

```

    phi = phi,
    sigma = sigma,
    phi_hat = phi_hat,
    sigma_hat = sigma_hat,
    delta_t = 1:5
  )
  plot(boot)
  plot(boot, type = "bc") # bias-corrected

# Methods -----
# BootDirectCentralStd has a number of methods including
# print, summary, confint, and plot
print(boot)
summary(boot)
confint(boot, level = 0.95)
print(boot, type = "bc") # bias-corrected
summary(boot, type = "bc")
confint(boot, level = 0.95, type = "bc")

## End(Not run)

```

---

|                     |                                                                                                                                      |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| BootIndirectCentral | <i>Bootstrap Sampling Distribution for the Indirect Effect Centrality Over a Specific Time Interval or a Range of Time Intervals</i> |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------|

---

## Description

This function generates a bootstrap method sampling distribution for the indirect effect centrality over a specific time interval  $\Delta t$  or a range of time intervals using the first-order stochastic differential equation model drift matrix  $\Phi$ .

## Usage

```
BootIndirectCentral(phi, phi_hat, delta_t, ncores = NULL, tol = 0.001)
```

## Arguments

|         |                                                                                                                                                                             |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phi     | List of numeric matrices. Each element of the list is a bootstrap estimate of the drift matrix ( $\Phi$ ).                                                                  |
| phi_hat | Numeric matrix. The estimated drift matrix ( $\hat{\Phi}$ ) from the original data set. phi_hat should have row and column names pertaining to the variables in the system. |
| delta_t | Numeric. Time interval ( $\Delta t$ ).                                                                                                                                      |
| ncores  | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of replications R is a large value.                |
| tol     | Numeric. Smallest possible time interval to allow.                                                                                                                          |

## Details

See [IndirectCentral\(\)](#) for more details.

## Value

Returns an object of class `ctmedboot` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("BootIndirectCentral").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**est** A vector of indirect effect centrality.

**thetahatstar** A matrix of bootstrap indirect effect centrality.

## Author(s)

Ivan Jacob Agaloos Pesigan

## References

Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. [doi:10.2307/271028](#)

Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. [doi:10.1080/10705511.2014.973960](#)

Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. [doi:10.1037/met0000779](#)

Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. [doi:10.1007/s11336021097670](#)

## See Also

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

**Examples**

```

## Not run:
library(bootStateSpace)
# prepare parameters
## number of individuals
n <- 50
## time points
time <- 100
delta_t <- 0.10
## dynamic structure
p <- 3
mu0 <- rep(x = 0, times = p)
sigma0 <- matrix(
  data = c(
    1.0,
    0.2,
    0.2,
    0.2,
    1.0,
    0.2,
    0.2,
    0.2,
    1.0
  ),
  nrow = p
)
sigma0_l <- t(chol(sigma0))
mu <- rep(x = 0, times = p)
phi <- matrix(
  data = c(
    -0.357,
    0.771,
    -0.450,
    0.0,
    -0.511,
    0.729,
    0,
    0,
    -0.693
  ),
  nrow = p
)
sigma <- matrix(
  data = c(
    0.24455556,
    0.02201587,
    -0.05004762,
    0.02201587,
    0.07067800,
    0.01539456,
    -0.05004762,
    0.01539456,

```

```

    0.07553061
  ),
  nrow = p
)
sigma_l <- t(chol(sigma))
## measurement model
k <- 3
nu <- rep(x = 0, times = k)
lambda <- diag(k)
theta <- 0.2 * diag(k)
theta_l <- t(chol(theta))

boot <- PBSSMOUFixed(
  R = 10L, # use at least 1000 in actual research
  path = getwd(),
  prefix = "ou",
  n = n,
  time = time,
  delta_t = delta_t,
  mu0 = mu0,
  sigma0_l = sigma0_l,
  mu = mu,
  phi = phi,
  sigma_l = sigma_l,
  nu = nu,
  lambda = lambda,
  theta_l = theta_l,
  ncores = NULL, # consider using multiple cores
  seed = 42
)
phi_hat <- phi
colnames(phi_hat) <- rownames(phi_hat) <- c("x", "m", "y")
phi <- extract(object = boot, what = "phi")

# Specific time interval -----
BootIndirectCentral(
  phi = phi,
  phi_hat = phi_hat,
  delta_t = 1
)

# Range of time intervals -----
boot <- BootIndirectCentral(
  phi = phi,
  phi_hat = phi_hat,
  delta_t = 1:5
)
plot(boot)
plot(boot, type = "bc") # bias-corrected

# Methods -----
# BootIndirectCentral has a number of methods including
# print, summary, confint, and plot

```

```

print(boot)
summary(boot)
confint(boot, level = 0.95)
print(boot, type = "bc") # bias-corrected
summary(boot, type = "bc")
confint(boot, level = 0.95, type = "bc")

## End(Not run)

```

---

## BootIndirectCentralStd

*Bootstrap Sampling Distribution for the Standardized Indirect Effect  
Centrality Over a Specific Time Interval or a Range of Time Intervals*

---

### Description

This function generates a bootstrap method sampling distribution for the standardized indirect effect centrality over a specific time interval  $\Delta t$  or a range of time intervals using the first-order stochastic differential equation model drift matrix  $\Phi$  and process noise covariance matrix  $\Sigma$ .

### Usage

```

BootIndirectCentralStd(
  phi,
  sigma,
  phi_hat,
  sigma_hat,
  delta_t,
  ncores = NULL,
  tol = 0.001
)

```

### Arguments

|           |                                                                                                                                                                             |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phi       | List of numeric matrices. Each element of the list is a bootstrap estimate of the drift matrix ( $\Phi$ ).                                                                  |
| sigma     | List of numeric matrices. Each element of the list is a bootstrap estimate of the process noise covariance matrix ( $\Sigma$ ).                                             |
| phi_hat   | Numeric matrix. The estimated drift matrix ( $\hat{\Phi}$ ) from the original data set. phi_hat should have row and column names pertaining to the variables in the system. |
| sigma_hat | Numeric matrix. The estimated process noise covariance matrix ( $\hat{\Sigma}$ ) from the original data set.                                                                |
| delta_t   | Numeric. Time interval ( $\Delta t$ ).                                                                                                                                      |
| ncores    | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of replications R is a large value.                |
| tol       | Numeric. Smallest possible time interval to allow.                                                                                                                          |

**Details**

See [IndirectCentralStd\(\)](#) for more details.

**Value**

Returns an object of class `ctmedboot` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("BootIndirectCentralStd").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**est** A vector of standardized indirect effect centrality.

**thetahatstar** A matrix of bootstrap standardized indirect effect centrality.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**See Also**

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

**Examples**

```
## Not run:
library(bootStateSpace)
# prepare parameters
## number of individuals
n <- 50
## time points
time <- 100
delta_t <- 0.10
## dynamic structure
p <- 3
mu0 <- rep(x = 0, times = p)
sigma0 <- matrix(
```



```

    data = c(
      1.0,
      0.2,
      0.2,
      0.2,
      1.0,
      0.2,
      0.2,
      0.2,
      1.0
    ),
    nrow = p
  )
  sigma0_l <- t(chol(sigma0))
  mu <- rep(x = 0, times = p)
  phi <- matrix(
    data = c(
      -0.357,
      0.771,
      -0.450,
      0.0,
      -0.511,
      0.729,
      0,
      0,
      -0.693
    ),
    nrow = p
  )
  sigma <- matrix(
    data = c(
      0.24455556,
      0.02201587,
      -0.05004762,
      0.02201587,
      0.07067800,
      0.01539456,
      -0.05004762,
      0.01539456,
      0.07553061
    ),
    nrow = p
  )
  sigma_l <- t(chol(sigma))
  ## measurement model
  k <- 3
  nu <- rep(x = 0, times = k)
  lambda <- diag(k)
  theta <- 0.2 * diag(k)
  theta_l <- t(chol(theta))

  boot <- PBSSMOUFixed(
    R = 10L, # use at least 1000 in actual research

```

```

path = getwd(),
prefix = "ou",
n = n,
time = time,
delta_t = delta_t,
mu0 = mu0,
sigma0_l = sigma0_l,
mu = mu,
phi = phi,
sigma_l = sigma_l,
nu = nu,
lambda = lambda,
theta_l = theta_l,
ncores = NULL, # consider using multiple cores
seed = 42
)
phi_hat <- phi
sigma_hat <- sigma
colnames(phi_hat) <- rownames(phi_hat) <- c("x", "m", "y")
phi <- extract(object = boot, what = "phi")
sigma <- extract(object = boot, what = "sigma")

# Specific time interval -----
BootIndirectCentralStd(
  phi = phi,
  sigma = sigma,
  phi_hat = phi_hat,
  sigma_hat = sigma_hat,
  delta_t = 1
)

# Range of time intervals -----
boot <- BootIndirectCentralStd(
  phi = phi,
  sigma = sigma,
  phi_hat = phi_hat,
  sigma_hat = sigma_hat,
  delta_t = 1:5
)
plot(boot)
plot(boot, type = "bc") # bias-corrected

# Methods -----
# BootIndirectCentralStd has a number of methods including
# print, summary, confint, and plot
print(boot)
summary(boot)
confint(boot, level = 0.95)
print(boot, type = "bc") # bias-corrected
summary(boot, type = "bc")
confint(boot, level = 0.95, type = "bc")

## End(Not run)

```

---

|         |                                                                                                                                                              |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BootMed | <i>Bootstrap Sampling Distribution of Total, Direct, and Indirect Effects of X on Y Through M Over a Specific Time Interval or a Range of Time Intervals</i> |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|

---

## Description

This function generates a bootstrap method sampling distribution of the total, direct and indirect effects of the independent variable  $X$  on the dependent variable  $Y$  through mediator variables  $\mathbf{m}$  over a specific time interval  $\Delta t$  or a range of time intervals using the first-order stochastic differential equation model drift matrix  $\Phi$ .

## Usage

```
BootMed(phi, phi_hat, delta_t, from, to, med, ncores = NULL, tol = 0.001)
```

## Arguments

|         |                                                                                                                                                                             |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phi     | List of numeric matrices. Each element of the list is a bootstrap estimate of the drift matrix ( $\Phi$ ).                                                                  |
| phi_hat | Numeric matrix. The estimated drift matrix ( $\hat{\Phi}$ ) from the original data set. phi_hat should have row and column names pertaining to the variables in the system. |
| delta_t | Numeric. Time interval ( $\Delta t$ ).                                                                                                                                      |
| from    | Character string. Name of the independent variable $X$ in phi.                                                                                                              |
| to      | Character string. Name of the dependent variable $Y$ in phi.                                                                                                                |
| med     | Character vector. Name/s of the mediator variable/s in phi.                                                                                                                 |
| ncores  | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of replications R is a large value.                |
| tol     | Numeric. Smallest possible time interval to allow.                                                                                                                          |

## Details

See [Total\(\)](#), [Direct\(\)](#), and [Indirect\(\)](#) for more details.

## Value

Returns an object of class `ctmedboot` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("BootMed").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**est** A vector of total, direct, and indirect effects.

**thetahatstar** A matrix of bootstrap total, direct, and indirect effects.

### Author(s)

Ivan Jacob Agaloos Pesigan

### References

Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. doi:[10.2307/271028](https://doi.org/10.2307/271028)

Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. doi:[10.1080/10705511.2014.973960](https://doi.org/10.1080/10705511.2014.973960)

Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. doi:[10.1037/met0000779](https://doi.org/10.1037/met0000779)

Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. doi:[10.1007/s11336021097670](https://doi.org/10.1007/s11336021097670)

### See Also

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

### Examples

```
## Not run:
library(bootStateSpace)
# prepare parameters
## number of individuals
n <- 50
## time points
time <- 100
delta_t <- 0.10
## dynamic structure
p <- 3
```

```

mu0 <- rep(x = 0, times = p)
sigma0 <- matrix(
  data = c(
    1.0,
    0.2,
    0.2,
    0.2,
    1.0,
    0.2,
    0.2,
    0.2,
    1.0
  ),
  nrow = p
)
sigma0_l <- t(chol(sigma0))
mu <- rep(x = 0, times = p)
phi <- matrix(
  data = c(
    -0.357,
    0.771,
    -0.450,
    0.0,
    -0.511,
    0.729,
    0,
    0,
    -0.693
  ),
  nrow = p
)
sigma <- matrix(
  data = c(
    0.24455556,
    0.02201587,
    -0.05004762,
    0.02201587,
    0.07067800,
    0.01539456,
    -0.05004762,
    0.01539456,
    0.07553061
  ),
  nrow = p
)
sigma_l <- t(chol(sigma))
## measurement model
k <- 3
nu <- rep(x = 0, times = k)
lambda <- diag(k)
theta <- 0.2 * diag(k)
theta_l <- t(chol(theta))

```

```

boot <- PBSSMOUFixed(
  R = 10L, # use at least 1000 in actual research
  path = getwd(),
  prefix = "ou",
  n = n,
  time = time,
  delta_t = delta_t,
  mu0 = mu0,
  sigma0_l = sigma0_l,
  mu = mu,
  phi = phi,
  sigma_l = sigma_l,
  nu = nu,
  lambda = lambda,
  theta_l = theta_l,
  ncores = NULL, # consider using multiple cores
  seed = 42
)
phi_hat <- phi
colnames(phi_hat) <- rownames(phi_hat) <- c("x", "m", "y")
phi <- extract(object = boot, what = "phi")

# Specific time interval -----
BootMed(
  phi = phi,
  phi_hat = phi_hat,
  delta_t = 1,
  from = "x",
  to = "y",
  med = "m"
)

# Range of time intervals -----
boot <- BootMed(
  phi = phi,
  phi_hat = phi_hat,
  delta_t = 1:5,
  from = "x",
  to = "y",
  med = "m"
)
plot(boot)
plot(boot, type = "bc") # bias-corrected

# Methods -----
# BootMed has a number of methods including
# print, summary, confint, and plot
print(boot)
summary(boot)
confint(boot, level = 0.95)
print(boot, type = "bc") # bias-corrected
summary(boot, type = "bc")
confint(boot, level = 0.95, type = "bc")

```

```
## End(Not run)
```

---

|            |                                                                                                                                                                           |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BootMedStd | <i>Bootstrap Sampling Distribution of Standardized Total, Direct, and Indirect Effects of X on Y Through M Over a Specific Time Interval or a Range of Time Intervals</i> |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

---

## Description

This function generates a bootstrap method sampling distribution of the standardized total, direct and indirect effects of the independent variable  $X$  on the dependent variable  $Y$  through mediator variables  $\mathbf{m}$  over a specific time interval  $\Delta t$  or a range of time intervals using the first-order stochastic differential equation model drift matrix  $\Phi$  and process noise covariance matrix  $\Sigma$ .

## Usage

```
BootMedStd(
  phi,
  sigma,
  phi_hat,
  sigma_hat,
  delta_t,
  from,
  to,
  med,
  ncores = NULL,
  tol = 0.001
)
```

## Arguments

|           |                                                                                                                                                                             |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phi       | List of numeric matrices. Each element of the list is a bootstrap estimate of the drift matrix ( $\Phi$ ).                                                                  |
| sigma     | List of numeric matrices. Each element of the list is a bootstrap estimate of the process noise covariance matrix ( $\Sigma$ ).                                             |
| phi_hat   | Numeric matrix. The estimated drift matrix ( $\hat{\Phi}$ ) from the original data set. phi_hat should have row and column names pertaining to the variables in the system. |
| sigma_hat | Numeric matrix. The estimated process noise covariance matrix ( $\hat{\Sigma}$ ) from the original data set.                                                                |
| delta_t   | Numeric. Time interval ( $\Delta t$ ).                                                                                                                                      |
| from      | Character string. Name of the independent variable $X$ in phi.                                                                                                              |
| to        | Character string. Name of the dependent variable $Y$ in phi.                                                                                                                |

|                     |                                                                                                                                                                                         |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>med</code>    | Character vector. Name/s of the mediator variable/s in <code>phi</code> .                                                                                                               |
| <code>ncores</code> | Positive integer. Number of cores to use. If <code>ncores = NULL</code> , use a single core. Consider using multiple cores when number of replications <code>R</code> is a large value. |
| <code>tol</code>    | Numeric. Smallest possible time interval to allow.                                                                                                                                      |

### Details

See [TotalStd\(\)](#), [DirectStd\(\)](#), and [IndirectStd\(\)](#) for more details.

### Value

Returns an object of class `ctmedboot` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("BootMedStd").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**est** A vector of standardized total, direct, and indirect effects.

**thetahatstar** A matrix of bootstrap standardized total, direct, and indirect effects.

### Author(s)

Ivan Jacob Agaloos Pesigan

### References

- Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. [doi:10.2307/271028](#)
- Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. [doi:10.1080/10705511.2014.973960](#)
- Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. [doi:10.1037/met0000779](#)
- Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. [doi:10.1007/s11336021097670](#)

### See Also

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#),



```

MCDirectCentral(), MCDirectCentralStd(), MCIndirectCentral(), MCIndirectCentralStd(),
MCMed(), MCMedStd(), MCPHi(), MCPHiSigma(), MCTotalCentral(), MCTotalCentralStd(), Med(),
MedStd(), PosteriorBeta(), PosteriorBetaStd(), PosteriorDirectCentral(), PosteriorDirectCentralStd(),
PosteriorIndirectCentral(), PosteriorIndirectCentralStd(), PosteriorMed(), PosteriorMedStd(),
PosteriorTotalCentral(), PosteriorTotalCentralStd(), Total(), TotalCentral(), TotalCentralStd(),
TotalStd(), Trajectory()

```

## Examples

```

## Not run:
library(bootStateSpace)
# prepare parameters
## number of individuals
n <- 50
## time points
time <- 100
delta_t <- 0.10
## dynamic structure
p <- 3
mu0 <- rep(x = 0, times = p)
sigma0 <- matrix(
  data = c(
    1.0,
    0.2,
    0.2,
    0.2,
    1.0,
    0.2,
    0.2,
    0.2,
    1.0
  ),
  nrow = p
)
sigma0_l <- t(chol(sigma0))
mu <- rep(x = 0, times = p)
phi <- matrix(
  data = c(
    -0.357,
    0.771,
    -0.450,
    0.0,
    -0.511,
    0.729,
    0,
    0,
    -0.693
  ),
  nrow = p
)
sigma <- matrix(
  data = c(

```

```

    0.24455556,
    0.02201587,
    -0.05004762,
    0.02201587,
    0.07067800,
    0.01539456,
    -0.05004762,
    0.01539456,
    0.07553061
  ),
  nrow = p
)
sigma_l <- t(chol(sigma))
## measurement model
k <- 3
nu <- rep(x = 0, times = k)
lambda <- diag(k)
theta <- 0.2 * diag(k)
theta_l <- t(chol(theta))

boot <- PBSSMOUFixed(
  R = 10L, # use at least 1000 in actual research
  path = getwd(),
  prefix = "ou",
  n = n,
  time = time,
  delta_t = delta_t,
  mu0 = mu0,
  sigma0_l = sigma0_l,
  mu = mu,
  phi = phi,
  sigma_l = sigma_l,
  nu = nu,
  lambda = lambda,
  theta_l = theta_l,
  ncores = NULL, # consider using multiple cores
  seed = 42
)
phi_hat <- phi
colnames(phi_hat) <- rownames(phi_hat) <- c("x", "m", "y")
sigma_hat <- sigma
phi <- extract(object = boot, what = "phi")
sigma <- extract(object = boot, what = "sigma")

# Specific time interval -----
BootMedStd(
  phi = phi,
  sigma = sigma,
  phi_hat = phi_hat,
  sigma_hat = sigma_hat,
  delta_t = 1,
  from = "x",
  to = "y",

```

```

    med = "m"
  )

  # Range of time intervals -----
  boot <- BootMedStd(
    phi = phi,
    sigma = sigma,
    phi_hat = phi_hat,
    sigma_hat = sigma_hat,
    delta_t = 1:5,
    from = "x",
    to = "y",
    med = "m"
  )
  plot(boot)
  plot(boot, type = "bc") # bias-corrected

  # Methods -----
  # BootMedStd has a number of methods including
  # print, summary, confint, and plot
  print(boot)
  summary(boot)
  confint(boot, level = 0.95)
  print(boot, type = "bc") # bias-corrected
  summary(boot, type = "bc")
  confint(boot, level = 0.95, type = "bc")

  ## End(Not run)

```

---

 BootTotalCentral

---

*Bootstrap Sampling Distribution for the Total Effect Centrality Over a Specific Time Interval or a Range of Time Intervals*


---

## Description

This function generates a bootstrap method sampling distribution for the total effect centrality over a specific time interval  $\Delta t$  or a range of time intervals using the first-order stochastic differential equation model drift matrix  $\Phi$ .

## Usage

```
BootTotalCentral(phi, phi_hat, delta_t, ncores = NULL, tol = 0.001)
```

## Arguments

**phi**                      List of numeric matrices. Each element of the list is a bootstrap estimate of the drift matrix ( $\Phi$ ).

|         |                                                                                                                                                                             |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phi_hat | Numeric matrix. The estimated drift matrix ( $\hat{\Phi}$ ) from the original data set. phi_hat should have row and column names pertaining to the variables in the system. |
| delta_t | Numeric. Time interval ( $\Delta t$ ).                                                                                                                                      |
| ncores  | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of replications R is a large value.                |
| tol     | Numeric. Smallest possible time interval to allow.                                                                                                                          |

## Details

See [TotalCentral\(\)](#) for more details.

## Value

Returns an object of class `ctmedboot` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("BootTotalCentral").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**est** A vector of total effect centrality.

**thetahatstar** A matrix of bootstrap total effect centrality.

## Author(s)

Ivan Jacob Agaloos Pesigan

## References

- Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. doi:10.2307/271028
- Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. doi:10.1080/10705511.2014.973960
- Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. doi:10.1037/met0000779
- Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. doi:10.1007/s11336021097670

**See Also**

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

**Examples**

```
## Not run:
library(bootStateSpace)
# prepare parameters
## number of individuals
n <- 50
## time points
time <- 100
delta_t <- 0.10
## dynamic structure
p <- 3
mu0 <- rep(x = 0, times = p)
sigma0 <- matrix(
  data = c(
    1.0,
    0.2,
    0.2,
    0.2,
    1.0,
    0.2,
    0.2,
    0.2,
    1.0
  ),
  nrow = p
)
sigma0_l <- t(chol(sigma0))
mu <- rep(x = 0, times = p)
phi <- matrix(
  data = c(
    -0.357,
    0.771,
    -0.450,
    0.0,
    -0.511,
    0.729,
  )
)
```

```

      0,
      0,
      -0.693
    ),
    nrow = p
  )
  sigma <- matrix(
    data = c(
      0.24455556,
      0.02201587,
      -0.05004762,
      0.02201587,
      0.07067800,
      0.01539456,
      -0.05004762,
      0.01539456,
      0.07553061
    ),
    nrow = p
  )
  sigma_l <- t(chol(sigma))
  ## measurement model
  k <- 3
  nu <- rep(x = 0, times = k)
  lambda <- diag(k)
  theta <- 0.2 * diag(k)
  theta_l <- t(chol(theta))

  boot <- PBSSMOUFixed(
    R = 10L, # use at least 1000 in actual research
    path = getwd(),
    prefix = "ou",
    n = n,
    time = time,
    delta_t = delta_t,
    mu0 = mu0,
    sigma0_l = sigma0_l,
    mu = mu,
    phi = phi,
    sigma_l = sigma_l,
    nu = nu,
    lambda = lambda,
    theta_l = theta_l,
    ncores = NULL, # consider using multiple cores
    seed = 42
  )
  phi_hat <- phi
  colnames(phi_hat) <- rownames(phi_hat) <- c("x", "m", "y")
  phi <- extract(object = boot, what = "phi")

  # Specific time interval -----
  BootTotalCentral(
    phi = phi,

```

```

    phi_hat = phi_hat,
    delta_t = 1
)

# Range of time intervals -----
boot <- BootTotalCentral(
  phi = phi,
  phi_hat = phi_hat,
  delta_t = 1:5
)
plot(boot)
plot(boot, type = "bc") # bias-corrected

# Methods -----
# BootTotalCentral has a number of methods including
# print, summary, confint, and plot
print(boot)
summary(boot)
confint(boot, level = 0.95)
print(boot, type = "bc") # bias-corrected
summary(boot, type = "bc")
confint(boot, level = 0.95, type = "bc")

## End(Not run)

```

---

|                     |                                                                                                                                                |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| BootTotalCentralStd | <i>Bootstrap Sampling Distribution for the Standardized Total Effect Centrality Over a Specific Time Interval or a Range of Time Intervals</i> |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------|

---

## Description

This function generates a bootstrap method sampling distribution for the standardized total effect centrality over a specific time interval  $\Delta t$  or a range of time intervals using the first-order stochastic differential equation model drift matrix  $\Phi$  and process noise covariance matrix  $\Sigma$ .

## Usage

```

BootTotalCentralStd(
  phi,
  sigma,
  phi_hat,
  sigma_hat,
  delta_t,
  ncores = NULL,
  tol = 0.001
)

```

**Arguments**

|           |                                                                                                                                                                             |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phi       | List of numeric matrices. Each element of the list is a bootstrap estimate of the drift matrix ( $\Phi$ ).                                                                  |
| sigma     | List of numeric matrices. Each element of the list is a bootstrap estimate of the process noise covariance matrix ( $\Sigma$ ).                                             |
| phi_hat   | Numeric matrix. The estimated drift matrix ( $\hat{\Phi}$ ) from the original data set. phi_hat should have row and column names pertaining to the variables in the system. |
| sigma_hat | Numeric matrix. The estimated process noise covariance matrix ( $\hat{\Sigma}$ ) from the original data set.                                                                |
| delta_t   | Numeric. Time interval ( $\Delta t$ ).                                                                                                                                      |
| ncores    | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of replications R is a large value.                |
| tol       | Numeric. Smallest possible time interval to allow.                                                                                                                          |

**Details**

See [TotalCentralStd\(\)](#) for more details.

**Value**

Returns an object of class `ctmedboot` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("BootTotalCentralStd").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**est** A vector of standardized total effect centrality.

**thetahatstar** A matrix of bootstrap standardized total effect centrality.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**References**

Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. [doi:10.2307/271028](#)

Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. [doi:10.1080/10705511.2014.973960](#)

Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. [doi:10.1037/met0000779](#)



Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. doi:[10.1007/s11336021097670](https://doi.org/10.1007/s11336021097670)

### See Also

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

### Examples

```
## Not run:
library(bootStateSpace)
# prepare parameters
## number of individuals
n <- 50
## time points
time <- 100
delta_t <- 0.10
## dynamic structure
p <- 3
mu0 <- rep(x = 0, times = p)
sigma0 <- matrix(
  data = c(
    1.0,
    0.2,
    0.2,
    0.2,
    1.0,
    0.2,
    0.2,
    0.2,
    1.0
  ),
  nrow = p
)
sigma0_l <- t(chol(sigma0))
mu <- rep(x = 0, times = p)
phi <- matrix(
  data = c(
    -0.357,
    0.771,
    -0.450,
```

```

    0.0,
    -0.511,
    0.729,
    0,
    0,
    -0.693
  ),
  nrow = p
)
sigma <- matrix(
  data = c(
    0.24455556,
    0.02201587,
    -0.05004762,
    0.02201587,
    0.07067800,
    0.01539456,
    -0.05004762,
    0.01539456,
    0.07553061
  ),
  nrow = p
)
sigma_l <- t(chol(sigma))
## measurement model
k <- 3
nu <- rep(x = 0, times = k)
lambda <- diag(k)
theta <- 0.2 * diag(k)
theta_l <- t(chol(theta))

boot <- PBSSMOUFixed(
  R = 10L, # use at least 1000 in actual research
  path = getwd(),
  prefix = "ou",
  n = n,
  time = time,
  delta_t = delta_t,
  mu0 = mu0,
  sigma0_l = sigma0_l,
  mu = mu,
  phi = phi,
  sigma_l = sigma_l,
  nu = nu,
  lambda = lambda,
  theta_l = theta_l,
  ncores = NULL, # consider using multiple cores
  seed = 42
)
phi_hat <- phi
sigma_hat <- sigma
colnames(phi_hat) <- rownames(phi_hat) <- c("x", "m", "y")
phi <- extract(object = boot, what = "phi")

```

```

sigma <- extract(object = boot, what = "sigma")

# Specific time interval -----
BootTotalCentralStd(
  phi = phi,
  sigma = sigma,
  phi_hat = phi_hat,
  sigma_hat = sigma_hat,
  delta_t = 1
)

# Range of time intervals -----
boot <- BootTotalCentralStd(
  phi = phi,
  sigma = sigma,
  phi_hat = phi_hat,
  sigma_hat = sigma_hat,
  delta_t = 1:5
)
plot(boot)
plot(boot, type = "bc") # bias-corrected

# Methods -----
# BootTotalCentralStd has a number of methods including
# print, summary, confint, and plot
print(boot)
summary(boot)
confint(boot, level = 0.95)
print(boot, type = "bc") # bias-corrected
summary(boot, type = "bc")
confint(boot, level = 0.95, type = "bc")

## End(Not run)

```

---

|                   |                                              |
|-------------------|----------------------------------------------|
| confint.ctmedboot | <i>Bootstrap Method Confidence Intervals</i> |
|-------------------|----------------------------------------------|

---

## Description

Bootstrap Method Confidence Intervals

## Usage

```

## S3 method for class 'ctmedboot'
confint(object, parm = NULL, level = 0.95, type = "pc", ...)

```

**Arguments**

|                     |                                                                                                                                                                       |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>object</code> | Object of class <code>ctmedboot</code> .                                                                                                                              |
| <code>parm</code>   | a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered. |
| <code>level</code>  | the confidence level required.                                                                                                                                        |
| <code>type</code>   | Character string. Confidence interval type, that is, <code>type = "pc"</code> for percentile; <code>type = "bc"</code> for bias corrected.                            |
| <code>...</code>    | additional arguments.                                                                                                                                                 |

**Value**

Returns a data frame of confidence intervals.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**Examples**

```
## Not run:
library(bootStateSpace)
# prepare parameters
## number of individuals
n <- 50
## time points
time <- 100
delta_t <- 0.10
## dynamic structure
p <- 3
mu0 <- rep(x = 0, times = p)
sigma0 <- matrix(
  data = c(
    1.0,
    0.2,
    0.2,
    0.2,
    1.0,
    0.2,
    0.2,
    0.2,
    1.0
  ),
  nrow = p
)
sigma0_1 <- t(chol(sigma0))
mu <- rep(x = 0, times = p)
phi <- matrix(
  data = c(
    -0.357,
```

```

      0.771,
      -0.450,
      0.0,
      -0.511,
      0.729,
      0,
      0,
      -0.693
    ),
    nrow = p
  )
sigma <- matrix(
  data = c(
    0.24455556,
    0.02201587,
    -0.05004762,
    0.02201587,
    0.07067800,
    0.01539456,
    -0.05004762,
    0.01539456,
    0.07553061
  ),
  nrow = p
)
sigma_l <- t(chol(sigma))
## measurement model
k <- 3
nu <- rep(x = 0, times = k)
lambda <- diag(k)
theta <- 0.2 * diag(k)
theta_l <- t(chol(theta))

boot <- PBSSMOUFixed(
  R = 1000L,
  path = getwd(),
  prefix = "ou",
  n = n,
  time = time,
  delta_t = delta_t,
  mu0 = mu0,
  sigma0_l = sigma0_l,
  mu = mu,
  phi = phi,
  sigma_l = sigma_l,
  nu = nu,
  lambda = lambda,
  theta_l = theta_l,
  ncores = parallel::detectCores() - 1,
  seed = 42
)
phi_hat <- phi
colnames(phi_hat) <- rownames(phi_hat) <- c("x", "m", "y")

```

```

phi <- extract(object = boot, what = "phi")

# Specific time interval -----
boot <- BootMed(
  phi = phi,
  phi_hat = phi_hat,
  delta_t = 1,
  from = "x",
  to = "y",
  med = "m"
)
confint(boot)
confint(boot, type = "bc") # bias-corrected

# Range of time intervals -----
boot <- BootMed(
  phi = phi,
  phi_hat = phi_hat,
  delta_t = 1:5,
  from = "x",
  to = "y",
  med = "m"
)
confint(boot)
confint(boot, type = "bc") # bias-corrected

## End(Not run)

```

---

confint.ctmeddelta      *Delta Method Confidence Intervals*

---

## Description

Delta Method Confidence Intervals

## Usage

```

## S3 method for class 'ctmeddelta'
confint(object, parm = NULL, level = 0.95, ...)

```

## Arguments

|        |                                                                                                                                                                       |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| object | Object of class ctmeddelta.                                                                                                                                           |
| parm   | a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered. |
| level  | the confidence level required.                                                                                                                                        |
| ...    | additional arguments.                                                                                                                                                 |

**Value**

Returns a data frame of confidence intervals.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**Examples**

```
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
vcov_phi_vec <- matrix(
  data = c(
    0.00843, 0.00040, -0.00151,
    -0.00600, -0.00033, 0.00110,
    0.00324, 0.00020, -0.00061,
    0.00040, 0.00374, 0.00016,
    -0.00022, -0.00273, -0.00016,
    0.00009, 0.00150, 0.00012,
    -0.00151, 0.00016, 0.00389,
    0.00103, -0.00007, -0.00283,
    -0.00050, 0.00000, 0.00156,
    -0.00600, -0.00022, 0.00103,
    0.00644, 0.00031, -0.00119,
    -0.00374, -0.00021, 0.00070,
    -0.00033, -0.00273, -0.00007,
    0.00031, 0.00287, 0.00013,
    -0.00014, -0.00170, -0.00012,
    0.00110, -0.00016, -0.00283,
    -0.00119, 0.00013, 0.00297,
    0.00063, -0.00004, -0.00177,
    0.00324, 0.00009, -0.00050,
    -0.00374, -0.00014, 0.00063,
    0.00495, 0.00024, -0.00093,
    0.00020, 0.00150, 0.00000,
    -0.00021, -0.00170, -0.00004,
    0.00024, 0.00214, 0.00012,
    -0.00061, 0.00012, 0.00156,
    0.00070, -0.00012, -0.00177,
    -0.00093, 0.00012, 0.00223
  ),
  nrow = 9
)

# Specific time interval -----
```

```

delta <- DeltaMed(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  delta_t = 1,
  from = "x",
  to = "y",
  med = "m"
)
confint(delta, level = 0.95)

# Range of time intervals -----
delta <- DeltaMed(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  delta_t = 1:5,
  from = "x",
  to = "y",
  med = "m"
)
confint(delta, level = 0.95)

```

---

confint.ctmedmc

*Monte Carlo Method Confidence Intervals*


---

## Description

Monte Carlo Method Confidence Intervals

## Usage

```

## S3 method for class 'ctmedmc'
confint(object, parm = NULL, level = 0.95, ...)

```

## Arguments

|        |                                                                                                                                                                       |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| object | Object of class ctmedmc.                                                                                                                                              |
| parm   | a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered. |
| level  | the confidence level required.                                                                                                                                        |
| ...    | additional arguments.                                                                                                                                                 |

## Value

Returns a data frame of confidence intervals.



**Author(s)**

Ivan Jacob Agaloos Pesigan

**Examples**

```

set.seed(42)
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
vcov_phi_vec <- matrix(
  data = c(
    0.00843, 0.00040, -0.00151,
    -0.00600, -0.00033, 0.00110,
    0.00324, 0.00020, -0.00061,
    0.00040, 0.00374, 0.00016,
    -0.00022, -0.00273, -0.00016,
    0.00009, 0.00150, 0.00012,
    -0.00151, 0.00016, 0.00389,
    0.00103, -0.00007, -0.00283,
    -0.00050, 0.00000, 0.00156,
    -0.00600, -0.00022, 0.00103,
    0.00644, 0.00031, -0.00119,
    -0.00374, -0.00021, 0.00070,
    -0.00033, -0.00273, -0.00007,
    0.00031, 0.00287, 0.00013,
    -0.00014, -0.00170, -0.00012,
    0.00110, -0.00016, -0.00283,
    -0.00119, 0.00013, 0.00297,
    0.00063, -0.00004, -0.00177,
    0.00324, 0.00009, -0.00050,
    -0.00374, -0.00014, 0.00063,
    0.00495, 0.00024, -0.00093,
    0.00020, 0.00150, 0.00000,
    -0.00021, -0.00170, -0.00004,
    0.00024, 0.00214, 0.00012,
    -0.00061, 0.00012, 0.00156,
    0.00070, -0.00012, -0.00177,
    -0.00093, 0.00012, 0.00223
  ),
  nrow = 9
)

# Specific time interval -----
mc <- MCMed(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,

```

```

    delta_t = 1,
    from = "x",
    to = "y",
    med = "m",
    R = 100L # use a large value for R in actual research
  )
  confint(mc, level = 0.95)

# Range of time intervals -----
mc <- MCMed(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  delta_t = 1:5,
  from = "x",
  to = "y",
  med = "m",
  R = 100L # use a large value for R in actual research
)
confint(mc, level = 0.95)

```

DeltaBeta

*Delta Method Sampling Variance-Covariance Matrix for the Elements of the Matrix of Lagged Coefficients Over a Specific Time Interval or a Range of Time Intervals*

## Description

This function computes the delta method sampling variance-covariance matrix for the elements of the matrix of lagged coefficients  $\beta$  over a specific time interval  $\Delta t$  or a range of time intervals using the first-order stochastic differential equation model's drift matrix  $\Phi$ .

## Usage

```
DeltaBeta(phi, vcov_phi_vec, delta_t, ncores = NULL, tol = 0.001)
```

## Arguments

|              |                                                                                                                                                  |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| phi          | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system.                     |
| vcov_phi_vec | Numeric matrix. The sampling variance-covariance matrix of $\text{vec}(\Phi)$ .                                                                  |
| delta_t      | Vector of positive numbers. Time interval ( $\Delta t$ ).                                                                                        |
| ncores       | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when the length of delta_t is long. |
| tol          | Numeric. Smallest possible time interval to allow.                                                                                               |

## Details

See `Total()`.

### Delta Method:

Let  $\theta$  be  $\text{vec}(\Phi)$ , that is, the elements of the  $\Phi$  matrix in vector form sorted column-wise. Let  $\hat{\theta}$  be  $\text{vec}(\hat{\Phi})$ . By the multivariate central limit theory, the function  $\mathbf{g}$  using  $\hat{\theta}$  as input can be expressed as:

$$\sqrt{n} \left( \mathbf{g}(\hat{\theta}) - \mathbf{g}(\theta) \right) \xrightarrow{D} \mathcal{N}(0, \mathbf{J}\mathbf{\Gamma}\mathbf{J}')$$

where  $\mathbf{J}$  is the matrix of first-order derivatives of the function  $\mathbf{g}$  with respect to the elements of  $\theta$  and  $\mathbf{\Gamma}$  is the asymptotic variance-covariance matrix of  $\hat{\theta}$ .

From the former, we can derive the distribution of  $\mathbf{g}(\hat{\theta})$  as follows:

$$\mathbf{g}(\hat{\theta}) \approx \mathcal{N}(\mathbf{g}(\theta), n^{-1}\mathbf{J}\mathbf{\Gamma}\mathbf{J}')$$

The uncertainty associated with the estimator  $\mathbf{g}(\hat{\theta})$  is, therefore, given by  $n^{-1}\mathbf{J}\mathbf{\Gamma}\mathbf{J}'$ . When  $\mathbf{\Gamma}$  is unknown, by substitution, we can use the estimated sampling variance-covariance matrix of  $\hat{\theta}$ , that is,  $\hat{\mathbf{V}}(\hat{\theta})$  for  $n^{-1}\mathbf{\Gamma}$ . Therefore, the sampling variance-covariance matrix of  $\mathbf{g}(\hat{\theta})$  is given by

$$\mathbf{g}(\hat{\theta}) \approx \mathcal{N}(\mathbf{g}(\theta), \mathbf{J}\hat{\mathbf{V}}(\hat{\theta})\mathbf{J}').$$

## Value

Returns an object of class `ctmeddelta` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("DeltaBeta").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**delta\_t** Time interval.

**jacobian** Jacobian matrix.

**est** Estimated elements of the matrix of lagged coefficients.

**vcov** Sampling variance-covariance matrix of estimated elements of the matrix of lagged coefficients.

## Author(s)

Ivan Jacob Agaloos Pesigan

## References

- Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. doi:10.2307/271028
- Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. doi:10.1080/10705511.2014.973960
- Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. doi:10.1037/met0000779
- Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. doi:10.1007/s11336021097670

## See Also

Other Continuous-Time Mediation Functions: `BootBeta()`, `BootBetaStd()`, `BootDirectCentral()`, `BootDirectCentralStd()`, `BootIndirectCentral()`, `BootIndirectCentralStd()`, `BootMed()`, `BootMedStd()`, `BootTotalCentral()`, `BootTotalCentralStd()`, `DeltaBetaStd()`, `DeltaDirectCentral()`, `DeltaDirectCentralStd()`, `DeltaIndirectCentral()`, `DeltaMed()`, `DeltaMedStd()`, `DeltaTotalCentral()`, `DeltaTotalCentralStd()`, `Direct()`, `DirectCentral()`, `DirectCentralStd()`, `DirectStd()`, `Indirect()`, `IndirectCentral()`, `IndirectCentralStd()`, `IndirectStd()`, `MCBeta()`, `MCBetaStd()`, `MCDirectCentral()`, `MCDirectCentralStd()`, `MCIndirectCentral()`, `MCIndirectCentralStd()`, `MCMed()`, `MCMedStd()`, `MCPhi()`, `MCPhiSigma()`, `MCTotalCentral()`, `MCTotalCentralStd()`, `Med()`, `MedStd()`, `PosteriorBeta()`, `PosteriorBetaStd()`, `PosteriorDirectCentral()`, `PosteriorDirectCentralStd()`, `PosteriorIndirectCentral()`, `PosteriorIndirectCentralStd()`, `PosteriorMed()`, `PosteriorMedStd()`, `PosteriorTotalCentral()`, `PosteriorTotalCentralStd()`, `Total()`, `TotalCentral()`, `TotalCentralStd()`, `TotalStd()`, `Trajectory()`

## Examples

```
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
vcov_phi_vec <- matrix(
  data = c(
    0.00843, 0.00040, -0.00151,
    -0.00600, -0.00033, 0.00110,
    0.00324, 0.00020, -0.00061,
    0.00040, 0.00374, 0.00016,
    -0.00022, -0.00273, -0.00016,
    0.00009, 0.00150, 0.00012,
    -0.00151, 0.00016, 0.00389,
    0.00103, -0.00007, -0.00283,
    -0.00050, 0.00000, 0.00156,
  )
)
```

```

-0.00600, -0.00022, 0.00103,
0.00644, 0.00031, -0.00119,
-0.00374, -0.00021, 0.00070,
-0.00033, -0.00273, -0.00007,
0.00031, 0.00287, 0.00013,
-0.00014, -0.00170, -0.00012,
0.00110, -0.00016, -0.00283,
-0.00119, 0.00013, 0.00297,
0.00063, -0.00004, -0.00177,
0.00324, 0.00009, -0.00050,
-0.00374, -0.00014, 0.00063,
0.00495, 0.00024, -0.00093,
0.00020, 0.00150, 0.00000,
-0.00021, -0.00170, -0.00004,
0.00024, 0.00214, 0.00012,
-0.00061, 0.00012, 0.00156,
0.00070, -0.00012, -0.00177,
-0.00093, 0.00012, 0.00223
),
nrow = 9
)

# Specific time interval -----
DeltaBeta(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  delta_t = 1
)

# Range of time intervals -----
delta <- DeltaBeta(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  delta_t = 1:5
)
plot(delta)

# Methods -----
# DeltaBeta has a number of methods including
# print, summary, confint, and plot
print(delta)
summary(delta)
confint(delta, level = 0.95)
plot(delta)

```

## Description

This function computes the delta method sampling variance-covariance matrix for the elements of the standardized matrix of lagged coefficients  $\beta$  over a specific time interval  $\Delta t$  or a range of time intervals using the first-order stochastic differential equation model's drift matrix  $\Phi$  and process noise covariance matrix  $\Sigma$ .

## Usage

```
DeltaBetaStd(
  phi,
  sigma,
  vcov_theta,
  delta_t,
  sigma_diag = FALSE,
  ncores = NULL,
  tol = 0.001
)
```

## Arguments

|            |                                                                                                                                                              |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phi        | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system.                                 |
| sigma      | Numeric matrix. The process noise covariance matrix ( $\Sigma$ ).                                                                                            |
| vcov_theta | Numeric matrix. The sampling variance-covariance matrix of $\text{vec}(\Phi)$ and $\text{vech}(\Sigma)$                                                      |
| delta_t    | Numeric. Time interval ( $\Delta t$ ).                                                                                                                       |
| sigma_diag | Logical. If sigma_diag = TRUE, treat $\Sigma$ as a diagonal matrix.                                                                                          |
| ncores     | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of replications R is a large value. |
| tol        | Numeric. Smallest possible time interval to allow.                                                                                                           |

## Details

See [TotalStd\(\)](#).

### Delta Method:

Let  $\theta$  be a vector that combines  $\text{vec}(\Phi)$ , that is, the elements of the  $\Phi$  matrix in vector form sorted column-wise and  $\text{vech}(\Sigma)$ , that is, the unique elements of the  $\Sigma$  matrix in vector form sorted column-wise. Let  $\hat{\theta}$  be a vector that combines  $\text{vec}(\hat{\Phi})$  and  $\text{vech}(\hat{\Sigma})$ . By the multivariate central limit theory, the function  $\mathbf{g}$  using  $\hat{\theta}$  as input can be expressed as:

$$\sqrt{n} \left( \mathbf{g}(\hat{\theta}) - \mathbf{g}(\theta) \right) \xrightarrow{D} \mathcal{N}(0, \mathbf{J}\mathbf{\Gamma}\mathbf{J}')$$

where  $\mathbf{J}$  is the matrix of first-order derivatives of the function  $\mathbf{g}$  with respect to the elements of  $\theta$  and  $\mathbf{\Gamma}$  is the asymptotic variance-covariance matrix of  $\hat{\theta}$ .

From the former, we can derive the distribution of  $\mathbf{g}(\hat{\theta})$  as follows:

$$\mathbf{g}(\hat{\boldsymbol{\theta}}) \approx \mathcal{N}(\mathbf{g}(\boldsymbol{\theta}), n^{-1} \mathbf{J} \boldsymbol{\Gamma} \mathbf{J}')$$

The uncertainty associated with the estimator  $\mathbf{g}(\hat{\boldsymbol{\theta}})$  is, therefore, given by  $n^{-1} \mathbf{J} \boldsymbol{\Gamma} \mathbf{J}'$ . When  $\boldsymbol{\Gamma}$  is unknown, by substitution, we can use the estimated sampling variance-covariance matrix of  $\hat{\boldsymbol{\theta}}$ , that is,  $\hat{\mathbf{V}}(\hat{\boldsymbol{\theta}})$  for  $n^{-1} \boldsymbol{\Gamma}$ . Therefore, the sampling variance-covariance matrix of  $\mathbf{g}(\hat{\boldsymbol{\theta}})$  is given by

$$\mathbf{g}(\hat{\boldsymbol{\theta}}) \approx \mathcal{N}(\mathbf{g}(\boldsymbol{\theta}), \mathbf{J} \hat{\mathbf{V}}(\hat{\boldsymbol{\theta}}) \mathbf{J}').$$

### Value

Returns an object of class `ctmeddelta` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("DeltaBetaStd").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**delta\_t** Time interval.

**jacobian** Jacobian matrix.

**est** Estimated elements of the standardized matrix of lagged coefficients.

**vcov** Sampling variance-covariance matrix of estimated elements of the standardized matrix of lagged coefficients.

### Author(s)

Ivan Jacob Agaloos Pesigan

### References

- Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. [doi:10.2307/271028](https://doi.org/10.2307/271028)
- Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. [doi:10.1080/10705511.2014.973960](https://doi.org/10.1080/10705511.2014.973960)
- Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. [doi:10.1037/met0000779](https://doi.org/10.1037/met0000779)
- Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. [doi:10.1007/s11336021097670](https://doi.org/10.1007/s11336021097670)

## See Also

Other Continuous-Time Mediation Functions: `BootBeta()`, `BootBetaStd()`, `BootDirectCentral()`, `BootDirectCentralStd()`, `BootIndirectCentral()`, `BootIndirectCentralStd()`, `BootMed()`, `BootMedStd()`, `BootTotalCentral()`, `BootTotalCentralStd()`, `DeltaBeta()`, `DeltaDirectCentral()`, `DeltaDirectCentralStd()`, `DeltaIndirectCentral()`, `DeltaMed()`, `DeltaMedStd()`, `DeltaTotalCentral()`, `DeltaTotalCentralStd()`, `Direct()`, `DirectCentral()`, `DirectCentralStd()`, `DirectStd()`, `Indirect()`, `IndirectCentral()`, `IndirectCentralStd()`, `IndirectStd()`, `MCBeta()`, `MCBetaStd()`, `MCDirectCentral()`, `MCDirectCentralStd()`, `MCIndirectCentral()`, `MCIndirectCentralStd()`, `MCMed()`, `MCMedStd()`, `MCPhi()`, `MCPhiSigma()`, `MCTotalCentral()`, `MCTotalCentralStd()`, `Med()`, `MedStd()`, `PosteriorBeta()`, `PosteriorBetaStd()`, `PosteriorDirectCentral()`, `PosteriorDirectCentralStd()`, `PosteriorIndirectCentral()`, `PosteriorIndirectCentralStd()`, `PosteriorMed()`, `PosteriorMedStd()`, `PosteriorTotalCentral()`, `PosteriorTotalCentralStd()`, `Total()`, `TotalCentral()`, `TotalCentralStd()`, `TotalStd()`, `Trajectory()`

## Examples

```
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
sigma <- matrix(
  data = c(
    0.24455556, 0.02201587, -0.05004762,
    0.02201587, 0.07067800, 0.01539456,
    -0.05004762, 0.01539456, 0.07553061
  ),
  nrow = 3
)
vcov_theta <- matrix(
  data = c(
    0.00843, 0.00040, -0.00151, -0.00600, -0.00033,
    0.00110, 0.00324, 0.00020, -0.00061, -0.00115,
    0.00011, 0.00015, 0.00001, -0.00002, -0.00001,
    0.00040, 0.00374, 0.00016, -0.00022, -0.00273,
    -0.00016, 0.00009, 0.00150, 0.00012, -0.00010,
    -0.00026, 0.00002, 0.00012, 0.00004, -0.00001,
    -0.00151, 0.00016, 0.00389, 0.00103, -0.00007,
    -0.00283, -0.00050, 0.00000, 0.00156, 0.00021,
    -0.00005, -0.00031, 0.00001, 0.00007, 0.00006,
    -0.00600, -0.00022, 0.00103, 0.00644, 0.00031,
    -0.00119, -0.00374, -0.00021, 0.00070, 0.00064,
    -0.00015, -0.00005, 0.00000, 0.00003, -0.00001,
    -0.00033, -0.00273, -0.00007, 0.00031, 0.00287,
    0.00013, -0.00014, -0.00170, -0.00012, 0.00006,
    0.00014, -0.00001, -0.00015, 0.00000, 0.00001,
    0.00110, -0.00016, -0.00283, -0.00119, 0.00013,
  ),
  nrow = 3
)
```



```

0.00297, 0.00063, -0.00004, -0.00177, -0.00013,
0.00005, 0.00017, -0.00002, -0.00008, 0.00001,
0.00324, 0.00009, -0.00050, -0.00374, -0.00014,
0.00063, 0.00495, 0.00024, -0.00093, -0.00020,
0.00006, -0.00010, 0.00000, -0.00001, 0.00004,
0.00020, 0.00150, 0.00000, -0.00021, -0.00170,
-0.00004, 0.00024, 0.00214, 0.00012, -0.00002,
-0.00004, 0.00000, 0.00006, -0.00005, -0.00001,
-0.00061, 0.00012, 0.00156, 0.00070, -0.00012,
-0.00177, -0.00093, 0.00012, 0.00223, 0.00004,
-0.00002, -0.00003, 0.00001, 0.00003, -0.00013,
-0.00115, -0.00010, 0.00021, 0.00064, 0.00006,
-0.00013, -0.00020, -0.00002, 0.00004, 0.00057,
0.00001, -0.00009, 0.00000, 0.00000, 0.00001,
0.00011, -0.00026, -0.00005, -0.00015, 0.00014,
0.00005, 0.00006, -0.00004, -0.00002, 0.00001,
0.00012, 0.00001, 0.00000, -0.00002, 0.00000,
0.00015, 0.00002, -0.00031, -0.00005, -0.00001,
0.00017, -0.00010, 0.00000, -0.00003, -0.00009,
0.00001, 0.00014, 0.00000, 0.00000, -0.00005,
0.00001, 0.00012, 0.00001, 0.00000, -0.00015,
-0.00002, 0.00000, 0.00006, 0.00001, 0.00000,
0.00000, 0.00000, 0.00010, 0.00001, 0.00000,
-0.00002, 0.00004, 0.00007, 0.00003, 0.00000,
-0.00008, -0.00001, -0.00005, 0.00003, 0.00000,
-0.00002, 0.00000, 0.00001, 0.00005, 0.00001,
-0.00001, -0.00001, 0.00006, -0.00001, 0.00001,
0.00001, 0.00004, -0.00001, -0.00013, 0.00001,
0.00000, -0.00005, 0.00000, 0.00001, 0.00012
),
nrow = 15
)

# Specific time interval -----
DeltaBetaStd(
  phi = phi,
  sigma = sigma,
  vcov_theta = vcov_theta,
  delta_t = 1
)

# Range of time intervals -----
delta <- DeltaBetaStd(
  phi = phi,
  sigma = sigma,
  vcov_theta = vcov_theta,
  delta_t = 1:5
)
plot(delta)

# Methods -----
# DeltaBetaStd has a number of methods including
# print, summary, confint, and plot

```

```
print(delta)
summary(delta)
confint(delta, level = 0.95)
plot(delta)
```

---

|                    |                                                                                                                                                     |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| DeltaDirectCentral | <i>Delta Method Sampling Variance-Covariance Matrix for the Direct Effect Centrality Over a Specific Time Interval or a Range of Time Intervals</i> |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|

---

### Description

This function computes the delta method sampling variance-covariance matrix for the direct effect centrality over a specific time interval  $\Delta t$  or a range of time intervals using the first-order stochastic differential equation model's drift matrix  $\Phi$ .

### Usage

```
DeltaDirectCentral(phi, vcov_phi_vec, delta_t, ncores = NULL, tol = 0.01)
```

### Arguments

|              |                                                                                                                                                  |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| phi          | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system.                     |
| vcov_phi_vec | Numeric matrix. The sampling variance-covariance matrix of $\text{vec}(\Phi)$ .                                                                  |
| delta_t      | Vector of positive numbers. Time interval ( $\Delta t$ ).                                                                                        |
| ncores       | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when the length of delta_t is long. |
| tol          | Numeric. Smallest possible time interval to allow.                                                                                               |

### Details

See [DirectCentral\(\)](#) for more details.

#### Delta Method:

Let  $\theta$  be  $\text{vec}(\Phi)$ , that is, the elements of the  $\Phi$  matrix in vector form sorted column-wise. Let  $\hat{\theta}$  be  $\text{vec}(\hat{\Phi})$ . By the multivariate central limit theory, the function  $g$  using  $\hat{\theta}$  as input can be expressed as:

$$\sqrt{n} \left( g(\hat{\theta}) - g(\theta) \right) \xrightarrow{D} \mathcal{N}(0, \mathbf{J} \mathbf{\Gamma} \mathbf{J}')$$

where  $\mathbf{J}$  is the matrix of first-order derivatives of the function  $g$  with respect to the elements of  $\theta$  and  $\mathbf{\Gamma}$  is the asymptotic variance-covariance matrix of  $\hat{\theta}$ .

From the former, we can derive the distribution of  $g(\hat{\theta})$  as follows:

$$\mathbf{g}(\hat{\boldsymbol{\theta}}) \approx \mathcal{N}(\mathbf{g}(\boldsymbol{\theta}), n^{-1} \mathbf{J} \boldsymbol{\Gamma} \mathbf{J}')$$

The uncertainty associated with the estimator  $\mathbf{g}(\hat{\boldsymbol{\theta}})$  is, therefore, given by  $n^{-1} \mathbf{J} \boldsymbol{\Gamma} \mathbf{J}'$ . When  $\boldsymbol{\Gamma}$  is unknown, by substitution, we can use the estimated sampling variance-covariance matrix of  $\hat{\boldsymbol{\theta}}$ , that is,  $\hat{\mathbf{V}}(\hat{\boldsymbol{\theta}})$  for  $n^{-1} \boldsymbol{\Gamma}$ . Therefore, the sampling variance-covariance matrix of  $\mathbf{g}(\hat{\boldsymbol{\theta}})$  is given by

$$\mathbf{g}(\hat{\boldsymbol{\theta}}) \approx \mathcal{N}(\mathbf{g}(\boldsymbol{\theta}), \mathbf{J} \hat{\mathbf{V}}(\hat{\boldsymbol{\theta}}) \mathbf{J}').$$

### Value

Returns an object of class `ctmeddelta` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("DeltaDirectCentral").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**delta\_t** Time interval.

**jacobian** Jacobian matrix.

**est** Estimated direct effect centrality.

**vcov** Sampling variance-covariance matrix of estimated direct effect centrality.

### Author(s)

Ivan Jacob Agaloos Pesigan

### References

- Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. [doi:10.2307/271028](https://doi.org/10.2307/271028)
- Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. [doi:10.1080/10705511.2014.973960](https://doi.org/10.1080/10705511.2014.973960)
- Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. [doi:10.1037/met0000779](https://doi.org/10.1037/met0000779)
- Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. [doi:10.1007/s11336021097670](https://doi.org/10.1007/s11336021097670)

## See Also

Other Continuous-Time Mediation Functions: `BootBeta()`, `BootBetaStd()`, `BootDirectCentral()`, `BootDirectCentralStd()`, `BootIndirectCentral()`, `BootIndirectCentralStd()`, `BootMed()`, `BootMedStd()`, `BootTotalCentral()`, `BootTotalCentralStd()`, `DeltaBeta()`, `DeltaBetaStd()`, `DeltaDirectCentralStd()`, `DeltaIndirectCentral()`, `DeltaMed()`, `DeltaMedStd()`, `DeltaTotalCentral()`, `DeltaTotalCentralStd()`, `Direct()`, `DirectCentral()`, `DirectCentralStd()`, `DirectStd()`, `Indirect()`, `IndirectCentral()`, `IndirectCentralStd()`, `IndirectStd()`, `MCBeta()`, `MCBetaStd()`, `MCDirectCentral()`, `MCDirectCentralStd()`, `MCIndirectCentral()`, `MCIndirectCentralStd()`, `MCMed()`, `MCMedStd()`, `MCPhi()`, `MCPhiSigma()`, `MCTotalCentral()`, `MCTotalCentralStd()`, `Med()`, `MedStd()`, `PosteriorBeta()`, `PosteriorBetaStd()`, `PosteriorDirectCentral()`, `PosteriorDirectCentralStd()`, `PosteriorIndirectCentral()`, `PosteriorIndirectCentralStd()`, `PosteriorMed()`, `PosteriorMedStd()`, `PosteriorTotalCentral()`, `PosteriorTotalCentralStd()`, `Total()`, `TotalCentral()`, `TotalCentralStd()`, `TotalStd()`, `Trajectory()`

## Examples

```
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
vcov_phi_vec <- matrix(
  data = c(
    0.002704274, -0.001475275, 0.000949122,
    -0.001619422, 0.000885122, -0.000569404,
    0.00085493, -0.000465824, 0.000297815,
    -0.001475275, 0.004428442, -0.002642303,
    0.000980573, -0.00271817, 0.001618805,
    -0.000586921, 0.001478421, -0.000871547,
    0.000949122, -0.002642303, 0.006402668,
    -0.000697798, 0.001813471, -0.004043138,
    0.000463086, -0.001120949, 0.002271711,
    -0.001619422, 0.000980573, -0.000697798,
    0.002079286, -0.001152501, 0.000753,
    -0.001528701, 0.000820587, -0.000517524,
    0.000885122, -0.00271817, 0.001813471,
    -0.001152501, 0.00342605, -0.002075005,
    0.000899165, -0.002532849, 0.001475579,
    -0.000569404, 0.001618805, -0.004043138,
    0.000753, -0.002075005, 0.004984032,
    -0.000622255, 0.001634917, -0.003705661,
    0.00085493, -0.000586921, 0.000463086,
    -0.001528701, 0.000899165, -0.000622255,
    0.002060076, -0.001096684, 0.000686386,
    -0.000465824, 0.001478421, -0.001120949,
    0.000820587, -0.002532849, 0.001634917,
    -0.001096684, 0.003328692, -0.001926088,
  )
)
```

```

      0.000297815, -0.000871547, 0.002271711,
      -0.000517524, 0.001475579, -0.003705661,
      0.000686386, -0.001926088, 0.004726235
    ),
    nrow = 9
  )

# Specific time interval -----
DeltaDirectCentral(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  delta_t = 1
)

# Range of time intervals -----
delta <- DeltaDirectCentral(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  delta_t = 1:5
)
plot(delta)

# Methods -----
# DeltaDirectCentral has a number of methods including
# print, summary, confint, and plot
print(delta)
summary(delta)
confint(delta, level = 0.95)
plot(delta)

```

---

DeltaDirectCentralStd *Delta Method Sampling Variance-Covariance Matrix for the Standardized Direct Effect Centrality Over a Specific Time Interval or a Range of Time Intervals*

---

## Description

This function computes the delta method sampling variance-covariance matrix for the standardized direct effect centrality over a specific time interval  $\Delta t$  or a range of time intervals using the first-order stochastic differential equation model's drift matrix  $\Phi$  and process noise covariance matrix  $\Sigma$ .

## Usage

```

DeltaDirectCentralStd(
  phi,
  sigma,
  vcov_theta,

```

```

    delta_t,
    sigma_diag = FALSE,
    ncores = NULL,
    tol = 0.001
)

```

### Arguments

|            |                                                                                                                                                  |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| phi        | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system.                     |
| sigma      | Numeric matrix. The process noise covariance matrix ( $\Sigma$ ).                                                                                |
| vcov_theta | Numeric matrix. The sampling variance-covariance matrix of $\text{vec}(\Phi)$ and $\text{vech}(\Sigma)$                                          |
| delta_t    | Vector of positive numbers. Time interval ( $\Delta t$ ).                                                                                        |
| sigma_diag | Logical. If sigma_diag = TRUE, treat $\Sigma$ as a diagonal matrix.                                                                              |
| ncores     | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when the length of delta_t is long. |
| tol        | Numeric. Smallest possible time interval to allow.                                                                                               |

### Details

See [DirectCentralStd\(\)](#) for more details.

#### Delta Method:

Let  $\theta$  be a vector that combines  $\text{vec}(\Phi)$ , that is, the elements of the  $\Phi$  matrix in vector form sorted column-wise and  $\text{vech}(\Sigma)$ , that is, the unique elements of the  $\Sigma$  matrix in vector form sorted column-wise. Let  $\hat{\theta}$  be a vector that combines  $\text{vec}(\hat{\Phi})$  and  $\text{vech}(\hat{\Sigma})$ . By the multivariate central limit theory, the function  $g$  using  $\hat{\theta}$  as input can be expressed as:

$$\sqrt{n} \left( g(\hat{\theta}) - g(\theta) \right) \xrightarrow{D} \mathcal{N}(0, \mathbf{J} \mathbf{\Gamma} \mathbf{J}')$$

where  $\mathbf{J}$  is the matrix of first-order derivatives of the function  $g$  with respect to the elements of  $\theta$  and  $\mathbf{\Gamma}$  is the asymptotic variance-covariance matrix of  $\hat{\theta}$ .

From the former, we can derive the distribution of  $g(\hat{\theta})$  as follows:

$$g(\hat{\theta}) \approx \mathcal{N}(g(\theta), n^{-1} \mathbf{J} \mathbf{\Gamma} \mathbf{J}')$$

The uncertainty associated with the estimator  $g(\hat{\theta})$  is, therefore, given by  $n^{-1} \mathbf{J} \mathbf{\Gamma} \mathbf{J}'$ . When  $\mathbf{\Gamma}$  is unknown, by substitution, we can use the estimated sampling variance-covariance matrix of  $\hat{\theta}$ , that is,  $\hat{\mathbf{V}}(\hat{\theta})$  for  $n^{-1} \mathbf{\Gamma}$ . Therefore, the sampling variance-covariance matrix of  $g(\hat{\theta})$  is given by

$$g(\hat{\theta}) \approx \mathcal{N}(g(\theta), \mathbf{J} \hat{\mathbf{V}}(\hat{\theta}) \mathbf{J}').$$

**Value**

Returns an object of class `ctmeddelta` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("DeltaDirectCentralStd").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**delta\_t** Time interval.

**jacobian** Jacobian matrix.

**est** Estimated standardized direct effect centrality.

**vcov** Sampling variance-covariance matrix of estimated standardized direct effect centrality.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**References**

Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. doi:[10.2307/271028](https://doi.org/10.2307/271028)

Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. doi:[10.1080/10705511.2014.973960](https://doi.org/10.1080/10705511.2014.973960)

Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. doi:[10.1037/met0000779](https://doi.org/10.1037/met0000779)

Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. doi:[10.1007/s11336021097670](https://doi.org/10.1007/s11336021097670)

**See Also**

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

**Examples**

```

phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
sigma <- matrix(
  data = c(
    0.24455556, 0.02201587, -0.05004762,
    0.02201587, 0.07067800, 0.01539456,
    -0.05004762, 0.01539456, 0.07553061
  ),
  nrow = 3
)
vcov_theta <- matrix(
  data = c(
    0.00843, 0.00040, -0.00151, -0.00600, -0.00033,
    0.00110, 0.00324, 0.00020, -0.00061, -0.00115,
    0.00011, 0.00015, 0.00001, -0.00002, -0.00001,
    0.00040, 0.00374, 0.00016, -0.00022, -0.00273,
    -0.00016, 0.00009, 0.00150, 0.00012, -0.00010,
    -0.00026, 0.00002, 0.00012, 0.00004, -0.00001,
    -0.00151, 0.00016, 0.00389, 0.00103, -0.00007,
    -0.00283, -0.00050, 0.00000, 0.00156, 0.00021,
    -0.00005, -0.00031, 0.00001, 0.00007, 0.00006,
    -0.00600, -0.00022, 0.00103, 0.00644, 0.00031,
    -0.00119, -0.00374, -0.00021, 0.00070, 0.00064,
    -0.00015, -0.00005, 0.00000, 0.00003, -0.00001,
    -0.00033, -0.00273, -0.00007, 0.00031, 0.00287,
    0.00013, -0.00014, -0.00170, -0.00012, 0.00006,
    0.00014, -0.00001, -0.00015, 0.00000, 0.00001,
    0.00110, -0.00016, -0.00283, -0.00119, 0.00013,
    0.00297, 0.00063, -0.00004, -0.00177, -0.00013,
    0.00005, 0.00017, -0.00002, -0.00008, 0.00001,
    0.00324, 0.00009, -0.00050, -0.00374, -0.00014,
    0.00063, 0.00495, 0.00024, -0.00093, -0.00020,
    0.00006, -0.00010, 0.00000, -0.00001, 0.00004,
    0.00020, 0.00150, 0.00000, -0.00021, -0.00170,
    -0.00004, 0.00024, 0.00214, 0.00012, -0.00002,
    -0.00004, 0.00000, 0.00006, -0.00005, -0.00001,
    -0.00061, 0.00012, 0.00156, 0.00070, -0.00012,
    -0.00177, -0.00093, 0.00012, 0.00223, 0.00004,
    -0.00002, -0.00003, 0.00001, 0.00003, -0.00013,
    -0.00115, -0.00010, 0.00021, 0.00064, 0.00006,
    -0.00013, -0.00020, -0.00002, 0.00004, 0.00057,
    0.00001, -0.00009, 0.00000, 0.00000, 0.00001,
    0.00011, -0.00026, -0.00005, -0.00015, 0.00014,
    0.00005, 0.00006, -0.00004, -0.00002, 0.00001,
  )

```



```

0.00012, 0.00001, 0.00000, -0.00002, 0.00000,
0.00015, 0.00002, -0.00031, -0.00005, -0.00001,
0.00017, -0.00010, 0.00000, -0.00003, -0.00009,
0.00001, 0.00014, 0.00000, 0.00000, -0.00005,
0.00001, 0.00012, 0.00001, 0.00000, -0.00015,
-0.00002, 0.00000, 0.00006, 0.00001, 0.00000,
0.00000, 0.00000, 0.00010, 0.00001, 0.00000,
-0.00002, 0.00004, 0.00007, 0.00003, 0.00000,
-0.00008, -0.00001, -0.00005, 0.00003, 0.00000,
-0.00002, 0.00000, 0.00001, 0.00005, 0.00001,
-0.00001, -0.00001, 0.00006, -0.00001, 0.00001,
0.00001, 0.00004, -0.00001, -0.00013, 0.00001,
0.00000, -0.00005, 0.00000, 0.00001, 0.00012
),
nrow = 15
)

# Specific time interval -----
DeltaDirectCentralStd(
  phi = phi,
  sigma = sigma,
  vcov_theta = vcov_theta,
  delta_t = 1
)

# Range of time intervals -----
delta <- DeltaDirectCentralStd(
  phi = phi,
  sigma = sigma,
  vcov_theta = vcov_theta,
  delta_t = 1:5
)
plot(delta)

# Methods -----
# DeltaDirectCentralStd has a number of methods including
# print, summary, confint, and plot
print(delta)
summary(delta)
confint(delta, level = 0.95)
plot(delta)

```

---

|                      |                                                                                                                                                       |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| DeltaIndirectCentral | <i>Delta Method Sampling Variance-Covariance Matrix for the Indirect Effect Centrality Over a Specific Time Interval or a Range of Time Intervals</i> |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|

---

### Description

This function computes the delta method sampling variance-covariance matrix for the indirect effect centrality over a specific time interval  $\Delta t$  or a range of time intervals using the first-order stochastic differential equation model's drift matrix  $\Phi$ .

### Usage

```
DeltaIndirectCentral(phi, vcov_phi_vec, delta_t, ncores = NULL, tol = 0.001)
```

### Arguments

|              |                                                                                                                                                  |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| phi          | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system.                     |
| vcov_phi_vec | Numeric matrix. The sampling variance-covariance matrix of $\text{vec}(\Phi)$ .                                                                  |
| delta_t      | Vector of positive numbers. Time interval ( $\Delta t$ ).                                                                                        |
| ncores       | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when the length of delta_t is long. |
| tol          | Numeric. Smallest possible time interval to allow.                                                                                               |

### Details

See [IndirectCentral\(\)](#) for more details.

#### Delta Method:

Let  $\theta$  be  $\text{vec}(\Phi)$ , that is, the elements of the  $\Phi$  matrix in vector form sorted column-wise. Let  $\hat{\theta}$  be  $\text{vec}(\hat{\Phi})$ . By the multivariate central limit theory, the function  $\mathbf{g}$  using  $\hat{\theta}$  as input can be expressed as:

$$\sqrt{n} \left( \mathbf{g}(\hat{\theta}) - \mathbf{g}(\theta) \right) \xrightarrow{D} \mathcal{N}(0, \mathbf{J}\mathbf{\Gamma}\mathbf{J}')$$

where  $\mathbf{J}$  is the matrix of first-order derivatives of the function  $\mathbf{g}$  with respect to the elements of  $\theta$  and  $\mathbf{\Gamma}$  is the asymptotic variance-covariance matrix of  $\hat{\theta}$ .

From the former, we can derive the distribution of  $\mathbf{g}(\hat{\theta})$  as follows:

$$\mathbf{g}(\hat{\theta}) \approx \mathcal{N}(\mathbf{g}(\theta), n^{-1}\mathbf{J}\mathbf{\Gamma}\mathbf{J}')$$

The uncertainty associated with the estimator  $\mathbf{g}(\hat{\theta})$  is, therefore, given by  $n^{-1}\mathbf{J}\mathbf{\Gamma}\mathbf{J}'$ . When  $\mathbf{\Gamma}$  is unknown, by substitution, we can use the estimated sampling variance-covariance matrix of  $\hat{\theta}$ , that is,  $\hat{\mathbf{V}}(\hat{\theta})$  for  $n^{-1}\mathbf{\Gamma}$ . Therefore, the sampling variance-covariance matrix of  $\mathbf{g}(\hat{\theta})$  is given by

$$\mathbf{g}(\hat{\theta}) \approx \mathcal{N}(\mathbf{g}(\theta), \mathbf{J}\hat{\mathbf{V}}(\hat{\theta})\mathbf{J}').$$

**Value**

Returns an object of class `ctmeddelta` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("DeltaIndirectCentral").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**delta\_t** Time interval.

**jacobian** Jacobian matrix.

**est** Estimated indirect effect centrality.

**vcov** Sampling variance-covariance matrix of estimated indirect effect centrality.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**References**

Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. doi:[10.2307/271028](https://doi.org/10.2307/271028)

Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. doi:[10.1080/10705511.2014.973960](https://doi.org/10.1080/10705511.2014.973960)

Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. doi:[10.1037/met0000779](https://doi.org/10.1037/met0000779)

Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. doi:[10.1007/s11336021097670](https://doi.org/10.1007/s11336021097670)

**See Also**

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

**Examples**

```

phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
vcov_phi_vec <- matrix(
  data = c(
    0.002704274, -0.001475275, 0.000949122,
    -0.001619422, 0.000885122, -0.000569404,
    0.00085493, -0.000465824, 0.000297815,
    -0.001475275, 0.004428442, -0.002642303,
    0.000980573, -0.00271817, 0.001618805,
    -0.000586921, 0.001478421, -0.000871547,
    0.000949122, -0.002642303, 0.006402668,
    -0.000697798, 0.001813471, -0.004043138,
    0.000463086, -0.001120949, 0.002271711,
    -0.001619422, 0.000980573, -0.000697798,
    0.002079286, -0.001152501, 0.000753,
    -0.001528701, 0.000820587, -0.000517524,
    0.000885122, -0.00271817, 0.001813471,
    -0.001152501, 0.00342605, -0.002075005,
    0.000899165, -0.002532849, 0.001475579,
    -0.000569404, 0.001618805, -0.004043138,
    0.000753, -0.002075005, 0.004984032,
    -0.000622255, 0.001634917, -0.003705661,
    0.00085493, -0.000586921, 0.000463086,
    -0.001528701, 0.000899165, -0.000622255,
    0.002060076, -0.001096684, 0.000686386,
    -0.000465824, 0.001478421, -0.001120949,
    0.000820587, -0.002532849, 0.001634917,
    -0.001096684, 0.003328692, -0.001926088,
    0.000297815, -0.000871547, 0.002271711,
    -0.000517524, 0.001475579, -0.003705661,
    0.000686386, -0.001926088, 0.004726235
  ),
  nrow = 9
)

# Specific time interval -----
DeltaIndirectCentral(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  delta_t = 1
)

# Range of time intervals -----
delta <- DeltaIndirectCentral(

```

```

    phi = phi,
    vcov_phi_vec = vcov_phi_vec,
    delta_t = 1:5
  )
plot(delta)

# Methods -----
# DeltaIndirectCentral has a number of methods including
# print, summary, confint, and plot
print(delta)
summary(delta)
confint(delta, level = 0.95)
plot(delta)

```

---

### DeltaIndirectCentralStd

*Delta Method Sampling Variance-Covariance Matrix for the Standardized Indirect Effect Centrality Over a Specific Time Interval or a Range of Time Intervals*

---

### Description

This function computes the delta method sampling variance-covariance matrix for the standardized indirect effect centrality over a specific time interval  $\Delta t$  or a range of time intervals using the first-order stochastic differential equation model's drift matrix  $\Phi$  and process noise covariance matrix  $\Sigma$ .

### Usage

```

DeltaIndirectCentralStd(
  phi,
  sigma,
  vcov_theta,
  delta_t,
  sigma_diag = FALSE,
  ncores = NULL,
  tol = 0.001
)

```

### Arguments

|            |                                                                                                                              |
|------------|------------------------------------------------------------------------------------------------------------------------------|
| phi        | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system. |
| sigma      | Numeric matrix. The process noise covariance matrix ( $\Sigma$ ).                                                            |
| vcov_theta | Numeric matrix. The sampling variance-covariance matrix of vec ( $\Phi$ ) and vech ( $\Sigma$ )                              |
| delta_t    | Vector of positive numbers. Time interval ( $\Delta t$ ).                                                                    |

|                         |                                                                                                                                                                             |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>sigma_diag</code> | Logical. If <code>sigma_diag = TRUE</code> , treat $\Sigma$ as a diagonal matrix.                                                                                           |
| <code>ncores</code>     | Positive integer. Number of cores to use. If <code>ncores = NULL</code> , use a single core. Consider using multiple cores when the length of <code>delta_t</code> is long. |
| <code>tol</code>        | Numeric. Smallest possible time interval to allow.                                                                                                                          |

### Details

See [IndirectCentralStd\(\)](#) for more details.

#### Delta Method:

Let  $\theta$  be a vector that combines  $\text{vec}(\Phi)$ , that is, the elements of the  $\Phi$  matrix in vector form sorted column-wise and  $\text{vech}(\Sigma)$ , that is, the unique elements of the  $\Sigma$  matrix in vector form sorted column-wise. Let  $\hat{\theta}$  be a vector that combines  $\text{vec}(\hat{\Phi})$  and  $\text{vech}(\hat{\Sigma})$ . By the multivariate central limit theory, the function  $g$  using  $\hat{\theta}$  as input can be expressed as:

$$\sqrt{n} \left( g(\hat{\theta}) - g(\theta) \right) \xrightarrow{D} \mathcal{N}(0, \mathbf{J} \mathbf{\Gamma} \mathbf{J}')$$

where  $\mathbf{J}$  is the matrix of first-order derivatives of the function  $g$  with respect to the elements of  $\theta$  and  $\mathbf{\Gamma}$  is the asymptotic variance-covariance matrix of  $\hat{\theta}$ .

From the former, we can derive the distribution of  $g(\hat{\theta})$  as follows:

$$g(\hat{\theta}) \approx \mathcal{N}(g(\theta), n^{-1} \mathbf{J} \mathbf{\Gamma} \mathbf{J}')$$

The uncertainty associated with the estimator  $g(\hat{\theta})$  is, therefore, given by  $n^{-1} \mathbf{J} \mathbf{\Gamma} \mathbf{J}'$ . When  $\mathbf{\Gamma}$  is unknown, by substitution, we can use the estimated sampling variance-covariance matrix of  $\hat{\theta}$ , that is,  $\hat{\mathbf{V}}(\hat{\theta})$  for  $n^{-1} \mathbf{\Gamma}$ . Therefore, the sampling variance-covariance matrix of  $g(\hat{\theta})$  is given by

$$g(\hat{\theta}) \approx \mathcal{N}(g(\theta), \mathbf{J} \hat{\mathbf{V}}(\hat{\theta}) \mathbf{J}').$$

### Value

Returns an object of class `ctmeddelta` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("DeltaIndirectCentralStd").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**delta\_t** Time interval.

**jacobian** Jacobian matrix.

**est** Estimated standardized indirect effect centrality.

**vcov** Sampling variance-covariance matrix of estimated standardized indirect effect centrality.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**References**

- Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. doi:10.2307/271028
- Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. doi:10.1080/10705511.2014.973960
- Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. doi:10.1037/met0000779
- Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. doi:10.1007/s11336021097670

**Examples**

```
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
sigma <- matrix(
  data = c(
    0.24455556, 0.02201587, -0.05004762,
    0.02201587, 0.07067800, 0.01539456,
    -0.05004762, 0.01539456, 0.07553061
  ),
  nrow = 3
)
vcov_theta <- matrix(
  data = c(
    0.00843, 0.00040, -0.00151, -0.00600, -0.00033,
    0.00110, 0.00324, 0.00020, -0.00061, -0.00115,
    0.00011, 0.00015, 0.00001, -0.00002, -0.00001,
    0.00040, 0.00374, 0.00016, -0.00022, -0.00273,
    -0.00016, 0.00009, 0.00150, 0.00012, -0.00010,
    -0.00026, 0.00002, 0.00012, 0.00004, -0.00001,
    -0.00151, 0.00016, 0.00389, 0.00103, -0.00007,
    -0.00283, -0.00050, 0.00000, 0.00156, 0.00021,
    -0.00005, -0.00031, 0.00001, 0.00007, 0.00006,
    -0.00600, -0.00022, 0.00103, 0.00644, 0.00031,
    -0.00119, -0.00374, -0.00021, 0.00070, 0.00064,
    -0.00015, -0.00005, 0.00000, 0.00003, -0.00001,
    -0.00033, -0.00273, -0.00007, 0.00031, 0.00287,
```

```

0.00013, -0.00014, -0.00170, -0.00012, 0.00006,
0.00014, -0.00001, -0.00015, 0.00000, 0.00001,
0.00110, -0.00016, -0.00283, -0.00119, 0.00013,
0.00297, 0.00063, -0.00004, -0.00177, -0.00013,
0.00005, 0.00017, -0.00002, -0.00008, 0.00001,
0.00324, 0.00009, -0.00050, -0.00374, -0.00014,
0.00063, 0.00495, 0.00024, -0.00093, -0.00020,
0.00006, -0.00010, 0.00000, -0.00001, 0.00004,
0.00020, 0.00150, 0.00000, -0.00021, -0.00170,
-0.00004, 0.00024, 0.00214, 0.00012, -0.00002,
-0.00004, 0.00000, 0.00006, -0.00005, -0.00001,
-0.00061, 0.00012, 0.00156, 0.00070, -0.00012,
-0.00177, -0.00093, 0.00012, 0.00223, 0.00004,
-0.00002, -0.00003, 0.00001, 0.00003, -0.00013,
-0.00115, -0.00010, 0.00021, 0.00064, 0.00006,
-0.00013, -0.00020, -0.00002, 0.00004, 0.00057,
0.00001, -0.00009, 0.00000, 0.00000, 0.00001,
0.00011, -0.00026, -0.00005, -0.00015, 0.00014,
0.00005, 0.00006, -0.00004, -0.00002, 0.00001,
0.00012, 0.00001, 0.00000, -0.00002, 0.00000,
0.00015, 0.00002, -0.00031, -0.00005, -0.00001,
0.00017, -0.00010, 0.00000, -0.00003, -0.00009,
0.00001, 0.00014, 0.00000, 0.00000, -0.00005,
0.00001, 0.00012, 0.00001, 0.00000, -0.00015,
-0.00002, 0.00000, 0.00006, 0.00001, 0.00000,
0.00000, 0.00000, 0.00010, 0.00001, 0.00000,
-0.00002, 0.00004, 0.00007, 0.00003, 0.00000,
-0.00008, -0.00001, -0.00005, 0.00003, 0.00000,
-0.00002, 0.00000, 0.00001, 0.00005, 0.00001,
-0.00001, -0.00001, 0.00006, -0.00001, 0.00001,
0.00001, 0.00004, -0.00001, -0.00013, 0.00001,
0.00000, -0.00005, 0.00000, 0.00001, 0.00012
),
nrow = 15
)

# Specific time interval -----
DeltaIndirectCentralStd(
  phi = phi,
  sigma = sigma,
  vcov_theta = vcov_theta,
  delta_t = 1
)

# Range of time intervals -----
delta <- DeltaIndirectCentralStd(
  phi = phi,
  sigma = sigma,
  vcov_theta = vcov_theta,
  delta_t = 1:5
)
plot(delta)

```



```
# Methods -----
# DeltaIndirectCentralStd has a number of methods including
# print, summary, confint, and plot
print(delta)
summary(delta)
confint(delta, level = 0.95)
plot(delta)
#' @family Continuous-Time Mediation Functions
```

---

|          |                                                                                                                                                                                    |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DeltaMed | <i>Delta Method Sampling Variance-Covariance Matrix for the Total, Direct, and Indirect Effects of X on Y Through M Over a Specific Time Interval or a Range of Time Intervals</i> |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

---

## Description

This function computes the delta method sampling variance-covariance matrix for the total, direct, and indirect effects of the independent variable  $X$  on the dependent variable  $Y$  through mediator variables  $\mathbf{m}$  over a specific time interval  $\Delta t$  or a range of time intervals using the first-order stochastic differential equation model's drift matrix  $\Phi$ .

## Usage

```
DeltaMed(phi, vcov_phi_vec, delta_t, from, to, med, ncores = NULL, tol = 0.001)
```

## Arguments

|              |                                                                                                                                                  |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| phi          | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system.                     |
| vcov_phi_vec | Numeric matrix. The sampling variance-covariance matrix of $\text{vec}(\Phi)$ .                                                                  |
| delta_t      | Vector of positive numbers. Time interval ( $\Delta t$ ).                                                                                        |
| from         | Character string. Name of the independent variable $X$ in phi.                                                                                   |
| to           | Character string. Name of the dependent variable $Y$ in phi.                                                                                     |
| med          | Character vector. Name/s of the mediator variable/s in phi.                                                                                      |
| ncores       | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when the length of delta_t is long. |
| tol          | Numeric. Smallest possible time interval to allow.                                                                                               |

## Details

See [Total\(\)](#), [Direct\(\)](#), and [Indirect\(\)](#) for more details.

### Delta Method:

Let  $\theta$  be  $\text{vec}(\Phi)$ , that is, the elements of the  $\Phi$  matrix in vector form sorted column-wise. Let  $\hat{\theta}$  be  $\text{vec}(\hat{\Phi})$ . By the multivariate central limit theory, the function `g` using  $\hat{\theta}$  as input can be expressed as:

$$\sqrt{n} \left( \mathbf{g}(\hat{\boldsymbol{\theta}}) - \mathbf{g}(\boldsymbol{\theta}) \right) \xrightarrow{D} \mathcal{N}(0, \mathbf{J}\boldsymbol{\Gamma}\mathbf{J}')$$

where  $\mathbf{J}$  is the matrix of first-order derivatives of the function  $\mathbf{g}$  with respect to the elements of  $\boldsymbol{\theta}$  and  $\boldsymbol{\Gamma}$  is the asymptotic variance-covariance matrix of  $\hat{\boldsymbol{\theta}}$ .

From the former, we can derive the distribution of  $\mathbf{g}(\hat{\boldsymbol{\theta}})$  as follows:

$$\mathbf{g}(\hat{\boldsymbol{\theta}}) \approx \mathcal{N}(\mathbf{g}(\boldsymbol{\theta}), n^{-1}\mathbf{J}\boldsymbol{\Gamma}\mathbf{J}')$$

The uncertainty associated with the estimator  $\mathbf{g}(\hat{\boldsymbol{\theta}})$  is, therefore, given by  $n^{-1}\mathbf{J}\boldsymbol{\Gamma}\mathbf{J}'$ . When  $\boldsymbol{\Gamma}$  is unknown, by substitution, we can use the estimated sampling variance-covariance matrix of  $\hat{\boldsymbol{\theta}}$ , that is,  $\hat{\mathbf{V}}(\hat{\boldsymbol{\theta}})$  for  $n^{-1}\boldsymbol{\Gamma}$ . Therefore, the sampling variance-covariance matrix of  $\mathbf{g}(\hat{\boldsymbol{\theta}})$  is given by

$$\mathbf{g}(\hat{\boldsymbol{\theta}}) \approx \mathcal{N}(\mathbf{g}(\boldsymbol{\theta}), \mathbf{J}\hat{\mathbf{V}}(\hat{\boldsymbol{\theta}})\mathbf{J}').$$

### Value

Returns an object of class `ctmeddelta` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("DeltaMed").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**delta\_t** Time interval.

**jacobian** Jacobian matrix.

**est** Estimated total, direct, and indirect effects.

**vcov** Sampling variance-covariance matrix of the estimated total, direct, and indirect effects.

### Author(s)

Ivan Jacob Agaloos Pesigan

### References

- Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. [doi:10.2307/271028](https://doi.org/10.2307/271028)
- Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. [doi:10.1080/10705511.2014.973960](https://doi.org/10.1080/10705511.2014.973960)
- Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. [doi:10.1037/met0000779](https://doi.org/10.1037/met0000779)
- Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. [doi:10.1007/s11336021097670](https://doi.org/10.1007/s11336021097670)

**See Also**

Other Continuous-Time Mediation Functions: `BootBeta()`, `BootBetaStd()`, `BootDirectCentral()`, `BootDirectCentralStd()`, `BootIndirectCentral()`, `BootIndirectCentralStd()`, `BootMed()`, `BootMedStd()`, `BootTotalCentral()`, `BootTotalCentralStd()`, `DeltaBeta()`, `DeltaBetaStd()`, `DeltaDirectCentral()`, `DeltaDirectCentralStd()`, `DeltaIndirectCentral()`, `DeltaMedStd()`, `DeltaTotalCentral()`, `DeltaTotalCentralStd()`, `Direct()`, `DirectCentral()`, `DirectCentralStd()`, `DirectStd()`, `Indirect()`, `IndirectCentral()`, `IndirectCentralStd()`, `IndirectStd()`, `MCBeta()`, `MCBetaStd()`, `MCDirectCentral()`, `MCDirectCentralStd()`, `MCIndirectCentral()`, `MCIndirectCentralStd()`, `MCMed()`, `MCMedStd()`, `MCPhi()`, `MCPhiSigma()`, `MCTotalCentral()`, `MCTotalCentralStd()`, `Med()`, `MedStd()`, `PosteriorBeta()`, `PosteriorBetaStd()`, `PosteriorDirectCentral()`, `PosteriorDirectCentralStd()`, `PosteriorIndirectCentral()`, `PosteriorIndirectCentralStd()`, `PosteriorMed()`, `PosteriorMedStd()`, `PosteriorTotalCentral()`, `PosteriorTotalCentralStd()`, `Total()`, `TotalCentral()`, `TotalCentralStd()`, `TotalStd()`, `Trajectory()`

**Examples**

```
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
vcov_phi_vec <- matrix(
  data = c(
    0.00843, 0.00040, -0.00151,
    -0.00600, -0.00033, 0.00110,
    0.00324, 0.00020, -0.00061,
    0.00040, 0.00374, 0.00016,
    -0.00022, -0.00273, -0.00016,
    0.00009, 0.00150, 0.00012,
    -0.00151, 0.00016, 0.00389,
    0.00103, -0.00007, -0.00283,
    -0.00050, 0.00000, 0.00156,
    -0.00600, -0.00022, 0.00103,
    0.00644, 0.00031, -0.00119,
    -0.00374, -0.00021, 0.00070,
    -0.00033, -0.00273, -0.00007,
    0.00031, 0.00287, 0.00013,
    -0.00014, -0.00170, -0.00012,
    0.00110, -0.00016, -0.00283,
    -0.00119, 0.00013, 0.00297,
    0.00063, -0.00004, -0.00177,
    0.00324, 0.00009, -0.00050,
    -0.00374, -0.00014, 0.00063,
    0.00495, 0.00024, -0.00093,
    0.00020, 0.00150, 0.00000,
    -0.00021, -0.00170, -0.00004,
    0.00024, 0.00214, 0.00012,
  )
)
```

```

      -0.00061, 0.00012, 0.00156,
      0.00070, -0.00012, -0.00177,
      -0.00093, 0.00012, 0.00223
    ),
    nrow = 9
  )

# Specific time interval -----
DeltaMed(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  delta_t = 1,
  from = "x",
  to = "y",
  med = "m"
)

# Range of time intervals -----
delta <- DeltaMed(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  delta_t = 1:5,
  from = "x",
  to = "y",
  med = "m"
)
plot(delta)

# Methods -----
# DeltaMed has a number of methods including
# print, summary, confint, and plot
print(delta)
summary(delta)
confint(delta, level = 0.95)
plot(delta)

```

DeltaMedStd

*Delta Method Sampling Variance-Covariance Matrix for the Standardized Total, Direct, and Indirect Effects of X on Y Through M Over a Specific Time Interval or a Range of Time Intervals*

## Description

This function computes the delta method sampling variance-covariance matrix for the standardized total, direct, and indirect effects of the independent variable  $X$  on the dependent variable  $Y$  through mediator variables  $\mathbf{m}$  over a specific time interval  $\Delta t$  or a range of time intervals using the first-order stochastic differential equation model's drift matrix  $\Phi$  and process noise covariance matrix  $\Sigma$ .

**Usage**

```
DeltaMedStd(
  phi,
  sigma,
  vcov_theta,
  delta_t,
  from,
  to,
  med,
  sigma_diag = FALSE,
  ncores = NULL,
  tol = 0.001
)
```

**Arguments**

|            |                                                                                                                                                              |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phi        | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system.                                 |
| sigma      | Numeric matrix. The process noise covariance matrix ( $\Sigma$ ).                                                                                            |
| vcov_theta | Numeric matrix. The sampling variance-covariance matrix of $\text{vec}(\Phi)$ and $\text{vech}(\Sigma)$                                                      |
| delta_t    | Numeric. Time interval ( $\Delta t$ ).                                                                                                                       |
| from       | Character string. Name of the independent variable $X$ in phi.                                                                                               |
| to         | Character string. Name of the dependent variable $Y$ in phi.                                                                                                 |
| med        | Character vector. Name/s of the mediator variable/s in phi.                                                                                                  |
| sigma_diag | Logical. If sigma_diag = TRUE, treat $\Sigma$ as a diagonal matrix.                                                                                          |
| ncores     | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of replications R is a large value. |
| tol        | Numeric. Smallest possible time interval to allow.                                                                                                           |

**Details**

See [TotalStd\(\)](#), [DirectStd\(\)](#), and [IndirectStd\(\)](#) for more details.

**Delta Method:**

Let  $\theta$  be a vector that combines  $\text{vec}(\Phi)$ , that is, the elements of the  $\Phi$  matrix in vector form sorted column-wise and  $\text{vech}(\Sigma)$ , that is, the unique elements of the  $\Sigma$  matrix in vector form sorted column-wise. Let  $\hat{\theta}$  be a vector that combines  $\text{vec}(\hat{\Phi})$  and  $\text{vech}(\hat{\Sigma})$ . By the multivariate central limit theory, the function  $g$  using  $\hat{\theta}$  as input can be expressed as:

$$\sqrt{n} \left( g(\hat{\theta}) - g(\theta) \right) \xrightarrow{D} \mathcal{N}(0, \mathbf{J} \mathbf{\Gamma} \mathbf{J}')$$

where  $\mathbf{J}$  is the matrix of first-order derivatives of the function  $g$  with respect to the elements of  $\theta$  and  $\mathbf{\Gamma}$  is the asymptotic variance-covariance matrix of  $\hat{\theta}$ .

From the former, we can derive the distribution of  $g(\hat{\theta})$  as follows:

$$\mathbf{g}(\hat{\boldsymbol{\theta}}) \approx \mathcal{N}(\mathbf{g}(\boldsymbol{\theta}), n^{-1} \mathbf{J} \boldsymbol{\Gamma} \mathbf{J}')$$

The uncertainty associated with the estimator  $\mathbf{g}(\hat{\boldsymbol{\theta}})$  is, therefore, given by  $n^{-1} \mathbf{J} \boldsymbol{\Gamma} \mathbf{J}'$ . When  $\boldsymbol{\Gamma}$  is unknown, by substitution, we can use the estimated sampling variance-covariance matrix of  $\hat{\boldsymbol{\theta}}$ , that is,  $\hat{\mathbf{V}}(\hat{\boldsymbol{\theta}})$  for  $n^{-1} \boldsymbol{\Gamma}$ . Therefore, the sampling variance-covariance matrix of  $\mathbf{g}(\hat{\boldsymbol{\theta}})$  is given by

$$\mathbf{g}(\hat{\boldsymbol{\theta}}) \approx \mathcal{N}(\mathbf{g}(\boldsymbol{\theta}), \mathbf{J} \hat{\mathbf{V}}(\hat{\boldsymbol{\theta}}) \mathbf{J}').$$

### Value

Returns an object of class `ctmeddelta` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("DeltaMedStd").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**delta\_t** Time interval.

**jacobian** Jacobian matrix.

**est** Estimated standardized total, direct, and indirect effects.

**vcov** Sampling variance-covariance matrix of the estimated standardized total, direct, and indirect effects.

### Author(s)

Ivan Jacob Agaloos Pesigan

### References

- Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. doi:10.2307/271028
- Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. doi:10.1080/10705511.2014.973960
- Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. doi:10.1037/met0000779
- Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. doi:10.1007/s11336021097670

**See Also**

Other Continuous-Time Mediation Functions: `BootBeta()`, `BootBetaStd()`, `BootDirectCentral()`, `BootDirectCentralStd()`, `BootIndirectCentral()`, `BootIndirectCentralStd()`, `BootMed()`, `BootMedStd()`, `BootTotalCentral()`, `BootTotalCentralStd()`, `DeltaBeta()`, `DeltaBetaStd()`, `DeltaDirectCentral()`, `DeltaDirectCentralStd()`, `DeltaIndirectCentral()`, `DeltaMed()`, `DeltaTotalCentral()`, `DeltaTotalCentralStd()`, `Direct()`, `DirectCentral()`, `DirectCentralStd()`, `DirectStd()`, `Indirect()`, `IndirectCentral()`, `IndirectCentralStd()`, `IndirectStd()`, `MCBeta()`, `MCBetaStd()`, `MCDirectCentral()`, `MCDirectCentralStd()`, `MCIndirectCentral()`, `MCIndirectCentralStd()`, `MCMed()`, `MCMedStd()`, `MCPhi()`, `MCPhiSigma()`, `MCTotalCentral()`, `MCTotalCentralStd()`, `Med()`, `MedStd()`, `PosteriorBeta()`, `PosteriorBetaStd()`, `PosteriorDirectCentral()`, `PosteriorDirectCentralStd()`, `PosteriorIndirectCentral()`, `PosteriorIndirectCentralStd()`, `PosteriorMed()`, `PosteriorMedStd()`, `PosteriorTotalCentral()`, `PosteriorTotalCentralStd()`, `Total()`, `TotalCentral()`, `TotalCentralStd()`, `TotalStd()`, `Trajectory()`

**Examples**

```
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
sigma <- matrix(
  data = c(
    0.24455556, 0.02201587, -0.05004762,
    0.02201587, 0.07067800, 0.01539456,
    -0.05004762, 0.01539456, 0.07553061
  ),
  nrow = 3
)
vcov_theta <- matrix(
  data = c(
    0.00843, 0.00040, -0.00151, -0.00600, -0.00033,
    0.00110, 0.00324, 0.00020, -0.00061, -0.00115,
    0.00011, 0.00015, 0.00001, -0.00002, -0.00001,
    0.00040, 0.00374, 0.00016, -0.00022, -0.00273,
    -0.00016, 0.00009, 0.00150, 0.00012, -0.00010,
    -0.00026, 0.00002, 0.00012, 0.00004, -0.00001,
    -0.00151, 0.00016, 0.00389, 0.00103, -0.00007,
    -0.00283, -0.00050, 0.00000, 0.00156, 0.00021,
    -0.00005, -0.00031, 0.00001, 0.00007, 0.00006,
    -0.00600, -0.00022, 0.00103, 0.00644, 0.00031,
    -0.00119, -0.00374, -0.00021, 0.00070, 0.00064,
    -0.00015, -0.00005, 0.00000, 0.00003, -0.00001,
    -0.00033, -0.00273, -0.00007, 0.00031, 0.00287,
    0.00013, -0.00014, -0.00170, -0.00012, 0.00006,
    0.00014, -0.00001, -0.00015, 0.00000, 0.00001,
    0.00110, -0.00016, -0.00283, -0.00119, 0.00013,
```

```

0.00297, 0.00063, -0.00004, -0.00177, -0.00013,
0.00005, 0.00017, -0.00002, -0.00008, 0.00001,
0.00324, 0.00009, -0.00050, -0.00374, -0.00014,
0.00063, 0.00495, 0.00024, -0.00093, -0.00020,
0.00006, -0.00010, 0.00000, -0.00001, 0.00004,
0.00020, 0.00150, 0.00000, -0.00021, -0.00170,
-0.00004, 0.00024, 0.00214, 0.00012, -0.00002,
-0.00004, 0.00000, 0.00006, -0.00005, -0.00001,
-0.00061, 0.00012, 0.00156, 0.00070, -0.00012,
-0.00177, -0.00093, 0.00012, 0.00223, 0.00004,
-0.00002, -0.00003, 0.00001, 0.00003, -0.00013,
-0.00115, -0.00010, 0.00021, 0.00064, 0.00006,
-0.00013, -0.00020, -0.00002, 0.00004, 0.00057,
0.00001, -0.00009, 0.00000, 0.00000, 0.00001,
0.00011, -0.00026, -0.00005, -0.00015, 0.00014,
0.00005, 0.00006, -0.00004, -0.00002, 0.00001,
0.00012, 0.00001, 0.00000, -0.00002, 0.00000,
0.00015, 0.00002, -0.00031, -0.00005, -0.00001,
0.00017, -0.00010, 0.00000, -0.00003, -0.00009,
0.00001, 0.00014, 0.00000, 0.00000, -0.00005,
0.00001, 0.00012, 0.00001, 0.00000, -0.00015,
-0.00002, 0.00000, 0.00006, 0.00001, 0.00000,
0.00000, 0.00000, 0.00010, 0.00001, 0.00000,
-0.00002, 0.00004, 0.00007, 0.00003, 0.00000,
-0.00008, -0.00001, -0.00005, 0.00003, 0.00000,
-0.00002, 0.00000, 0.00001, 0.00005, 0.00001,
-0.00001, -0.00001, 0.00006, -0.00001, 0.00001,
0.00001, 0.00004, -0.00001, -0.00013, 0.00001,
0.00000, -0.00005, 0.00000, 0.00001, 0.00012
),
nrow = 15
)

# Specific time interval -----
DeltaMedStd(
  phi = phi,
  sigma = sigma,
  vcov_theta = vcov_theta,
  delta_t = 1,
  from = "x",
  to = "y",
  med = "m"
)

# Range of time intervals -----
delta <- DeltaMedStd(
  phi = phi,
  sigma = sigma,
  vcov_theta = vcov_theta,
  delta_t = 1:5,
  from = "x",
  to = "y",
  med = "m"
)

```



```

)
plot(delta)

# Methods -----
# DeltaMedStd has a number of methods including
# print, summary, confint, and plot
print(delta)
summary(delta)
confint(delta, level = 0.95)
plot(delta)

```

---

|                   |                                                                                                                                                    |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| DeltaTotalCentral | <i>Delta Method Sampling Variance-Covariance Matrix for the Total Effect Centrality Over a Specific Time Interval or a Range of Time Intervals</i> |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|

---

### Description

This function computes the delta method sampling variance-covariance matrix for the total effect centrality over a specific time interval  $\Delta t$  or a range of time intervals using the first-order stochastic differential equation model's drift matrix  $\Phi$ .

### Usage

```
DeltaTotalCentral(phi, vcov_phi_vec, delta_t, ncores = NULL, tol = 0.001)
```

### Arguments

|              |                                                                                                                                                  |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| phi          | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system.                     |
| vcov_phi_vec | Numeric matrix. The sampling variance-covariance matrix of $\text{vec}(\Phi)$ .                                                                  |
| delta_t      | Vector of positive numbers. Time interval ( $\Delta t$ ).                                                                                        |
| ncores       | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when the length of delta_t is long. |
| tol          | Numeric. Smallest possible time interval to allow.                                                                                               |

### Details

See [TotalCentral\(\)](#) for more details.

#### Delta Method:

Let  $\theta$  be  $\text{vec}(\Phi)$ , that is, the elements of the  $\Phi$  matrix in vector form sorted column-wise. Let  $\hat{\theta}$  be  $\text{vec}(\hat{\Phi})$ . By the multivariate central limit theory, the function  $g$  using  $\hat{\theta}$  as input can be expressed as:

$$\sqrt{n} \left( g(\hat{\theta}) - g(\theta) \right) \xrightarrow{D} \mathcal{N}(0, \mathbf{J} \mathbf{\Gamma} \mathbf{J}')$$

where  $\mathbf{J}$  is the matrix of first-order derivatives of the function  $\mathbf{g}$  with respect to the elements of  $\boldsymbol{\theta}$  and  $\boldsymbol{\Gamma}$  is the asymptotic variance-covariance matrix of  $\hat{\boldsymbol{\theta}}$ .

From the former, we can derive the distribution of  $\mathbf{g}(\hat{\boldsymbol{\theta}})$  as follows:

$$\mathbf{g}(\hat{\boldsymbol{\theta}}) \approx \mathcal{N}(\mathbf{g}(\boldsymbol{\theta}), n^{-1}\mathbf{J}\boldsymbol{\Gamma}\mathbf{J}')$$

The uncertainty associated with the estimator  $\mathbf{g}(\hat{\boldsymbol{\theta}})$  is, therefore, given by  $n^{-1}\mathbf{J}\boldsymbol{\Gamma}\mathbf{J}'$ . When  $\boldsymbol{\Gamma}$  is unknown, by substitution, we can use the estimated sampling variance-covariance matrix of  $\hat{\boldsymbol{\theta}}$ , that is,  $\hat{\mathbf{V}}(\hat{\boldsymbol{\theta}})$  for  $n^{-1}\boldsymbol{\Gamma}$ . Therefore, the sampling variance-covariance matrix of  $\mathbf{g}(\hat{\boldsymbol{\theta}})$  is given by

$$\mathbf{g}(\hat{\boldsymbol{\theta}}) \approx \mathcal{N}(\mathbf{g}(\boldsymbol{\theta}), \mathbf{J}\hat{\mathbf{V}}(\hat{\boldsymbol{\theta}})\mathbf{J}').$$

### Value

Returns an object of class `ctmeddelta` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("DeltaTotalCentral").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**delta\_t** Time interval.

**jacobian** Jacobian matrix.

**est** Estimated total effect centrality.

**vcov** Sampling variance-covariance matrix of estimated total effect centrality.

### Author(s)

Ivan Jacob Agaloos Pesigan

### References

- Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. doi:10.2307/271028
- Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. doi:10.1080/10705511.2014.973960
- Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. doi:10.1037/met0000779
- Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. doi:10.1007/s11336021097670

**See Also**

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

**Examples**

```
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
vcov_phi_vec <- matrix(
  data = c(
    0.00843, 0.00040, -0.00151,
    -0.00600, -0.00033, 0.00110,
    0.00324, 0.00020, -0.00061,
    0.00040, 0.00374, 0.00016,
    -0.00022, -0.00273, -0.00016,
    0.00009, 0.00150, 0.00012,
    -0.00151, 0.00016, 0.00389,
    0.00103, -0.00007, -0.00283,
    -0.00050, 0.00000, 0.00156,
    -0.00600, -0.00022, 0.00103,
    0.00644, 0.00031, -0.00119,
    -0.00374, -0.00021, 0.00070,
    -0.00033, -0.00273, -0.00007,
    0.00031, 0.00287, 0.00013,
    -0.00014, -0.00170, -0.00012,
    0.00110, -0.00016, -0.00283,
    -0.00119, 0.00013, 0.00297,
    0.00063, -0.00004, -0.00177,
    0.00324, 0.00009, -0.00050,
    -0.00374, -0.00014, 0.00063,
    0.00495, 0.00024, -0.00093,
    0.00020, 0.00150, 0.00000,
    -0.00021, -0.00170, -0.00004,
    0.00024, 0.00214, 0.00012,
  )
)
```

```

      -0.00061, 0.00012, 0.00156,
      0.00070, -0.00012, -0.00177,
      -0.00093, 0.00012, 0.00223
    ),
    nrow = 9
  )

# Specific time interval -----
DeltaTotalCentral(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  delta_t = 1
)

# Range of time intervals -----
delta <- DeltaTotalCentral(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  delta_t = 1:5
)
plot(delta)

# Methods -----
# DeltaTotalCentral has a number of methods including
# print, summary, confint, and plot
print(delta)
summary(delta)
confint(delta, level = 0.95)
plot(delta)

```

---

|                      |                                                                                                                                                                 |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DeltaTotalCentralStd | <i>Delta Method Sampling Variance-Covariance Matrix for the Standardized Total Effect Centrality Over a Specific Time Interval or a Range of Time Intervals</i> |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|

---

## Description

This function computes the delta method sampling variance-covariance matrix for the standardized total effect centrality over a specific time interval  $\Delta t$  or a range of time intervals using the first-order stochastic differential equation model's drift matrix  $\Phi$  and process noise covariance matrix  $\Sigma$ .

## Usage

```

DeltaTotalCentralStd(
  phi,
  sigma,
  vcov_theta,
  delta_t,

```

```

    sigma_diag = FALSE,
    ncores = NULL,
    tol = 0.001
)

```

### Arguments

|            |                                                                                                                                                  |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| phi        | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system.                     |
| sigma      | Numeric matrix. The process noise covariance matrix ( $\Sigma$ ).                                                                                |
| vcov_theta | Numeric matrix. The sampling variance-covariance matrix of $\text{vec}(\Phi)$ and $\text{vech}(\Sigma)$                                          |
| delta_t    | Vector of positive numbers. Time interval ( $\Delta t$ ).                                                                                        |
| sigma_diag | Logical. If sigma_diag = TRUE, treat $\Sigma$ as a diagonal matrix.                                                                              |
| ncores     | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when the length of delta_t is long. |
| tol        | Numeric. Smallest possible time interval to allow.                                                                                               |

### Details

See [TotalCentralStd\(\)](#) for more details.

#### Delta Method:

Let  $\theta$  be a vector that combines  $\text{vec}(\Phi)$ , that is, the elements of the  $\Phi$  matrix in vector form sorted column-wise and  $\text{vech}(\Sigma)$ , that is, the unique elements of the  $\Sigma$  matrix in vector form sorted column-wise. Let  $\hat{\theta}$  be a vector that combines  $\text{vec}(\hat{\Phi})$  and  $\text{vech}(\hat{\Sigma})$ . By the multivariate central limit theory, the function  $g$  using  $\hat{\theta}$  as input can be expressed as:

$$\sqrt{n} \left( g(\hat{\theta}) - g(\theta) \right) \xrightarrow{D} \mathcal{N}(0, \mathbf{J} \mathbf{\Gamma} \mathbf{J}')$$

where  $\mathbf{J}$  is the matrix of first-order derivatives of the function  $g$  with respect to the elements of  $\theta$  and  $\mathbf{\Gamma}$  is the asymptotic variance-covariance matrix of  $\hat{\theta}$ .

From the former, we can derive the distribution of  $g(\hat{\theta})$  as follows:

$$g(\hat{\theta}) \approx \mathcal{N}(g(\theta), n^{-1} \mathbf{J} \mathbf{\Gamma} \mathbf{J}')$$

The uncertainty associated with the estimator  $g(\hat{\theta})$  is, therefore, given by  $n^{-1} \mathbf{J} \mathbf{\Gamma} \mathbf{J}'$ . When  $\mathbf{\Gamma}$  is unknown, by substitution, we can use the estimated sampling variance-covariance matrix of  $\hat{\theta}$ , that is,  $\hat{\mathbf{V}}(\hat{\theta})$  for  $n^{-1} \mathbf{\Gamma}$ . Therefore, the sampling variance-covariance matrix of  $g(\hat{\theta})$  is given by

$$g(\hat{\theta}) \approx \mathcal{N}(g(\theta), \mathbf{J} \hat{\mathbf{V}}(\hat{\theta}) \mathbf{J}').$$

**Value**

Returns an object of class `ctmeddelta` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("DeltaTotalCentralStd").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**delta\_t** Time interval.

**jacobian** Jacobian matrix.

**est** Estimated standardized total effect centrality.

**vcov** Sampling variance-covariance matrix of estimated standardized total effect centrality.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**References**

Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. doi:[10.2307/271028](https://doi.org/10.2307/271028)

Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. doi:[10.1080/10705511.2014.973960](https://doi.org/10.1080/10705511.2014.973960)

Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. doi:[10.1037/met0000779](https://doi.org/10.1037/met0000779)

Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. doi:[10.1007/s11336021097670](https://doi.org/10.1007/s11336021097670)

**See Also**

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

**Examples**

```

phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
sigma <- matrix(
  data = c(
    0.24455556, 0.02201587, -0.05004762,
    0.02201587, 0.07067800, 0.01539456,
    -0.05004762, 0.01539456, 0.07553061
  ),
  nrow = 3
)
vcov_theta <- matrix(
  data = c(
    0.00843, 0.00040, -0.00151, -0.00600, -0.00033,
    0.00110, 0.00324, 0.00020, -0.00061, -0.00115,
    0.00011, 0.00015, 0.00001, -0.00002, -0.00001,
    0.00040, 0.00374, 0.00016, -0.00022, -0.00273,
    -0.00016, 0.00009, 0.00150, 0.00012, -0.00010,
    -0.00026, 0.00002, 0.00012, 0.00004, -0.00001,
    -0.00151, 0.00016, 0.00389, 0.00103, -0.00007,
    -0.00283, -0.00050, 0.00000, 0.00156, 0.00021,
    -0.00005, -0.00031, 0.00001, 0.00007, 0.00006,
    -0.00600, -0.00022, 0.00103, 0.00644, 0.00031,
    -0.00119, -0.00374, -0.00021, 0.00070, 0.00064,
    -0.00015, -0.00005, 0.00000, 0.00003, -0.00001,
    -0.00033, -0.00273, -0.00007, 0.00031, 0.00287,
    0.00013, -0.00014, -0.00170, -0.00012, 0.00006,
    0.00014, -0.00001, -0.00015, 0.00000, 0.00001,
    0.00110, -0.00016, -0.00283, -0.00119, 0.00013,
    0.00297, 0.00063, -0.00004, -0.00177, -0.00013,
    0.00005, 0.00017, -0.00002, -0.00008, 0.00001,
    0.00324, 0.00009, -0.00050, -0.00374, -0.00014,
    0.00063, 0.00495, 0.00024, -0.00093, -0.00020,
    0.00006, -0.00010, 0.00000, -0.00001, 0.00004,
    0.00020, 0.00150, 0.00000, -0.00021, -0.00170,
    -0.00004, 0.00024, 0.00214, 0.00012, -0.00002,
    -0.00004, 0.00000, 0.00006, -0.00005, -0.00001,
    -0.00061, 0.00012, 0.00156, 0.00070, -0.00012,
    -0.00177, -0.00093, 0.00012, 0.00223, 0.00004,
    -0.00002, -0.00003, 0.00001, 0.00003, -0.00013,
    -0.00115, -0.00010, 0.00021, 0.00064, 0.00006,
    -0.00013, -0.00020, -0.00002, 0.00004, 0.00057,
    0.00001, -0.00009, 0.00000, 0.00000, 0.00001,
    0.00011, -0.00026, -0.00005, -0.00015, 0.00014,
    0.00005, 0.00006, -0.00004, -0.00002, 0.00001,

```

```

0.00012, 0.00001, 0.00000, -0.00002, 0.00000,
0.00015, 0.00002, -0.00031, -0.00005, -0.00001,
0.00017, -0.00010, 0.00000, -0.00003, -0.00009,
0.00001, 0.00014, 0.00000, 0.00000, -0.00005,
0.00001, 0.00012, 0.00001, 0.00000, -0.00015,
-0.00002, 0.00000, 0.00006, 0.00001, 0.00000,
0.00000, 0.00000, 0.00010, 0.00001, 0.00000,
-0.00002, 0.00004, 0.00007, 0.00003, 0.00000,
-0.00008, -0.00001, -0.00005, 0.00003, 0.00000,
-0.00002, 0.00000, 0.00001, 0.00005, 0.00001,
-0.00001, -0.00001, 0.00006, -0.00001, 0.00001,
0.00001, 0.00004, -0.00001, -0.00013, 0.00001,
0.00000, -0.00005, 0.00000, 0.00001, 0.00012
),
nrow = 15
)

# Specific time interval -----
DeltaTotalCentralStd(
  phi = phi,
  sigma = sigma,
  vcov_theta = vcov_theta,
  delta_t = 1
)

# Range of time intervals -----
delta <- DeltaTotalCentralStd(
  phi = phi,
  sigma = sigma,
  vcov_theta = vcov_theta,
  delta_t = 1:5
)
plot(delta)

# Methods -----
# DeltaTotalCentralStd has a number of methods including
# print, summary, confint, and plot
print(delta)
summary(delta)
confint(delta, level = 0.95)
plot(delta)

```

---

Direct

---

*Direct Effect of X on Y Over a Specific Time Interval*


---

## Description

This function computes the direct effect of the independent variable  $X$  on the dependent variable  $Y$  through mediator variables  $\mathbf{m}$  over a specific time interval  $\Delta t$  using the first-order stochastic differential equation model's drift matrix  $\Phi$ .



**Usage**

```
Direct(phi, delta_t, from, to, med)
```

**Arguments**

|         |                                                                                                                              |
|---------|------------------------------------------------------------------------------------------------------------------------------|
| phi     | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system. |
| delta_t | Numeric. Time interval ( $\Delta t$ ).                                                                                       |
| from    | Character string. Name of the independent variable $X$ in phi.                                                               |
| to      | Character string. Name of the dependent variable $Y$ in phi.                                                                 |
| med     | Character vector. Name/s of the mediator variable/s in phi.                                                                  |

**Details**

The direct effect of the independent variable  $X$  on the dependent variable  $Y$  relative to some mediator variables  $\mathbf{m}$  is given by

$$\text{Direct}_{\Delta t, i, j} = \exp(\Delta t \mathbf{D} \Phi \mathbf{D})_{i, j}$$

where  $\Phi$  denotes the drift matrix,  $\mathbf{D}$  a diagonal matrix where the diagonal elements corresponding to mediator variables  $\mathbf{m}$  are set to zero and the rest to one,  $i$  the row index of  $Y$  in  $\Phi$ ,  $j$  the column index of  $X$  in  $\Phi$ , and  $\Delta t$  the time interval.

**Linear Stochastic Differential Equation Model:**

The measurement model is given by

$$\mathbf{y}_{i,t} = \boldsymbol{\nu} + \mathbf{\Lambda} \boldsymbol{\eta}_{i,t} + \boldsymbol{\varepsilon}_{i,t}, \quad \text{with} \quad \boldsymbol{\varepsilon}_{i,t} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Theta})$$

where  $\mathbf{y}_{i,t}$ ,  $\boldsymbol{\eta}_{i,t}$ , and  $\boldsymbol{\varepsilon}_{i,t}$  are random variables and  $\boldsymbol{\nu}$ ,  $\mathbf{\Lambda}$ , and  $\boldsymbol{\Theta}$  are model parameters.  $\mathbf{y}_{i,t}$  represents a vector of observed random variables,  $\boldsymbol{\eta}_{i,t}$  a vector of latent random variables, and  $\boldsymbol{\varepsilon}_{i,t}$  a vector of random measurement errors, at time  $t$  and individual  $i$ .  $\boldsymbol{\nu}$  denotes a vector of intercepts,  $\mathbf{\Lambda}$  a matrix of factor loadings, and  $\boldsymbol{\Theta}$  the covariance matrix of  $\boldsymbol{\varepsilon}$ .

An alternative representation of the measurement error is given by

$$\boldsymbol{\varepsilon}_{i,t} = \boldsymbol{\Theta}^{\frac{1}{2}} \mathbf{z}_{i,t}, \quad \text{with} \quad \mathbf{z}_{i,t} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

where  $\mathbf{z}_{i,t}$  is a vector of independent standard normal random variables and  $\left(\boldsymbol{\Theta}^{\frac{1}{2}}\right) \left(\boldsymbol{\Theta}^{\frac{1}{2}}\right)' = \boldsymbol{\Theta}$ .

The dynamic structure is given by

$$d\boldsymbol{\eta}_{i,t} = (\boldsymbol{\iota} + \Phi \boldsymbol{\eta}_{i,t}) dt + \boldsymbol{\Sigma}^{\frac{1}{2}} d\mathbf{W}_{i,t}$$

where  $\boldsymbol{\iota}$  is a term which is unobserved and constant over time,  $\Phi$  is the drift matrix which represents the rate of change of the solution in the absence of any random fluctuations,  $\boldsymbol{\Sigma}$  is the matrix of volatility or randomness in the process, and  $d\mathbf{W}$  is a Wiener process or Brownian motion, which represents random fluctuations.

**Value**

Returns an object of class `ctmedeffect` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("Direct").

**output** The direct effect.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**References**

Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. doi:[10.2307/271028](https://doi.org/10.2307/271028)

Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. doi:[10.1080/10705511.2014.973960](https://doi.org/10.1080/10705511.2014.973960)

Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. doi:[10.1037/met0000779](https://doi.org/10.1037/met0000779)

Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. doi:[10.1007/s11336021097670](https://doi.org/10.1007/s11336021097670)

**See Also**

Other Continuous-Time Mediation Functions: `BootBeta()`, `BootBetaStd()`, `BootDirectCentral()`, `BootDirectCentralStd()`, `BootIndirectCentral()`, `BootIndirectCentralStd()`, `BootMed()`, `BootMedStd()`, `BootTotalCentral()`, `BootTotalCentralStd()`, `DeltaBeta()`, `DeltaBetaStd()`, `DeltaDirectCentral()`, `DeltaDirectCentralStd()`, `DeltaIndirectCentral()`, `DeltaMed()`, `DeltaMedStd()`, `DeltaTotalCentral()`, `DeltaTotalCentralStd()`, `DirectCentral()`, `DirectCentralStd()`, `DirectStd()`, `Indirect()`, `IndirectCentral()`, `IndirectCentralStd()`, `IndirectStd()`, `MCBeta()`, `MCBetaStd()`, `MCDirectCentral()`, `MCDirectCentralStd()`, `MCIndirectCentral()`, `MCIndirectCentralStd()`, `MCMed()`, `MCMedStd()`, `MCPhi()`, `MCPhiSigma()`, `MCTotalCentral()`, `MCTotalCentralStd()`, `Med()`, `MedStd()`, `PosteriorBeta()`, `PosteriorBetaStd()`, `PosteriorDirectCentral()`, `PosteriorDirectCentralStd()`, `PosteriorIndirectCentral()`, `PosteriorIndirectCentralStd()`, `PosteriorMed()`, `PosteriorMedStd()`, `PosteriorTotalCentral()`, `PosteriorTotalCentralStd()`, `Total()`, `TotalCentral()`, `TotalCentralStd()`, `TotalStd()`, `Trajectory()`

**Examples**

```
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
```

```

    nrow = 3
  )
  colnames(phi) <- rownames(phi) <- c("x", "m", "y")
  delta_t <- 1
  Direct(
    phi = phi,
    delta_t = delta_t,
    from = "x",
    to = "y",
    med = "m"
  )
  phi <- matrix(
    data = c(
      -6, 5.5, 0, 0,
      1.25, -2.5, 5.9, -7.3,
      0, 0, -6, 2.5,
      5, 0, 0, -6
    ),
    nrow = 4
  )
  colnames(phi) <- rownames(phi) <- paste0("y", 1:4)
  Direct(
    phi = phi,
    delta_t = delta_t,
    from = "y2",
    to = "y4",
    med = c("y1", "y3")
  )

```

---

|               |                                 |
|---------------|---------------------------------|
| DirectCentral | <i>Direct Effect Centrality</i> |
|---------------|---------------------------------|

---

## Description

Direct Effect Centrality

## Usage

```
DirectCentral(phi, delta_t, tol = 0.001)
```

## Arguments

|         |                                                                                                                              |
|---------|------------------------------------------------------------------------------------------------------------------------------|
| phi     | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system. |
| delta_t | Vector of positive numbers. Time interval ( $\Delta t$ ).                                                                    |
| tol     | Numeric. Smallest possible time interval to allow.                                                                           |

## Details

Direct effect centrality is the sum of all possible direct effects between different pairs of variables in which a specific variable serves as the only mediator.

## Value

Returns an object of class `ctmedmed` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("DirectCentral").

**output** A matrix of direct effect centrality.

## Author(s)

Ivan Jacob Agaloos Pesigan

## References

Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. [doi:10.2307/271028](https://doi.org/10.2307/271028)

Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. [doi:10.1080/10705511.2014.973960](https://doi.org/10.1080/10705511.2014.973960)

Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. [doi:10.1037/met0000779](https://doi.org/10.1037/met0000779)

Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. [doi:10.1007/s11336021097670](https://doi.org/10.1007/s11336021097670)

## See Also

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

**Examples**

```

phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")

# Specific time interval -----
DirectCentral(
  phi = phi,
  delta_t = 1
)

# Range of time intervals -----
direct_central <- DirectCentral(
  phi = phi,
  delta_t = 1:30
)
plot(direct_central)

# Methods -----
# DirectCentral has a number of methods including
# print, summary, and plot
direct_central <- DirectCentral(
  phi = phi,
  delta_t = 1:5
)
print(direct_central)
summary(direct_central)
plot(direct_central)

```

---

DirectCentralStd

*Standardized Direct Effect Centrality*


---

**Description**

Standardized Direct Effect Centrality

**Usage**

```
DirectCentralStd(phi, sigma, delta_t, tol = 0.001)
```

**Arguments**

|         |                                                                                                                              |
|---------|------------------------------------------------------------------------------------------------------------------------------|
| phi     | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system. |
| sigma   | Numeric matrix. The process noise covariance matrix ( $\Sigma$ ).                                                            |
| delta_t | Vector of positive numbers. Time interval ( $\Delta t$ ).                                                                    |
| tol     | Numeric. Smallest possible time interval to allow.                                                                           |

**Details**

Standardized direct effect centrality is the sum of all possible standardized direct effects between different pairs of variables in which a specific variable serves as the only mediator.

**Value**

Returns an object of class `ctmedmed` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("DirectCentralStd").

**output** A matrix of standardized direct effect centrality.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**See Also**

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

**Examples**

```
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
```

```

    nrow = 3
  )
  colnames(phi) <- rownames(phi) <- c("x", "m", "y")
  sigma <- matrix(
    data = c(
      0.24455556, 0.02201587, -0.05004762,
      0.02201587, 0.07067800, 0.01539456,
      -0.05004762, 0.01539456, 0.07553061
    ),
    nrow = 3
  )

  # Specific time interval -----
  DirectCentralStd(
    phi = phi,
    sigma = sigma,
    delta_t = 1
  )

  # Range of time intervals -----
  direct_central_std <- DirectCentralStd(
    phi = phi,
    sigma = sigma,
    delta_t = 1:30
  )
  plot(direct_central_std)

  # Methods -----
  # DirectCentralStd has a number of methods including
  # print, summary, and plot
  direct_central_std <- DirectCentralStd(
    phi = phi,
    sigma = sigma,
    delta_t = 1:5
  )
  print(direct_central_std)
  summary(direct_central_std)
  plot(direct_central_std)

```

---

DirectStd

---

Standardized Direct Effect of  $X$  on  $Y$  Over a Specific Time Interval

---

### Description

This function computes the standardized direct effect of the independent variable  $X$  on the dependent variable  $Y$  through mediator variables  $\mathbf{m}$  over a specific time interval  $\Delta t$  using the first-order stochastic differential equation model's drift matrix  $\Phi$  and process noise covariance matrix  $\Sigma$ .

**Usage**

```
DirectStd(phi, sigma, delta_t, from, to, med)
```

**Arguments**

|         |                                                                                                                              |
|---------|------------------------------------------------------------------------------------------------------------------------------|
| phi     | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system. |
| sigma   | Numeric matrix. The process noise covariance matrix ( $\Sigma$ ).                                                            |
| delta_t | Numeric. Time interval ( $\Delta t$ ).                                                                                       |
| from    | Character string. Name of the independent variable $X$ in phi.                                                               |
| to      | Character string. Name of the dependent variable $Y$ in phi.                                                                 |
| med     | Character vector. Name/s of the mediator variable/s in phi.                                                                  |

**Details**

The standardized direct effect of the independent variable  $X$  on the dependent variable  $Y$  relative to some mediator variables  $\mathbf{m}$  is given by

$$\text{Direct}_{\Delta t_{i,j}}^* = \text{Direct}_{\Delta t_{i,j}} \left( \frac{\sigma_{x_j}}{\sigma_{y_i}} \right)$$

where  $\Phi$  denotes the drift matrix,  $\sigma_{x_j}$  and  $\sigma_{y_i}$  are the steady-state model-implied standard deviations of the state independent and dependent variables, respectively, and  $\Delta t$  the time interval.

**Value**

Returns an object of class `ctmedeffect` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("DirectStd").

**output** The standardized direct effect.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**References**

- Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. [doi:10.2307/271028](https://doi.org/10.2307/271028)
- Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. [doi:10.1080/10705511.2014.973960](https://doi.org/10.1080/10705511.2014.973960)
- Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. [doi:10.1037/met0000779](https://doi.org/10.1037/met0000779)
- Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. [doi:10.1007/s11336021097670](https://doi.org/10.1007/s11336021097670)



**See Also**

Other Continuous-Time Mediation Functions: `BootBeta()`, `BootBetaStd()`, `BootDirectCentral()`, `BootDirectCentralStd()`, `BootIndirectCentral()`, `BootIndirectCentralStd()`, `BootMed()`, `BootMedStd()`, `BootTotalCentral()`, `BootTotalCentralStd()`, `DeltaBeta()`, `DeltaBetaStd()`, `DeltaDirectCentral()`, `DeltaDirectCentralStd()`, `DeltaIndirectCentral()`, `DeltaMed()`, `DeltaMedStd()`, `DeltaTotalCentral()`, `DeltaTotalCentralStd()`, `Direct()`, `DirectCentral()`, `DirectCentralStd()`, `Indirect()`, `IndirectCentral()`, `IndirectCentralStd()`, `IndirectStd()`, `MCBeta()`, `MCBetaStd()`, `MCDirectCentral()`, `MCDirectCentralStd()`, `MCIndirectCentral()`, `MCIndirectCentralStd()`, `MCMed()`, `MCMedStd()`, `MCPhi()`, `MCPhiSigma()`, `MCTotalCentral()`, `MCTotalCentralStd()`, `Med()`, `MedStd()`, `PosteriorBeta()`, `PosteriorBetaStd()`, `PosteriorDirectCentral()`, `PosteriorDirectCentralStd()`, `PosteriorIndirectCentral()`, `PosteriorIndirectCentralStd()`, `PosteriorMed()`, `PosteriorMedStd()`, `PosteriorTotalCentral()`, `PosteriorTotalCentralStd()`, `Total()`, `TotalCentral()`, `TotalCentralStd()`, `TotalStd()`, `Trajectory()`

**Examples**

```
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
sigma <- matrix(
  data = c(
    0.24455556, 0.02201587, -0.05004762,
    0.02201587, 0.07067800, 0.01539456,
    -0.05004762, 0.01539456, 0.07553061
  ),
  nrow = 3
)
delta_t <- 1
DirectStd(
  phi = phi,
  sigma = sigma,
  delta_t = delta_t,
  from = "x",
  to = "y",
  med = "m"
)
```

### Description

This function computes the indirect effect of the independent variable  $X$  on the dependent variable  $Y$  through mediator variables  $\mathbf{m}$  over a specific time interval  $\Delta t$  using the first-order stochastic differential equation model's drift matrix  $\Phi$ .

### Usage

```
Indirect(phi, delta_t, from, to, med)
```

### Arguments

|         |                                                                                                                              |
|---------|------------------------------------------------------------------------------------------------------------------------------|
| phi     | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system. |
| delta_t | Numeric. Time interval ( $\Delta t$ ).                                                                                       |
| from    | Character string. Name of the independent variable $X$ in phi.                                                               |
| to      | Character string. Name of the dependent variable $Y$ in phi.                                                                 |
| med     | Character vector. Name/s of the mediator variable/s in phi.                                                                  |

### Details

The indirect effect of the independent variable  $X$  on the dependent variable  $Y$  relative to some mediator variables  $\mathbf{m}$  over a specific time interval  $\Delta t$  is given by

$$\text{Indirect}_{\Delta t, i, j} = \exp(\Delta t \Phi)_{i, j} - \exp(\Delta t \mathbf{D}_m \Phi \mathbf{D}_m)_{i, j}$$

where  $\Phi$  denotes the drift matrix,  $\mathbf{D}_m$  a matrix where the off diagonal elements are zeros and the diagonal elements are zero for the index/indices of mediator variables  $\mathbf{m}$  and one otherwise,  $i$  the row index of  $Y$  in  $\Phi$ ,  $j$  the column index of  $X$  in  $\Phi$ , and  $\Delta t$  the time interval.

#### Linear Stochastic Differential Equation Model:

The measurement model is given by

$$\mathbf{y}_{i,t} = \boldsymbol{\nu} + \mathbf{\Lambda} \boldsymbol{\eta}_{i,t} + \boldsymbol{\varepsilon}_{i,t}, \quad \text{with} \quad \boldsymbol{\varepsilon}_{i,t} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Theta})$$

where  $\mathbf{y}_{i,t}$ ,  $\boldsymbol{\eta}_{i,t}$ , and  $\boldsymbol{\varepsilon}_{i,t}$  are random variables and  $\boldsymbol{\nu}$ ,  $\mathbf{\Lambda}$ , and  $\boldsymbol{\Theta}$  are model parameters.  $\mathbf{y}_{i,t}$  represents a vector of observed random variables,  $\boldsymbol{\eta}_{i,t}$  a vector of latent random variables, and  $\boldsymbol{\varepsilon}_{i,t}$  a vector of random measurement errors, at time  $t$  and individual  $i$ .  $\boldsymbol{\nu}$  denotes a vector of intercepts,  $\mathbf{\Lambda}$  a matrix of factor loadings, and  $\boldsymbol{\Theta}$  the covariance matrix of  $\boldsymbol{\varepsilon}$ .

An alternative representation of the measurement error is given by

$$\boldsymbol{\varepsilon}_{i,t} = \boldsymbol{\Theta}^{\frac{1}{2}} \mathbf{z}_{i,t}, \quad \text{with} \quad \mathbf{z}_{i,t} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

where  $\mathbf{z}_{i,t}$  is a vector of independent standard normal random variables and  $\left(\boldsymbol{\Theta}^{\frac{1}{2}}\right) \left(\boldsymbol{\Theta}^{\frac{1}{2}}\right)' = \boldsymbol{\Theta}$ .

The dynamic structure is given by

$$d\boldsymbol{\eta}_{i,t} = (\boldsymbol{\iota} + \Phi \boldsymbol{\eta}_{i,t}) dt + \boldsymbol{\Sigma}^{\frac{1}{2}} d\mathbf{W}_{i,t}$$

where  $\boldsymbol{\iota}$  is a term which is unobserved and constant over time,  $\Phi$  is the drift matrix which represents the rate of change of the solution in the absence of any random fluctuations,  $\boldsymbol{\Sigma}$  is the matrix of volatility or randomness in the process, and  $d\mathbf{W}$  is a Wiener process or Brownian motion, which represents random fluctuations.

**Value**

Returns an object of class `ctmedeffect` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("Indirect").

**output** The indirect effect.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**References**

Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. doi:[10.2307/271028](https://doi.org/10.2307/271028)

Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. doi:[10.1080/10705511.2014.973960](https://doi.org/10.1080/10705511.2014.973960)

Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. doi:[10.1037/met0000779](https://doi.org/10.1037/met0000779)

Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. doi:[10.1007/s11336021097670](https://doi.org/10.1007/s11336021097670)

**See Also**

Other Continuous-Time Mediation Functions: `BootBeta()`, `BootBetaStd()`, `BootDirectCentral()`, `BootDirectCentralStd()`, `BootIndirectCentral()`, `BootIndirectCentralStd()`, `BootMed()`, `BootMedStd()`, `BootTotalCentral()`, `BootTotalCentralStd()`, `DeltaBeta()`, `DeltaBetaStd()`, `DeltaDirectCentral()`, `DeltaDirectCentralStd()`, `DeltaIndirectCentral()`, `DeltaMed()`, `DeltaMedStd()`, `DeltaTotalCentral()`, `DeltaTotalCentralStd()`, `Direct()`, `DirectCentral()`, `DirectCentralStd()`, `DirectStd()`, `IndirectCentral()`, `IndirectCentralStd()`, `IndirectStd()`, `MCBeta()`, `MCBetaStd()`, `MCDirectCentral()`, `MCDirectCentralStd()`, `MCIndirectCentral()`, `MCIndirectCentralStd()`, `MCMed()`, `MCMedStd()`, `MCPPhi()`, `MCPPhiSigma()`, `MCTotalCentral()`, `MCTotalCentralStd()`, `Med()`, `MedStd()`, `PosteriorBeta()`, `PosteriorBetaStd()`, `PosteriorDirectCentral()`, `PosteriorDirectCentralStd()`, `PosteriorIndirectCentral()`, `PosteriorIndirectCentralStd()`, `PosteriorMed()`, `PosteriorMedStd()`, `PosteriorTotalCentral()`, `PosteriorTotalCentralStd()`, `Total()`, `TotalCentral()`, `TotalCentralStd()`, `TotalStd()`, `Trajectory()`

**Examples**

```
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
```

```
nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
delta_t <- 1
Indirect(
  phi = phi,
  delta_t = delta_t,
  from = "x",
  to = "y",
  med = "m"
)
phi <- matrix(
  data = c(
    -6, 5.5, 0, 0,
    1.25, -2.5, 5.9, -7.3,
    0, 0, -6, 2.5,
    5, 0, 0, -6
  ),
  nrow = 4
)
colnames(phi) <- rownames(phi) <- paste0("y", 1:4)
Indirect(
  phi = phi,
  delta_t = delta_t,
  from = "y2",
  to = "y4",
  med = c("y1", "y3")
)
```

---

|                 |                                   |
|-----------------|-----------------------------------|
| IndirectCentral | <i>Indirect Effect Centrality</i> |
|-----------------|-----------------------------------|

---

**Description**

Indirect Effect Centrality

**Usage**

```
IndirectCentral(phi, delta_t, tol = 0.001)
```

**Arguments**

|         |                                                                                                                              |
|---------|------------------------------------------------------------------------------------------------------------------------------|
| phi     | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system. |
| delta_t | Vector of positive numbers. Time interval ( $\Delta t$ ).                                                                    |
| tol     | Numeric. Smallest possible time interval to allow.                                                                           |

## Details

Indirect effect centrality is the sum of all possible indirect effects between different pairs of variables in which a specific variable serves as the only mediator.

## Value

Returns an object of class `ctmedmed` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("IndirectCentral").

**output** A matrix of indirect effect centrality.

## Author(s)

Ivan Jacob Agaloos Pesigan

## References

Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. [doi:10.2307/271028](https://doi.org/10.2307/271028)

Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. [doi:10.1080/10705511.2014.973960](https://doi.org/10.1080/10705511.2014.973960)

Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. [doi:10.1037/met0000779](https://doi.org/10.1037/met0000779)

Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. [doi:10.1007/s11336021097670](https://doi.org/10.1007/s11336021097670)

## See Also

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

**Examples**

```

phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")

# Specific time interval -----
IndirectCentral(
  phi = phi,
  delta_t = 1
)

# Range of time intervals -----
indirect_central <- IndirectCentral(
  phi = phi,
  delta_t = 1:30
)
plot(indirect_central)

# Methods -----
# IndirectCentral has a number of methods including
# print, summary, and plot
indirect_central <- IndirectCentral(
  phi = phi,
  delta_t = 1:5
)
print(indirect_central)
summary(indirect_central)
plot(indirect_central)

```

---

|                    |                                                |
|--------------------|------------------------------------------------|
| IndirectCentralStd | <i>Standardized Indirect Effect Centrality</i> |
|--------------------|------------------------------------------------|

---

**Description**

Standardized Indirect Effect Centrality

**Usage**

```
IndirectCentralStd(phi, sigma, delta_t, tol = 0.001)
```

**Arguments**

|         |                                                                                                                              |
|---------|------------------------------------------------------------------------------------------------------------------------------|
| phi     | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system. |
| sigma   | Numeric matrix. The process noise covariance matrix ( $\Sigma$ ).                                                            |
| delta_t | Vector of positive numbers. Time interval ( $\Delta t$ ).                                                                    |
| tol     | Numeric. Smallest possible time interval to allow.                                                                           |

**Details**

Standardized indirect effect centrality is the sum of all possible standardized indirect effects between different pairs of variables in which a specific variable serves as the only mediator.

**Value**

Returns an object of class `ctmedmed` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("IndirectCentralStd").

**output** A matrix of standardized indirect effect centrality.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**See Also**

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

**Examples**

```
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
```

```

    nrow = 3
  )
  colnames(phi) <- rownames(phi) <- c("x", "m", "y")
  sigma <- matrix(
    data = c(
      0.24455556, 0.02201587, -0.05004762,
      0.02201587, 0.07067800, 0.01539456,
      -0.05004762, 0.01539456, 0.07553061
    ),
    nrow = 3
  )

# Specific time interval -----
IndirectCentralStd(
  phi = phi,
  sigma = sigma,
  delta_t = 1
)

# Range of time intervals -----
indirect_central_std <- IndirectCentralStd(
  phi = phi,
  sigma = sigma,
  delta_t = 1:30
)
plot(indirect_central_std)

# Methods -----
# IndirectCentralStd has a number of methods including
# print, summary, and plot
indirect_central_std <- IndirectCentralStd(
  phi = phi,
  sigma = sigma,
  delta_t = 1:5
)
print(indirect_central_std)
summary(indirect_central_std)
plot(indirect_central_std)

```

---

IndirectStd

---

Standardized Indirect Effect of  $X$  on  $Y$  Through  $M$  Over a Specific Time Interval

---

### Description

This function computes the standardized indirect effect of the independent variable  $X$  on the dependent variable  $Y$  through mediator variables  $\mathbf{m}$  over a specific time interval  $\Delta t$  using the first-order stochastic differential equation model's drift matrix  $\Phi$  and process noise covariance matrix  $\Sigma$ .



**Usage**

```
IndirectStd(phi, sigma, delta_t, from, to, med)
```

**Arguments**

|         |                                                                                                                              |
|---------|------------------------------------------------------------------------------------------------------------------------------|
| phi     | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system. |
| sigma   | Numeric matrix. The process noise covariance matrix ( $\Sigma$ ).                                                            |
| delta_t | Numeric. Time interval ( $\Delta t$ ).                                                                                       |
| from    | Character string. Name of the independent variable $X$ in phi.                                                               |
| to      | Character string. Name of the dependent variable $Y$ in phi.                                                                 |
| med     | Character vector. Name/s of the mediator variable/s in phi.                                                                  |

**Details**

The standardized indirect effect of the independent variable  $X$  on the dependent variable  $Y$  relative to some mediator variables  $\mathbf{m}$  over a specific time interval  $\Delta t$  is given by

$$\text{Indirect}^*_{\Delta t, i, j} = \text{Total}^*_{\Delta t, i, j} - \text{Direct}^*_{\Delta t, i, j}$$

where  $\text{Total}^*_{\Delta t}$  and  $\text{Direct}^*_{\Delta t}$  are standardized total and direct effects for time interval  $\Delta t$ .

**Value**

Returns an object of class `ctmedeffect` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("IndirectStd").

**output** The standardized indirect effect.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**References**

- Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. [doi:10.2307/271028](https://doi.org/10.2307/271028)
- Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. [doi:10.1080/10705511.2014.973960](https://doi.org/10.1080/10705511.2014.973960)
- Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. [doi:10.1037/met0000779](https://doi.org/10.1037/met0000779)
- Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. [doi:10.1007/s11336021097670](https://doi.org/10.1007/s11336021097670)

**See Also**

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

**Examples**

```
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
sigma <- matrix(
  data = c(
    0.24455556, 0.02201587, -0.05004762,
    0.02201587, 0.07067800, 0.01539456,
    -0.05004762, 0.01539456, 0.07553061
  ),
  nrow = 3
)
delta_t <- 1
IndirectStd(
  phi = phi,
  sigma = sigma,
  delta_t = delta_t,
  from = "x",
  to = "y",
  med = "m"
)
```

## Description

This function generates a Monte Carlo method sampling distribution for the elements of the matrix of lagged coefficients  $\beta$  over a specific time interval  $\Delta t$  or a range of time intervals using the first-order stochastic differential equation model drift matrix  $\Phi$ .

## Usage

```
MCBeta(
  phi,
  vcov_phi_vec,
  delta_t,
  R,
  test_phi = TRUE,
  ncores = NULL,
  seed = NULL,
  tol = 0.001
)
```

## Arguments

|              |                                                                                                                                                                                                                                             |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phi          | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system.                                                                                                                |
| vcov_phi_vec | Numeric matrix. The sampling variance-covariance matrix of vec( $\Phi$ ).                                                                                                                                                                   |
| delta_t      | Numeric. Time interval ( $\Delta t$ ).                                                                                                                                                                                                      |
| R            | Positive integer. Number of replications.                                                                                                                                                                                                   |
| test_phi     | Logical. If test_phi = TRUE, the function tests the stability of the generated drift matrix $\Phi$ . If the test returns FALSE, the function generates a new drift matrix $\Phi$ and runs the test recursively until the test returns TRUE. |
| ncores       | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of replications R is a large value.                                                                                |
| seed         | Random seed.                                                                                                                                                                                                                                |
| tol          | Numeric. Smallest possible time interval to allow.                                                                                                                                                                                          |

## Details

See [Total\(\)](#).

### Monte Carlo Method:

Let  $\theta$  be vec( $\Phi$ ), that is, the elements of the  $\Phi$  matrix in vector form sorted column-wise. Let  $\hat{\theta}$  be vec( $\hat{\Phi}$ ). Based on the asymptotic properties of maximum likelihood estimators, we can assume that estimators are normally distributed around the population parameters.

$$\hat{\theta} \sim \mathcal{N}(\theta, \mathbb{V}(\hat{\theta}))$$

Using this distributional assumption, a sampling distribution of  $\hat{\theta}$  which we refer to as  $\hat{\theta}^*$  can be generated by replacing the population parameters with sample estimates, that is,

$$\hat{\theta}^* \sim \mathcal{N}(\hat{\theta}, \hat{\mathbb{V}}(\hat{\theta})).$$

Let  $g(\hat{\theta})$  be a parameter that is a function of the estimated parameters. A sampling distribution of  $g(\hat{\theta})$ , which we refer to as  $g(\hat{\theta}^*)$ , can be generated by using the simulated estimates to calculate  $g$ . The standard deviations of the simulated estimates are the standard errors. Percentiles corresponding to  $100(1 - \alpha)\%$  are the confidence intervals.

### Value

Returns an object of class `ctmedmc` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("MCBeta").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**est** Estimated elements of the matrix of lagged coefficients.

**thetahatstar** A matrix of Monte Carlo elements of the matrix of lagged coefficients.

### Author(s)

Ivan Jacob Agaloos Pesigan

### References

Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. doi:10.2307/271028

Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. doi:10.1080/10705511.2014.973960

Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. doi:10.1037/met0000779

Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. doi:10.1007/s11336021097670

### See Also

Other Continuous-Time Mediation Functions: `BootBeta()`, `BootBetaStd()`, `BootDirectCentral()`, `BootDirectCentralStd()`, `BootIndirectCentral()`, `BootIndirectCentralStd()`, `BootMed()`, `BootMedStd()`, `BootTotalCentral()`, `BootTotalCentralStd()`, `DeltaBeta()`, `DeltaBetaStd()`, `DeltaDirectCentral()`, `DeltaDirectCentralStd()`, `DeltaIndirectCentral()`, `DeltaMed()`, `DeltaMedStd()`, `DeltaTotalCentral()`, `DeltaTotalCentralStd()`, `Direct()`, `DirectCentral()`, `DirectCentralStd()`, `DirectStd()`, `Indirect()`, `IndirectCentral()`, `IndirectCentralStd()`, `IndirectStd()`, `MCBetaStd()`, `MCDirectCentral()`, `MCDirectCentralStd()`, `MCIndirectCentral()`, `MCIndirectCentralStd()`, `MCMed()`, `MCMedStd()`, `MCPhi()`, `MCPhiSigma()`, `MCTotalCentral()`, `MCTotalCentralStd()`, `Med()`, `MedStd()`, `PosteriorBeta()`, `PosteriorBetaStd()`, `PosteriorDirectCentral()`,

PosteriorDirectCentralStd(), PosteriorIndirectCentral(), PosteriorIndirectCentralStd(),  
 PosteriorMed(), PosteriorMedStd(), PosteriorTotalCentral(), PosteriorTotalCentralStd(),  
 Total(), TotalCentral(), TotalCentralStd(), TotalStd(), Trajectory()

## Examples

```
set.seed(42)
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
vcov_phi_vec <- matrix(
  data = c(
    0.00843, 0.00040, -0.00151,
    -0.00600, -0.00033, 0.00110,
    0.00324, 0.00020, -0.00061,
    0.00040, 0.00374, 0.00016,
    -0.00022, -0.00273, -0.00016,
    0.00009, 0.00150, 0.00012,
    -0.00151, 0.00016, 0.00389,
    0.00103, -0.00007, -0.00283,
    -0.00050, 0.00000, 0.00156,
    -0.00600, -0.00022, 0.00103,
    0.00644, 0.00031, -0.00119,
    -0.00374, -0.00021, 0.00070,
    -0.00033, -0.00273, -0.00007,
    0.00031, 0.00287, 0.00013,
    -0.00014, -0.00170, -0.00012,
    0.00110, -0.00016, -0.00283,
    -0.00119, 0.00013, 0.00297,
    0.00063, -0.00004, -0.00177,
    0.00324, 0.00009, -0.00050,
    -0.00374, -0.00014, 0.00063,
    0.00495, 0.00024, -0.00093,
    0.00020, 0.00150, 0.00000,
    -0.00021, -0.00170, -0.00004,
    0.00024, 0.00214, 0.00012,
    -0.00061, 0.00012, 0.00156,
    0.00070, -0.00012, -0.00177,
    -0.00093, 0.00012, 0.00223
  ),
  nrow = 9
)

# Specific time interval -----
MCBeta(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
```

```

    delta_t = 1,
    R = 100L # use a large value for R in actual research
  )

# Range of time intervals -----
mc <- MCBeta(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  delta_t = 1:5,
  R = 100L # use a large value for R in actual research
)
plot(mc)

# Methods -----
# MCBeta has a number of methods including
# print, summary, confint, and plot
print(mc)
summary(mc)
confint(mc, level = 0.95)
plot(mc)

```

---

MCBetaStd

*Monte Carlo Sampling Distribution for the Elements of the Standardized Matrix of Lagged Coefficients Over a Specific Time Interval or a Range of Time Intervals*

---

## Description

This function generates a Monte Carlo method sampling distribution for the elements of the standardized matrix of lagged coefficients  $\beta$  over a specific time interval  $\Delta t$  or a range of time intervals using the first-order stochastic differential equation model drift matrix  $\Phi$  and process noise covariance matrix  $\Sigma$ .

## Usage

```

MCBetaStd(
  phi,
  sigma,
  vcov_theta,
  delta_t,
  R,
  test_phi = TRUE,
  sigma_diag = FALSE,
  ncores = NULL,
  seed = NULL,
  tol = 0.001
)

```

**Arguments**

|            |                                                                                                                                                                                                                                             |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phi        | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system.                                                                                                                |
| sigma      | Numeric matrix. The process noise covariance matrix ( $\Sigma$ ).                                                                                                                                                                           |
| vcov_theta | Numeric matrix. The sampling variance-covariance matrix of $\text{vec}(\Phi)$ and $\text{vech}(\Sigma)$                                                                                                                                     |
| delta_t    | Numeric. Time interval ( $\Delta t$ ).                                                                                                                                                                                                      |
| R          | Positive integer. Number of replications.                                                                                                                                                                                                   |
| test_phi   | Logical. If test_phi = TRUE, the function tests the stability of the generated drift matrix $\Phi$ . If the test returns FALSE, the function generates a new drift matrix $\Phi$ and runs the test recursively until the test returns TRUE. |
| sigma_diag | Logical. If sigma_diag = TRUE, treat $\Sigma$ as a diagonal matrix.                                                                                                                                                                         |
| ncores     | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of replications R is a large value.                                                                                |
| seed       | Random seed.                                                                                                                                                                                                                                |
| tol        | Numeric. Smallest possible time interval to allow.                                                                                                                                                                                          |

**Details**

See [TotalStd\(\)](#).

**Monte Carlo Method:**

Let  $\theta$  be a vector that combines  $\text{vec}(\Phi)$ , that is, the elements of the  $\Phi$  matrix in vector form sorted column-wise and  $\text{vech}(\Sigma)$ , that is, the unique elements of the  $\Sigma$  matrix in vector form sorted column-wise. Let  $\hat{\theta}$  be a vector that combines  $\text{vec}(\hat{\Phi})$  and  $\text{vech}(\hat{\Sigma})$ . Based on the asymptotic properties of maximum likelihood estimators, we can assume that estimators are normally distributed around the population parameters.

$$\hat{\theta} \sim \mathcal{N}(\theta, \mathbb{V}(\hat{\theta}))$$

Using this distributional assumption, a sampling distribution of  $\hat{\theta}$  which we refer to as  $\hat{\theta}^*$  can be generated by replacing the population parameters with sample estimates, that is,

$$\hat{\theta}^* \sim \mathcal{N}(\hat{\theta}, \hat{\mathbb{V}}(\hat{\theta})).$$

Let  $\mathbf{g}(\hat{\theta})$  be a parameter that is a function of the estimated parameters. A sampling distribution of  $\mathbf{g}(\hat{\theta})$ , which we refer to as  $\mathbf{g}(\hat{\theta}^*)$ , can be generated by using the simulated estimates to calculate  $\mathbf{g}$ . The standard deviations of the simulated estimates are the standard errors. Percentiles corresponding to  $100(1 - \alpha)\%$  are the confidence intervals.

**Value**

Returns an object of class `ctmedmc` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("MCBetaStd").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**est** Estimated elements of the standardized matrix of lagged coefficients.

**thetahatstar** A matrix of Monte Carlo elements of the standardized matrix of lagged coefficients.

### Author(s)

Ivan Jacob Agaloos Pesigan

### References

Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. doi:[10.2307/271028](https://doi.org/10.2307/271028)

Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. doi:[10.1080/10705511.2014.973960](https://doi.org/10.1080/10705511.2014.973960)

Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. doi:[10.1037/met0000779](https://doi.org/10.1037/met0000779)

Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. doi:[10.1007/s11336021097670](https://doi.org/10.1007/s11336021097670)

### See Also

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

### Examples

```
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
```



```

nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
sigma <- matrix(
  data = c(
    0.24455556, 0.02201587, -0.05004762,
    0.02201587, 0.07067800, 0.01539456,
    -0.05004762, 0.01539456, 0.07553061
  ),
  nrow = 3
)
vcov_theta <- matrix(
  data = c(
    0.00843, 0.00040, -0.00151, -0.00600, -0.00033,
    0.00110, 0.00324, 0.00020, -0.00061, -0.00115,
    0.00011, 0.00015, 0.00001, -0.00002, -0.00001,
    0.00040, 0.00374, 0.00016, -0.00022, -0.00273,
    -0.00016, 0.00009, 0.00150, 0.00012, -0.00010,
    -0.00026, 0.00002, 0.00012, 0.00004, -0.00001,
    -0.00151, 0.00016, 0.00389, 0.00103, -0.00007,
    -0.00283, -0.00050, 0.00000, 0.00156, 0.00021,
    -0.00005, -0.00031, 0.00001, 0.00007, 0.00006,
    -0.00600, -0.00022, 0.00103, 0.00644, 0.00031,
    -0.00119, -0.00374, -0.00021, 0.00070, 0.00064,
    -0.00015, -0.00005, 0.00000, 0.00003, -0.00001,
    -0.00033, -0.00273, -0.00007, 0.00031, 0.00287,
    0.00013, -0.00014, -0.00170, -0.00012, 0.00006,
    0.00014, -0.00001, -0.00015, 0.00000, 0.00001,
    0.00110, -0.00016, -0.00283, -0.00119, 0.00013,
    0.00297, 0.00063, -0.00004, -0.00177, -0.00013,
    0.00005, 0.00017, -0.00002, -0.00008, 0.00001,
    0.00324, 0.00009, -0.00050, -0.00374, -0.00014,
    0.00063, 0.00495, 0.00024, -0.00093, -0.00020,
    0.00006, -0.00010, 0.00000, -0.00001, 0.00004,
    0.00020, 0.00150, 0.00000, -0.00021, -0.00170,
    -0.00004, 0.00024, 0.00214, 0.00012, -0.00002,
    -0.00004, 0.00000, 0.00006, -0.00005, -0.00001,
    -0.00061, 0.00012, 0.00156, 0.00070, -0.00012,
    -0.00177, -0.00093, 0.00012, 0.00223, 0.00004,
    -0.00002, -0.00003, 0.00001, 0.00003, -0.00013,
    -0.00115, -0.00010, 0.00021, 0.00064, 0.00006,
    -0.00013, -0.00020, -0.00002, 0.00004, 0.00057,
    0.00001, -0.00009, 0.00000, 0.00000, 0.00001,
    0.00011, -0.00026, -0.00005, -0.00015, 0.00014,
    0.00005, 0.00006, -0.00004, -0.00002, 0.00001,
    0.00012, 0.00001, 0.00000, -0.00002, 0.00000,
    0.00015, 0.00002, -0.00031, -0.00005, -0.00001,
    0.00017, -0.00010, 0.00000, -0.00003, -0.00009,
    0.00001, 0.00014, 0.00000, 0.00000, -0.00005,
    0.00001, 0.00012, 0.00001, 0.00000, -0.00015,
    -0.00002, 0.00000, 0.00006, 0.00001, 0.00000,
    0.00000, 0.00000, 0.00010, 0.00001, 0.00000,
    -0.00002, 0.00004, 0.00007, 0.00003, 0.00000,
  )

```

```

-0.00008, -0.00001, -0.00005, 0.00003, 0.00000,
-0.00002, 0.00000, 0.00001, 0.00005, 0.00001,
-0.00001, -0.00001, 0.00006, -0.00001, 0.00001,
0.00001, 0.00004, -0.00001, -0.00013, 0.00001,
0.00000, -0.00005, 0.00000, 0.00001, 0.00012
),
nrow = 15
)

# Specific time interval -----
MCBetaStd(
  phi = phi,
  sigma = sigma,
  vcov_theta = vcov_theta,
  delta_t = 1,
  R = 100L # use a large value for R in actual research
)

# Range of time intervals -----
mc <- MCBetaStd(
  phi = phi,
  sigma = sigma,
  vcov_theta = vcov_theta,
  delta_t = 1:5,
  R = 100L # use a large value for R in actual research
)
plot(mc)

# Methods -----
# MCBetaStd has a number of methods including
# print, summary, confint, and plot
print(mc)
summary(mc)
confint(mc, level = 0.95)
plot(mc)

```

---

MCDirectCentral

---

*Monte Carlo Sampling Distribution of Direct Effect Centrality Over a Specific Time Interval or a Range of Time Intervals*


---

## Description

This function generates a Monte Carlo method sampling distribution of the direct effect centrality at a particular time interval  $\Delta t$  using the first-order stochastic differential equation model drift matrix  $\Phi$ .

## Usage

```
MCDirectCentral(
```

```

    phi,
    vcov_phi_vec,
    delta_t,
    R,
    test_phi = TRUE,
    ncores = NULL,
    seed = NULL,
    tol = 0.001
)

```

### Arguments

|              |                                                                                                                                                                                                                                             |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phi          | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system.                                                                                                                |
| vcov_phi_vec | Numeric matrix. The sampling variance-covariance matrix of $\text{vec}(\Phi)$ .                                                                                                                                                             |
| delta_t      | Numeric. Time interval ( $\Delta t$ ).                                                                                                                                                                                                      |
| R            | Positive integer. Number of replications.                                                                                                                                                                                                   |
| test_phi     | Logical. If test_phi = TRUE, the function tests the stability of the generated drift matrix $\Phi$ . If the test returns FALSE, the function generates a new drift matrix $\Phi$ and runs the test recursively until the test returns TRUE. |
| ncores       | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of replications R is a large value.                                                                                |
| seed         | Random seed.                                                                                                                                                                                                                                |
| tol          | Numeric. Smallest possible time interval to allow.                                                                                                                                                                                          |

### Details

See [DirectCentral\(\)](#) for more details.

#### Monte Carlo Method:

Let  $\theta$  be  $\text{vec}(\Phi)$ , that is, the elements of the  $\Phi$  matrix in vector form sorted column-wise. Let  $\hat{\theta}$  be  $\text{vec}(\hat{\Phi})$ . Based on the asymptotic properties of maximum likelihood estimators, we can assume that estimators are normally distributed around the population parameters.

$$\hat{\theta} \sim \mathcal{N}(\theta, \mathbb{V}(\hat{\theta}))$$

Using this distributional assumption, a sampling distribution of  $\hat{\theta}$  which we refer to as  $\hat{\theta}^*$  can be generated by replacing the population parameters with sample estimates, that is,

$$\hat{\theta}^* \sim \mathcal{N}(\hat{\theta}, \hat{\mathbb{V}}(\hat{\theta})).$$

Let  $\mathbf{g}(\hat{\theta})$  be a parameter that is a function of the estimated parameters. A sampling distribution of  $\mathbf{g}(\hat{\theta})$ , which we refer to as  $\mathbf{g}(\hat{\theta}^*)$ , can be generated by using the simulated estimates to calculate  $\mathbf{g}$ . The standard deviations of the simulated estimates are the standard errors. Percentiles corresponding to  $100(1 - \alpha)\%$  are the confidence intervals.

**Value**

Returns an object of class `ctmedmc` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("MCDirectCentral").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**est** A vector of direct effect centrality.

**thetahatstar** A matrix of Monte Carlo direct effect centrality.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**References**

Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. [doi:10.2307/271028](https://doi.org/10.2307/271028)

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Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. [doi:10.1037/met0000779](https://doi.org/10.1037/met0000779)

Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. [doi:10.1007/s11336021097670](https://doi.org/10.1007/s11336021097670)

**See Also**

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

**Examples**

```

set.seed(42)
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
vcov_phi_vec <- matrix(
  data = c(
    0.00843, 0.00040, -0.00151,
    -0.00600, -0.00033, 0.00110,
    0.00324, 0.00020, -0.00061,
    0.00040, 0.00374, 0.00016,
    -0.00022, -0.00273, -0.00016,
    0.00009, 0.00150, 0.00012,
    -0.00151, 0.00016, 0.00389,
    0.00103, -0.00007, -0.00283,
    -0.00050, 0.00000, 0.00156,
    -0.00600, -0.00022, 0.00103,
    0.00644, 0.00031, -0.00119,
    -0.00374, -0.00021, 0.00070,
    -0.00033, -0.00273, -0.00007,
    0.00031, 0.00287, 0.00013,
    -0.00014, -0.00170, -0.00012,
    0.00110, -0.00016, -0.00283,
    -0.00119, 0.00013, 0.00297,
    0.00063, -0.00004, -0.00177,
    0.00324, 0.00009, -0.00050,
    -0.00374, -0.00014, 0.00063,
    0.00495, 0.00024, -0.00093,
    0.00020, 0.00150, 0.00000,
    -0.00021, -0.00170, -0.00004,
    0.00024, 0.00214, 0.00012,
    -0.00061, 0.00012, 0.00156,
    0.00070, -0.00012, -0.00177,
    -0.00093, 0.00012, 0.00223
  ),
  nrow = 9
)

# Specific time interval -----
MCDirectCentral(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  delta_t = 1,
  R = 100L # use a large value for R in actual research
)

```

```

# Range of time intervals -----
mc <- MCDirectCentral(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  delta_t = 1:5,
  R = 100L # use a large value for R in actual research
)
plot(mc)

# Methods -----
# MCDirectCentral has a number of methods including
# print, summary, confint, and plot
print(mc)
summary(mc)
confint(mc, level = 0.95)
plot(mc)

```

---

|                    |                                                                                                                                              |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| MCDirectCentralStd | <i>Monte Carlo Sampling Distribution of Standardized Direct Effect Centrality Over a Specific Time Interval or a Range of Time Intervals</i> |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------|

---

## Description

This function generates a Monte Carlo method sampling distribution of the standardized direct effect centrality at a particular time interval  $\Delta t$  using the first-order stochastic differential equation model drift matrix  $\Phi$  and process noise covariance matrix  $\Sigma$ .

## Usage

```

MCDirectCentralStd(
  phi,
  sigma,
  vcov_theta,
  delta_t,
  R,
  test_phi = TRUE,
  sigma_diag = FALSE,
  ncores = NULL,
  seed = NULL,
  tol = 0.001
)

```

## Arguments

|       |                                                                                                                              |
|-------|------------------------------------------------------------------------------------------------------------------------------|
| phi   | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system. |
| sigma | Numeric matrix. The process noise covariance matrix ( $\Sigma$ ).                                                            |

|            |                                                                                                                                                                                                                                             |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| vcov_theta | Numeric matrix. The sampling variance-covariance matrix of $\text{vec}(\Phi)$ and $\text{vech}(\Sigma)$                                                                                                                                     |
| delta_t    | Numeric. Time interval ( $\Delta t$ ).                                                                                                                                                                                                      |
| R          | Positive integer. Number of replications.                                                                                                                                                                                                   |
| test_phi   | Logical. If test_phi = TRUE, the function tests the stability of the generated drift matrix $\Phi$ . If the test returns FALSE, the function generates a new drift matrix $\Phi$ and runs the test recursively until the test returns TRUE. |
| sigma_diag | Logical. If sigma_diag = TRUE, treat $\Sigma$ as a diagonal matrix.                                                                                                                                                                         |
| ncores     | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of replications R is a large value.                                                                                |
| seed       | Random seed.                                                                                                                                                                                                                                |
| tol        | Numeric. Smallest possible time interval to allow.                                                                                                                                                                                          |

### Details

See [DirectCentralStd\(\)](#) for more details.

#### Monte Carlo Method:

Let  $\theta$  be a vector that combines  $\text{vec}(\Phi)$ , that is, the elements of the  $\Phi$  matrix in vector form sorted column-wise and  $\text{vech}(\Sigma)$ , that is, the unique elements of the  $\Sigma$  matrix in vector form sorted column-wise. Let  $\hat{\theta}$  be a vector that combines  $\text{vec}(\hat{\Phi})$  and  $\text{vech}(\hat{\Sigma})$ . Based on the asymptotic properties of maximum likelihood estimators, we can assume that estimators are normally distributed around the population parameters.

$$\hat{\theta} \sim \mathcal{N}(\theta, \mathbb{V}(\hat{\theta}))$$

Using this distributional assumption, a sampling distribution of  $\hat{\theta}$  which we refer to as  $\hat{\theta}^*$  can be generated by replacing the population parameters with sample estimates, that is,

$$\hat{\theta}^* \sim \mathcal{N}(\hat{\theta}, \hat{\mathbb{V}}(\hat{\theta})).$$

Let  $g(\hat{\theta})$  be a parameter that is a function of the estimated parameters. A sampling distribution of  $g(\hat{\theta})$ , which we refer to as  $g(\hat{\theta}^*)$ , can be generated by using the simulated estimates to calculate  $g$ . The standard deviations of the simulated estimates are the standard errors. Percentiles corresponding to  $100(1 - \alpha)\%$  are the confidence intervals.

### Value

Returns an object of class `ctmedmc` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("MCDirectCentralStd").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**est** A vector of standardized direct effect centrality.

**thetahatstar** A matrix of Monte Carlo standardized direct effect centrality.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**References**

- Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. doi:10.2307/271028
- Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. doi:10.1080/10705511.2014.973960
- Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. doi:10.1037/met0000779
- Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. doi:10.1007/s11336021097670

**See Also**

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

**Examples**

```
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
sigma <- matrix(
  data = c(
    0.24455556, 0.02201587, -0.05004762,
    0.02201587, 0.07067800, 0.01539456,
    -0.05004762, 0.01539456, 0.07553061
  ),
  nrow = 3
)
```



```

)
vcov_theta <- matrix(
  data = c(
    0.00843, 0.00040, -0.00151, -0.00600, -0.00033,
    0.00110, 0.00324, 0.00020, -0.00061, -0.00115,
    0.00011, 0.00015, 0.00001, -0.00002, -0.00001,
    0.00040, 0.00374, 0.00016, -0.00022, -0.00273,
    -0.00016, 0.00009, 0.00150, 0.00012, -0.00010,
    -0.00026, 0.00002, 0.00012, 0.00004, -0.00001,
    -0.00151, 0.00016, 0.00389, 0.00103, -0.00007,
    -0.00283, -0.00050, 0.00000, 0.00156, 0.00021,
    -0.00005, -0.00031, 0.00001, 0.00007, 0.00006,
    -0.00600, -0.00022, 0.00103, 0.00644, 0.00031,
    -0.00119, -0.00374, -0.00021, 0.00070, 0.00064,
    -0.00015, -0.00005, 0.00000, 0.00003, -0.00001,
    -0.00033, -0.00273, -0.00007, 0.00031, 0.00287,
    0.00013, -0.00014, -0.00170, -0.00012, 0.00006,
    0.00014, -0.00001, -0.00015, 0.00000, 0.00001,
    0.00110, -0.00016, -0.00283, -0.00119, 0.00013,
    0.00297, 0.00063, -0.00004, -0.00177, -0.00013,
    0.00005, 0.00017, -0.00002, -0.00008, 0.00001,
    0.00324, 0.00009, -0.00050, -0.00374, -0.00014,
    0.00063, 0.00495, 0.00024, -0.00093, -0.00020,
    0.00006, -0.00010, 0.00000, -0.00001, 0.00004,
    0.00020, 0.00150, 0.00000, -0.00021, -0.00170,
    -0.00004, 0.00024, 0.00214, 0.00012, -0.00002,
    -0.00004, 0.00000, 0.00006, -0.00005, -0.00001,
    -0.00061, 0.00012, 0.00156, 0.00070, -0.00012,
    -0.00177, -0.00093, 0.00012, 0.00223, 0.00004,
    -0.00002, -0.00003, 0.00001, 0.00003, -0.00013,
    -0.00115, -0.00010, 0.00021, 0.00064, 0.00006,
    -0.00013, -0.00020, -0.00002, 0.00004, 0.00057,
    0.00001, -0.00009, 0.00000, 0.00000, 0.00001,
    0.00011, -0.00026, -0.00005, -0.00015, 0.00014,
    0.00005, 0.00006, -0.00004, -0.00002, 0.00001,
    0.00012, 0.00001, 0.00000, -0.00002, 0.00000,
    0.00015, 0.00002, -0.00031, -0.00005, -0.00001,
    0.00017, -0.00010, 0.00000, -0.00003, -0.00009,
    0.00001, 0.00014, 0.00000, 0.00000, -0.00005,
    0.00001, 0.00012, 0.00001, 0.00000, -0.00015,
    -0.00002, 0.00000, 0.00006, 0.00001, 0.00000,
    0.00000, 0.00000, 0.00010, 0.00001, 0.00000,
    -0.00002, 0.00004, 0.00007, 0.00003, 0.00000,
    -0.00008, -0.00001, -0.00005, 0.00003, 0.00000,
    -0.00002, 0.00000, 0.00001, 0.00005, 0.00001,
    -0.00001, -0.00001, 0.00006, -0.00001, 0.00001,
    0.00001, 0.00004, -0.00001, -0.00013, 0.00001,
    0.00000, -0.00005, 0.00000, 0.00001, 0.00012
  ),
  nrow = 15
)

# Specific time interval -----

```

```

MCDirectCentralStd(
  phi = phi,
  sigma = sigma,
  vcov_theta = vcov_theta,
  delta_t = 1,
  R = 100L # use a large value for R in actual research
)

# Range of time intervals -----
mc <- MCDirectCentralStd(
  phi = phi,
  sigma = sigma,
  vcov_theta = vcov_theta,
  delta_t = 1:5,
  R = 100L # use a large value for R in actual research
)
plot(mc)

# Methods -----
# MCDirectCentralStd has a number of methods including
# print, summary, confint, and plot
print(mc)
summary(mc)
confint(mc, level = 0.95)
plot(mc)

```

---

MCIndirectCentral

---

*Monte Carlo Sampling Distribution of Indirect Effect Centrality Over  
a Specific Time Interval or a Range of Time Intervals*


---

## Description

This function generates a Monte Carlo method sampling distribution of the indirect effect centrality at a particular time interval  $\Delta t$  using the first-order stochastic differential equation model drift matrix  $\Phi$ .

## Usage

```

MCIndirectCentral(
  phi,
  vcov_phi_vec,
  delta_t,
  R,
  test_phi = TRUE,
  ncores = NULL,
  seed = NULL,
  tol = 0.001
)

```

### Arguments

|              |                                                                                                                                                                                                                                             |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phi          | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system.                                                                                                                |
| vcov_phi_vec | Numeric matrix. The sampling variance-covariance matrix of $\text{vec}(\Phi)$ .                                                                                                                                                             |
| delta_t      | Numeric. Time interval ( $\Delta t$ ).                                                                                                                                                                                                      |
| R            | Positive integer. Number of replications.                                                                                                                                                                                                   |
| test_phi     | Logical. If test_phi = TRUE, the function tests the stability of the generated drift matrix $\Phi$ . If the test returns FALSE, the function generates a new drift matrix $\Phi$ and runs the test recursively until the test returns TRUE. |
| ncores       | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of replications R is a large value.                                                                                |
| seed         | Random seed.                                                                                                                                                                                                                                |
| tol          | Numeric. Smallest possible time interval to allow.                                                                                                                                                                                          |

### Details

See [IndirectCentral\(\)](#) for more details.

#### Monte Carlo Method:

Let  $\theta$  be  $\text{vec}(\Phi)$ , that is, the elements of the  $\Phi$  matrix in vector form sorted column-wise. Let  $\hat{\theta}$  be  $\text{vec}(\hat{\Phi})$ . Based on the asymptotic properties of maximum likelihood estimators, we can assume that estimators are normally distributed around the population parameters.

$$\hat{\theta} \sim \mathcal{N}(\theta, \mathbb{V}(\hat{\theta}))$$

Using this distributional assumption, a sampling distribution of  $\hat{\theta}$  which we refer to as  $\hat{\theta}^*$  can be generated by replacing the population parameters with sample estimates, that is,

$$\hat{\theta}^* \sim \mathcal{N}(\hat{\theta}, \hat{\mathbb{V}}(\hat{\theta})).$$

Let  $g(\hat{\theta})$  be a parameter that is a function of the estimated parameters. A sampling distribution of  $g(\hat{\theta})$ , which we refer to as  $g(\hat{\theta}^*)$ , can be generated by using the simulated estimates to calculate  $g$ . The standard deviations of the simulated estimates are the standard errors. Percentiles corresponding to  $100(1 - \alpha)\%$  are the confidence intervals.

### Value

Returns an object of class `ctmedmc` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("MCIndirectCentral").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**est** A vector of indirect effect centrality.

**thetahatstar** A matrix of Monte Carlo indirect effect centrality.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**References**

- Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. doi:10.2307/271028
- Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. doi:10.1080/10705511.2014.973960
- Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. doi:10.1037/met0000779
- Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. doi:10.1007/s11336021097670

**See Also**

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

**Examples**

```
set.seed(42)
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
vcov_phi_vec <- matrix(
  data = c(
    0.00843, 0.00040, -0.00151,
    -0.00600, -0.00033, 0.00110,
    0.00324, 0.00020, -0.00061,
    0.00040, 0.00374, 0.00016,
```

```

-0.00022, -0.00273, -0.00016,
0.00009, 0.00150, 0.00012,
-0.00151, 0.00016, 0.00389,
0.00103, -0.00007, -0.00283,
-0.00050, 0.00000, 0.00156,
-0.00600, -0.00022, 0.00103,
0.00644, 0.00031, -0.00119,
-0.00374, -0.00021, 0.00070,
-0.00033, -0.00273, -0.00007,
0.00031, 0.00287, 0.00013,
-0.00014, -0.00170, -0.00012,
0.00110, -0.00016, -0.00283,
-0.00119, 0.00013, 0.00297,
0.00063, -0.00004, -0.00177,
0.00324, 0.00009, -0.00050,
-0.00374, -0.00014, 0.00063,
0.00495, 0.00024, -0.00093,
0.00020, 0.00150, 0.00000,
-0.00021, -0.00170, -0.00004,
0.00024, 0.00214, 0.00012,
-0.00061, 0.00012, 0.00156,
0.00070, -0.00012, -0.00177,
-0.00093, 0.00012, 0.00223
),
nrow = 9
)

# Specific time interval -----
MCIndirectCentral(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  delta_t = 1,
  R = 100L # use a large value for R in actual research
)

# Range of time intervals -----
mc <- MCIndirectCentral(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  delta_t = 1:5,
  R = 100L # use a large value for R in actual research
)
plot(mc)

# Methods -----
# MCIndirectCentral has a number of methods including
# print, summary, confint, and plot
print(mc)
summary(mc)
confint(mc, level = 0.95)
plot(mc)

```

---

|                      |                                                                                                                                                |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| MCIndirectCentralStd | <i>Monte Carlo Sampling Distribution of Standardized Indirect Effect Centrality Over a Specific Time Interval or a Range of Time Intervals</i> |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------|

---

### Description

This function generates a Monte Carlo method sampling distribution of the standardized indirect effect centrality at a particular time interval  $\Delta t$  using the first-order stochastic differential equation model drift matrix  $\Phi$  and process noise covariance matrix  $\Sigma$ .

### Usage

```
MCIndirectCentralStd(
  phi,
  sigma,
  vcov_theta,
  delta_t,
  R,
  test_phi = TRUE,
  sigma_diag = FALSE,
  ncores = NULL,
  seed = NULL,
  tol = 0.001
)
```

### Arguments

|            |                                                                                                                                                                                                                                             |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phi        | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system.                                                                                                                |
| sigma      | Numeric matrix. The process noise covariance matrix ( $\Sigma$ ).                                                                                                                                                                           |
| vcov_theta | Numeric matrix. The sampling variance-covariance matrix of vec ( $\Phi$ ) and vech ( $\Sigma$ )                                                                                                                                             |
| delta_t    | Numeric. Time interval ( $\Delta t$ ).                                                                                                                                                                                                      |
| R          | Positive integer. Number of replications.                                                                                                                                                                                                   |
| test_phi   | Logical. If test_phi = TRUE, the function tests the stability of the generated drift matrix $\Phi$ . If the test returns FALSE, the function generates a new drift matrix $\Phi$ and runs the test recursively until the test returns TRUE. |
| sigma_diag | Logical. If sigma_diag = TRUE, treat $\Sigma$ as a diagonal matrix.                                                                                                                                                                         |
| ncores     | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of replications R is a large value.                                                                                |
| seed       | Random seed.                                                                                                                                                                                                                                |
| tol        | Numeric. Smallest possible time interval to allow.                                                                                                                                                                                          |

## Details

See `IndirectCentralStd()` for more details.

### Monte Carlo Method:

Let  $\theta$  be a vector that combines  $\text{vec}(\Phi)$ , that is, the elements of the  $\Phi$  matrix in vector form sorted column-wise and  $\text{vech}(\Sigma)$ , that is, the unique elements of the  $\Sigma$  matrix in vector form sorted column-wise. Let  $\hat{\theta}$  be a vector that combines  $\text{vec}(\hat{\Phi})$  and  $\text{vech}(\hat{\Sigma})$ . Based on the asymptotic properties of maximum likelihood estimators, we can assume that estimators are normally distributed around the population parameters.

$$\hat{\theta} \sim \mathcal{N}(\theta, \mathbb{V}(\hat{\theta}))$$

Using this distributional assumption, a sampling distribution of  $\hat{\theta}$  which we refer to as  $\hat{\theta}^*$  can be generated by replacing the population parameters with sample estimates, that is,

$$\hat{\theta}^* \sim \mathcal{N}(\hat{\theta}, \hat{\mathbb{V}}(\hat{\theta})).$$

Let  $g(\hat{\theta})$  be a parameter that is a function of the estimated parameters. A sampling distribution of  $g(\hat{\theta})$ , which we refer to as  $g(\hat{\theta}^*)$ , can be generated by using the simulated estimates to calculate  $g$ . The standard deviations of the simulated estimates are the standard errors. Percentiles corresponding to  $100(1 - \alpha)\%$  are the confidence intervals.

## Value

Returns an object of class `ctmedmc` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("MCIndirectCentralStd").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**est** A vector of standardized indirect effect centrality.

**thetahatstar** A matrix of Monte Carlo standardized indirect effect centrality.

## Author(s)

Ivan Jacob Agaloos Pesigan

## References

- Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. doi:10.2307/271028
- Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. doi:10.1080/10705511.2014.973960

Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. doi:10.1037/met0000779

Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. doi:10.1007/s11336021097670

### See Also

Other Continuous-Time Mediation Functions: `BootBeta()`, `BootBetaStd()`, `BootDirectCentral()`, `BootDirectCentralStd()`, `BootIndirectCentral()`, `BootIndirectCentralStd()`, `BootMed()`, `BootMedStd()`, `BootTotalCentral()`, `BootTotalCentralStd()`, `DeltaBeta()`, `DeltaBetaStd()`, `DeltaDirectCentral()`, `DeltaDirectCentralStd()`, `DeltaIndirectCentral()`, `DeltaMed()`, `DeltaMedStd()`, `DeltaTotalCentral()`, `DeltaTotalCentralStd()`, `Direct()`, `DirectCentral()`, `DirectCentralStd()`, `DirectStd()`, `Indirect()`, `IndirectCentral()`, `IndirectCentralStd()`, `IndirectStd()`, `MCBeta()`, `MCBetaStd()`, `MCDirectCentral()`, `MCDirectCentralStd()`, `MCIndirectCentral()`, `MCIndirectStd()`, `MCMed()`, `MCMedStd()`, `MCPPhi()`, `MCPPhiSigma()`, `MCTotalCentral()`, `MCTotalCentralStd()`, `Med()`, `MedStd()`, `PosteriorBeta()`, `PosteriorBetaStd()`, `PosteriorDirectCentral()`, `PosteriorDirectCentralStd()`, `PosteriorIndirectCentral()`, `PosteriorIndirectCentralStd()`, `PosteriorMed()`, `PosteriorMedStd()`, `PosteriorTotalCentral()`, `PosteriorTotalCentralStd()`, `Total()`, `TotalCentral()`, `TotalCentralStd()`, `TotalStd()`, `Trajectory()`

### Examples

```
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
sigma <- matrix(
  data = c(
    0.24455556, 0.02201587, -0.05004762,
    0.02201587, 0.07067800, 0.01539456,
    -0.05004762, 0.01539456, 0.07553061
  ),
  nrow = 3
)
vcov_theta <- matrix(
  data = c(
    0.00843, 0.00040, -0.00151, -0.00600, -0.00033,
    0.00110, 0.00324, 0.00020, -0.00061, -0.00115,
    0.00011, 0.00015, 0.00001, -0.00002, -0.00001,
    0.00040, 0.00374, 0.00016, -0.00022, -0.00273,
    -0.00016, 0.00009, 0.00150, 0.00012, -0.00010,
    -0.00026, 0.00002, 0.00012, 0.00004, -0.00001,
    -0.00151, 0.00016, 0.00389, 0.00103, -0.00007,
    -0.00283, -0.00050, 0.00000, 0.00156, 0.00021,
    -0.00005, -0.00031, 0.00001, 0.00007, 0.00006,
```



```

-0.00600, -0.00022, 0.00103, 0.00644, 0.00031,
-0.00119, -0.00374, -0.00021, 0.00070, 0.00064,
-0.00015, -0.00005, 0.00000, 0.00003, -0.00001,
-0.00033, -0.00273, -0.00007, 0.00031, 0.00287,
0.00013, -0.00014, -0.00170, -0.00012, 0.00006,
0.00014, -0.00001, -0.00015, 0.00000, 0.00001,
0.00110, -0.00016, -0.00283, -0.00119, 0.00013,
0.00297, 0.00063, -0.00004, -0.00177, -0.00013,
0.00005, 0.00017, -0.00002, -0.00008, 0.00001,
0.00324, 0.00009, -0.00050, -0.00374, -0.00014,
0.00063, 0.00495, 0.00024, -0.00093, -0.00020,
0.00006, -0.00010, 0.00000, -0.00001, 0.00004,
0.00020, 0.00150, 0.00000, -0.00021, -0.00170,
-0.00004, 0.00024, 0.00214, 0.00012, -0.00002,
-0.00004, 0.00000, 0.00006, -0.00005, -0.00001,
-0.00061, 0.00012, 0.00156, 0.00070, -0.00012,
-0.00177, -0.00093, 0.00012, 0.00223, 0.00004,
-0.00002, -0.00003, 0.00001, 0.00003, -0.00013,
-0.00115, -0.00010, 0.00021, 0.00064, 0.00006,
-0.00013, -0.00020, -0.00002, 0.00004, 0.00057,
0.00001, -0.00009, 0.00000, 0.00000, 0.00001,
0.00011, -0.00026, -0.00005, -0.00015, 0.00014,
0.00005, 0.00006, -0.00004, -0.00002, 0.00001,
0.00012, 0.00001, 0.00000, -0.00002, 0.00000,
0.00015, 0.00002, -0.00031, -0.00005, -0.00001,
0.00017, -0.00010, 0.00000, -0.00003, -0.00009,
0.00001, 0.00014, 0.00000, 0.00000, -0.00005,
0.00001, 0.00012, 0.00001, 0.00000, -0.00015,
-0.00002, 0.00000, 0.00006, 0.00001, 0.00000,
0.00000, 0.00000, 0.00010, 0.00001, 0.00000,
-0.00002, 0.00004, 0.00007, 0.00003, 0.00000,
-0.00008, -0.00001, -0.00005, 0.00003, 0.00000,
-0.00002, 0.00000, 0.00001, 0.00005, 0.00001,
-0.00001, -0.00001, 0.00006, -0.00001, 0.00001,
0.00001, 0.00004, -0.00001, -0.00013, 0.00001,
0.00000, -0.00005, 0.00000, 0.00001, 0.00012
),
nrow = 15
)

# Specific time interval -----
MCIndirectCentralStd(
  phi = phi,
  sigma = sigma,
  vcov_theta = vcov_theta,
  delta_t = 1,
  R = 100L # use a large value for R in actual research
)

# Range of time intervals -----
mc <- MCIndirectCentralStd(
  phi = phi,
  sigma = sigma,

```

```

vcov_theta = vcov_theta,
delta_t = 1:5,
R = 100L # use a large value for R in actual research
)
plot(mc)

# Methods -----
# MCIndirectCentralStd has a number of methods including
# print, summary, confint, and plot
print(mc)
summary(mc)
confint(mc, level = 0.95)
plot(mc)

```

MCMed

*Monte Carlo Sampling Distribution of Total, Direct, and Indirect Effects of  $X$  on  $Y$  Through  $M$  Over a Specific Time Interval or a Range of Time Intervals*

## Description

This function generates a Monte Carlo method sampling distribution of the total, direct and indirect effects of the independent variable  $X$  on the dependent variable  $Y$  through mediator variables  $\mathbf{m}$  over a specific time interval  $\Delta t$  or a range of time intervals using the first-order stochastic differential equation model drift matrix  $\Phi$ .

## Usage

```

MCMed(
  phi,
  vcov_phi_vec,
  delta_t,
  from,
  to,
  med,
  R,
  test_phi = TRUE,
  ncores = NULL,
  seed = NULL,
  tol = 0.001
)

```

## Arguments

|                           |                                                                                                                                           |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <code>phi</code>          | Numeric matrix. The drift matrix ( $\Phi$ ). <code>phi</code> should have row and column names pertaining to the variables in the system. |
| <code>vcov_phi_vec</code> | Numeric matrix. The sampling variance-covariance matrix of $\text{vec}(\Phi)$ .                                                           |

|          |                                                                                                                                                                                                                                             |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| delta_t  | Numeric. Time interval ( $\Delta t$ ).                                                                                                                                                                                                      |
| from     | Character string. Name of the independent variable $X$ in phi.                                                                                                                                                                              |
| to       | Character string. Name of the dependent variable $Y$ in phi.                                                                                                                                                                                |
| med      | Character vector. Name/s of the mediator variable/s in phi.                                                                                                                                                                                 |
| R        | Positive integer. Number of replications.                                                                                                                                                                                                   |
| test_phi | Logical. If test_phi = TRUE, the function tests the stability of the generated drift matrix $\Phi$ . If the test returns FALSE, the function generates a new drift matrix $\Phi$ and runs the test recursively until the test returns TRUE. |
| ncores   | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of replications R is a large value.                                                                                |
| seed     | Random seed.                                                                                                                                                                                                                                |
| tol      | Numeric. Smallest possible time interval to allow.                                                                                                                                                                                          |

### Details

See [Total\(\)](#), [Direct\(\)](#), and [Indirect\(\)](#) for more details.

#### Monte Carlo Method:

Let  $\theta$  be  $\text{vec}(\Phi)$ , that is, the elements of the  $\Phi$  matrix in vector form sorted column-wise. Let  $\hat{\theta}$  be  $\text{vec}(\hat{\Phi})$ . Based on the asymptotic properties of maximum likelihood estimators, we can assume that estimators are normally distributed around the population parameters.

$$\hat{\theta} \sim \mathcal{N}(\theta, \mathbb{V}(\hat{\theta}))$$

Using this distributional assumption, a sampling distribution of  $\hat{\theta}$  which we refer to as  $\hat{\theta}^*$  can be generated by replacing the population parameters with sample estimates, that is,

$$\hat{\theta}^* \sim \mathcal{N}(\hat{\theta}, \hat{\mathbb{V}}(\hat{\theta})).$$

Let  $g(\hat{\theta})$  be a parameter that is a function of the estimated parameters. A sampling distribution of  $g(\hat{\theta})$ , which we refer to as  $g(\hat{\theta}^*)$ , can be generated by using the simulated estimates to calculate  $g$ . The standard deviations of the simulated estimates are the standard errors. Percentiles corresponding to  $100(1 - \alpha)\%$  are the confidence intervals.

### Value

Returns an object of class `ctmedmc` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("MCMed").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**est** A vector of total, direct, and indirect effects.

**thetahatstar** A matrix of Monte Carlo total, direct, and indirect effects.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**References**

Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. doi:10.2307/271028

Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. doi:10.1080/10705511.2014.973960

Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. doi:10.1037/met0000779

Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. doi:10.1007/s11336021097670

**See Also**

Other Continuous-Time Mediation Functions: `BootBeta()`, `BootBetaStd()`, `BootDirectCentral()`, `BootDirectCentralStd()`, `BootIndirectCentral()`, `BootIndirectCentralStd()`, `BootMed()`, `BootMedStd()`, `BootTotalCentral()`, `BootTotalCentralStd()`, `DeltaBeta()`, `DeltaBetaStd()`, `DeltaDirectCentral()`, `DeltaDirectCentralStd()`, `DeltaIndirectCentral()`, `DeltaMed()`, `DeltaMedStd()`, `DeltaTotalCentral()`, `DeltaTotalCentralStd()`, `Direct()`, `DirectCentral()`, `DirectCentralStd()`, `DirectStd()`, `Indirect()`, `IndirectCentral()`, `IndirectCentralStd()`, `IndirectStd()`, `MCBeta()`, `MCBetaStd()`, `MCDirectCentral()`, `MCDirectCentralStd()`, `MCIndirectCentral()`, `MCIndirectCentralStd()`, `MCMedStd()`, `MCPhi()`, `MCPhiSigma()`, `MCTotalCentral()`, `MCTotalCentralStd()`, `Med()`, `MedStd()`, `PosteriorBeta()`, `PosteriorBetaStd()`, `PosteriorDirectCentral()`, `PosteriorDirectCentralStd()`, `PosteriorIndirectCentral()`, `PosteriorIndirectCentralStd()`, `PosteriorMed()`, `PosteriorMedStd()`, `PosteriorTotalCentral()`, `PosteriorTotalCentralStd()`, `Total()`, `TotalCentral()`, `TotalCentralStd()`, `TotalStd()`, `Trajectory()`

**Examples**

```
set.seed(42)
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
vcov_phi_vec <- matrix(
  data = c(
    0.00843, 0.00040, -0.00151,
    -0.00600, -0.00033, 0.00110,
    0.00324, 0.00020, -0.00061,
    0.00040, 0.00374, 0.00016,
```

```

-0.00022, -0.00273, -0.00016,
0.00009, 0.00150, 0.00012,
-0.00151, 0.00016, 0.00389,
0.00103, -0.00007, -0.00283,
-0.00050, 0.00000, 0.00156,
-0.00600, -0.00022, 0.00103,
0.00644, 0.00031, -0.00119,
-0.00374, -0.00021, 0.00070,
-0.00033, -0.00273, -0.00007,
0.00031, 0.00287, 0.00013,
-0.00014, -0.00170, -0.00012,
0.00110, -0.00016, -0.00283,
-0.00119, 0.00013, 0.00297,
0.00063, -0.00004, -0.00177,
0.00324, 0.00009, -0.00050,
-0.00374, -0.00014, 0.00063,
0.00495, 0.00024, -0.00093,
0.00020, 0.00150, 0.00000,
-0.00021, -0.00170, -0.00004,
0.00024, 0.00214, 0.00012,
-0.00061, 0.00012, 0.00156,
0.00070, -0.00012, -0.00177,
-0.00093, 0.00012, 0.00223
),
nrow = 9
)

# Specific time interval -----
MCMed(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  delta_t = 1,
  from = "x",
  to = "y",
  med = "m",
  R = 100L # use a large value for R in actual research
)

# Range of time intervals -----
mc <- MCMed(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  delta_t = 1:5,
  from = "x",
  to = "y",
  med = "m",
  R = 100L # use a large value for R in actual research
)
plot(mc)

# Methods -----
# MCMed has a number of methods including
# print, summary, confint, and plot

```

```
print(mc)
summary(mc)
confint(mc, level = 0.95)
```

MCMedStd

*Monte Carlo Sampling Distribution of Standardized Total, Direct, and Indirect Effects of X on Y Through M Over a Specific Time Interval or a Range of Time Intervals*

## Description

This function generates a Monte Carlo method sampling distribution of the standardized total, direct and indirect effects of the independent variable  $X$  on the dependent variable  $Y$  through mediator variables  $\mathbf{m}$  over a specific time interval  $\Delta t$  or a range of time intervals using the first-order stochastic differential equation model drift matrix  $\Phi$  and process noise covariance matrix  $\Sigma$ .

## Usage

```
MCMedStd(
  phi,
  sigma,
  vcov_theta,
  delta_t,
  from,
  to,
  med,
  R,
  test_phi = TRUE,
  sigma_diag = FALSE,
  ncores = NULL,
  seed = NULL,
  tol = 0.001
)
```

## Arguments

|            |                                                                                                                              |
|------------|------------------------------------------------------------------------------------------------------------------------------|
| phi        | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system. |
| sigma      | Numeric matrix. The process noise covariance matrix ( $\Sigma$ ).                                                            |
| vcov_theta | Numeric matrix. The sampling variance-covariance matrix of vec ( $\Phi$ ) and vech ( $\Sigma$ )                              |
| delta_t    | Numeric. Time interval ( $\Delta t$ ).                                                                                       |
| from       | Character string. Name of the independent variable $X$ in phi.                                                               |
| to         | Character string. Name of the dependent variable $Y$ in phi.                                                                 |
| med        | Character vector. Name/s of the mediator variable/s in phi.                                                                  |

|            |                                                                                                                                                                                                                                             |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R          | Positive integer. Number of replications.                                                                                                                                                                                                   |
| test_phi   | Logical. If test_phi = TRUE, the function tests the stability of the generated drift matrix $\Phi$ . If the test returns FALSE, the function generates a new drift matrix $\Phi$ and runs the test recursively until the test returns TRUE. |
| sigma_diag | Logical. If sigma_diag = TRUE, treat $\Sigma$ as a diagonal matrix.                                                                                                                                                                         |
| ncores     | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of replications R is a large value.                                                                                |
| seed       | Random seed.                                                                                                                                                                                                                                |
| tol        | Numeric. Smallest possible time interval to allow.                                                                                                                                                                                          |

### Details

See [TotalStd\(\)](#), [DirectStd\(\)](#), and [IndirectStd\(\)](#) for more details.

#### Monte Carlo Method:

Let  $\theta$  be a vector that combines  $\text{vec}(\Phi)$ , that is, the elements of the  $\Phi$  matrix in vector form sorted column-wise and  $\text{vech}(\Sigma)$ , that is, the unique elements of the  $\Sigma$  matrix in vector form sorted column-wise. Let  $\hat{\theta}$  be a vector that combines  $\text{vec}(\hat{\Phi})$  and  $\text{vech}(\hat{\Sigma})$ . Based on the asymptotic properties of maximum likelihood estimators, we can assume that estimators are normally distributed around the population parameters.

$$\hat{\theta} \sim \mathcal{N}(\theta, \mathbb{V}(\hat{\theta}))$$

Using this distributional assumption, a sampling distribution of  $\hat{\theta}$  which we refer to as  $\hat{\theta}^*$  can be generated by replacing the population parameters with sample estimates, that is,

$$\hat{\theta}^* \sim \mathcal{N}(\hat{\theta}, \hat{\mathbb{V}}(\hat{\theta})).$$

Let  $g(\hat{\theta})$  be a parameter that is a function of the estimated parameters. A sampling distribution of  $g(\hat{\theta})$ , which we refer to as  $g(\hat{\theta}^*)$ , can be generated by using the simulated estimates to calculate  $g$ . The standard deviations of the simulated estimates are the standard errors. Percentiles corresponding to  $100(1 - \alpha)\%$  are the confidence intervals.

### Value

Returns an object of class `ctmedmc` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("MCMedStd").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**est** A vector of standardized total, direct, and indirect effects.

**thetahatstar** A matrix of Monte Carlo standardized total, direct, and indirect effects.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**References**

- Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. doi:10.2307/271028
- Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. doi:10.1080/10705511.2014.973960
- Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. doi:10.1037/met0000779
- Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. doi:10.1007/s11336021097670

**See Also**

Other Continuous-Time Mediation Functions: `BootBeta()`, `BootBetaStd()`, `BootDirectCentral()`, `BootDirectCentralStd()`, `BootIndirectCentral()`, `BootIndirectCentralStd()`, `BootMed()`, `BootMedStd()`, `BootTotalCentral()`, `BootTotalCentralStd()`, `DeltaBeta()`, `DeltaBetaStd()`, `DeltaDirectCentral()`, `DeltaDirectCentralStd()`, `DeltaIndirectCentral()`, `DeltaMed()`, `DeltaMedStd()`, `DeltaTotalCentral()`, `DeltaTotalCentralStd()`, `Direct()`, `DirectCentral()`, `DirectCentralStd()`, `DirectStd()`, `Indirect()`, `IndirectCentral()`, `IndirectCentralStd()`, `IndirectStd()`, `MCBeta()`, `MCBetaStd()`, `MCDirectCentral()`, `MCDirectCentralStd()`, `MCIndirectCentral()`, `MCIndirectCentralStd()`, `MCMed()`, `MCPhi()`, `MCPhiSigma()`, `MCTotalCentral()`, `MCTotalCentralStd()`, `Med()`, `MedStd()`, `PosteriorBeta()`, `PosteriorBetaStd()`, `PosteriorDirectCentral()`, `PosteriorDirectCentralStd()`, `PosteriorIndirectCentral()`, `PosteriorIndirectCentralStd()`, `PosteriorMed()`, `PosteriorMedStd()`, `PosteriorTotalCentral()`, `PosteriorTotalCentralStd()`, `Total()`, `TotalCentral()`, `TotalCentralStd()`, `TotalStd()`, `Trajectory()`

**Examples**

```
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
sigma <- matrix(
  data = c(
    0.24455556, 0.02201587, -0.05004762,
    0.02201587, 0.07067800, 0.01539456,
    -0.05004762, 0.01539456, 0.07553061
  ),
  nrow = 3
)
```



```

)
vcov_theta <- matrix(
  data = c(
    0.00843, 0.00040, -0.00151, -0.00600, -0.00033,
    0.00110, 0.00324, 0.00020, -0.00061, -0.00115,
    0.00011, 0.00015, 0.00001, -0.00002, -0.00001,
    0.00040, 0.00374, 0.00016, -0.00022, -0.00273,
    -0.00016, 0.00009, 0.00150, 0.00012, -0.00010,
    -0.00026, 0.00002, 0.00012, 0.00004, -0.00001,
    -0.00151, 0.00016, 0.00389, 0.00103, -0.00007,
    -0.00283, -0.00050, 0.00000, 0.00156, 0.00021,
    -0.00005, -0.00031, 0.00001, 0.00007, 0.00006,
    -0.00600, -0.00022, 0.00103, 0.00644, 0.00031,
    -0.00119, -0.00374, -0.00021, 0.00070, 0.00064,
    -0.00015, -0.00005, 0.00000, 0.00003, -0.00001,
    -0.00033, -0.00273, -0.00007, 0.00031, 0.00287,
    0.00013, -0.00014, -0.00170, -0.00012, 0.00006,
    0.00014, -0.00001, -0.00015, 0.00000, 0.00001,
    0.00110, -0.00016, -0.00283, -0.00119, 0.00013,
    0.00297, 0.00063, -0.00004, -0.00177, -0.00013,
    0.00005, 0.00017, -0.00002, -0.00008, 0.00001,
    0.00324, 0.00009, -0.00050, -0.00374, -0.00014,
    0.00063, 0.00495, 0.00024, -0.00093, -0.00020,
    0.00006, -0.00010, 0.00000, -0.00001, 0.00004,
    0.00020, 0.00150, 0.00000, -0.00021, -0.00170,
    -0.00004, 0.00024, 0.00214, 0.00012, -0.00002,
    -0.00004, 0.00000, 0.00006, -0.00005, -0.00001,
    -0.00061, 0.00012, 0.00156, 0.00070, -0.00012,
    -0.00177, -0.00093, 0.00012, 0.00223, 0.00004,
    -0.00002, -0.00003, 0.00001, 0.00003, -0.00013,
    -0.00115, -0.00010, 0.00021, 0.00064, 0.00006,
    -0.00013, -0.00020, -0.00002, 0.00004, 0.00057,
    0.00001, -0.00009, 0.00000, 0.00000, 0.00001,
    0.00011, -0.00026, -0.00005, -0.00015, 0.00014,
    0.00005, 0.00006, -0.00004, -0.00002, 0.00001,
    0.00012, 0.00001, 0.00000, -0.00002, 0.00000,
    0.00015, 0.00002, -0.00031, -0.00005, -0.00001,
    0.00017, -0.00010, 0.00000, -0.00003, -0.00009,
    0.00001, 0.00014, 0.00000, 0.00000, -0.00005,
    0.00001, 0.00012, 0.00001, 0.00000, -0.00015,
    -0.00002, 0.00000, 0.00006, 0.00001, 0.00000,
    0.00000, 0.00000, 0.00010, 0.00001, 0.00000,
    -0.00002, 0.00004, 0.00007, 0.00003, 0.00000,
    -0.00008, -0.00001, -0.00005, 0.00003, 0.00000,
    -0.00002, 0.00000, 0.00001, 0.00005, 0.00001,
    -0.00001, -0.00001, 0.00006, -0.00001, 0.00001,
    0.00001, 0.00004, -0.00001, -0.00013, 0.00001,
    0.00000, -0.00005, 0.00000, 0.00001, 0.00012
  ),
  nrow = 15
)

# Specific time interval -----

```

```

MCMedStd(
  phi = phi,
  sigma = sigma,
  vcov_theta = vcov_theta,
  delta_t = 1,
  from = "x",
  to = "y",
  med = "m",
  R = 100L # use a large value for R in actual research
)

# Range of time intervals -----
mc <- MCMedStd(
  phi = phi,
  sigma = sigma,
  vcov_theta = vcov_theta,
  delta_t = 1:5,
  from = "x",
  to = "y",
  med = "m",
  R = 100L # use a large value for R in actual research
)
plot(mc)

# Methods -----
# MCMedStd has a number of methods including
# print, summary, confint, and plot
print(mc)
summary(mc)
confint(mc, level = 0.95)

```

---

MCPhi

---

*Generate Random Drift Matrices Using the Monte Carlo Method*


---

## Description

This function generates random drift matrices  $\Phi$  using the Monte Carlo method.

## Usage

```
MCPhi(phi, vcov_phi_vec, R, test_phi = TRUE, ncores = NULL, seed = NULL)
```

## Arguments

|              |                                                                                                                              |
|--------------|------------------------------------------------------------------------------------------------------------------------------|
| phi          | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system. |
| vcov_phi_vec | Numeric matrix. The sampling variance-covariance matrix of $\text{vec}(\Phi)$ .                                              |
| R            | Positive integer. Number of replications.                                                                                    |

|          |                                                                                                                                                                                                                                             |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| test_phi | Logical. If test_phi = TRUE, the function tests the stability of the generated drift matrix $\Phi$ . If the test returns FALSE, the function generates a new drift matrix $\Phi$ and runs the test recursively until the test returns TRUE. |
| ncores   | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of replications R is a large value.                                                                                |
| seed     | Random seed.                                                                                                                                                                                                                                |

## Details

### Monte Carlo Method:

Let  $\theta$  be  $\text{vec}(\Phi)$ , that is, the elements of the  $\Phi$  matrix in vector form sorted column-wise. Let  $\hat{\theta}$  be  $\text{vec}(\hat{\Phi})$ . Based on the asymptotic properties of maximum likelihood estimators, we can assume that estimators are normally distributed around the population parameters.

$$\hat{\theta} \sim \mathcal{N}(\theta, \mathbb{V}(\hat{\theta}))$$

Using this distributional assumption, a sampling distribution of  $\hat{\theta}$  which we refer to as  $\hat{\theta}^*$  can be generated by replacing the population parameters with sample estimates, that is,

$$\hat{\theta}^* \sim \mathcal{N}(\hat{\theta}, \hat{\mathbb{V}}(\hat{\theta})).$$

## Value

Returns an object of class `ctmedmcphi` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("MCPhi").

**output** A list of simulated drift matrices.

## Author(s)

Ivan Jacob Agaloos Pesigan

## See Also

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

**Examples**

```

set.seed(42)
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
MCPhi(
  phi = phi,
  vcov_phi_vec = 0.1 * diag(9),
  R = 100L # use a large value for R in actual research
)
phi <- matrix(
  data = c(
    -6, 5.5, 0, 0,
    1.25, -2.5, 5.9, -7.3,
    0, 0, -6, 2.5,
    5, 0, 0, -6
  ),
  nrow = 4
)
colnames(phi) <- rownames(phi) <- paste0("y", 1:4)
MCPhi(
  phi = phi,
  vcov_phi_vec = 0.1 * diag(16),
  R = 100L, # use a large value for R in actual research
  test_phi = FALSE
)

```

---

MCPhiSigma

---

*Generate Random Drift Matrices and Process Noise Covariance Matrices Using the Monte Carlo Method*


---

**Description**

This function generates random drift matrices  $\Phi$  and process noise covariance matrices  $\Sigma$  using the Monte Carlo method.

**Usage**

```

MCPhiSigma(
  phi,
  sigma,
  vcov_theta,

```

```

R,
test_phi = TRUE,
sigma_diag = FALSE,
ncores = NULL,
seed = NULL
)

```

### Arguments

|            |                                                                                                                                                                                                                                             |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phi        | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system.                                                                                                                |
| sigma      | Numeric matrix. The process noise covariance matrix ( $\Sigma$ ).                                                                                                                                                                           |
| vcov_theta | Numeric matrix. The sampling variance-covariance matrix of $\text{vec}(\Phi)$ and $\text{vech}(\Sigma)$                                                                                                                                     |
| R          | Positive integer. Number of replications.                                                                                                                                                                                                   |
| test_phi   | Logical. If test_phi = TRUE, the function tests the stability of the generated drift matrix $\Phi$ . If the test returns FALSE, the function generates a new drift matrix $\Phi$ and runs the test recursively until the test returns TRUE. |
| sigma_diag | Logical. If sigma_diag = TRUE, treat $\Sigma$ as a diagonal matrix.                                                                                                                                                                         |
| ncores     | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of replications R is a large value.                                                                                |
| seed       | Random seed.                                                                                                                                                                                                                                |

### Details

#### Monte Carlo Method:

Let  $\theta$  be a vector that combines  $\text{vec}(\Phi)$ , that is, the elements of the  $\Phi$  matrix in vector form sorted column-wise and  $\text{vech}(\Sigma)$ , that is, the unique elements of the  $\Sigma$  matrix in vector form sorted column-wise. Let  $\hat{\theta}$  be a vector that combines  $\text{vec}(\hat{\Phi})$  and  $\text{vech}(\hat{\Sigma})$ . Based on the asymptotic properties of maximum likelihood estimators, we can assume that estimators are normally distributed around the population parameters.

$$\hat{\theta} \sim \mathcal{N}(\theta, \mathbb{V}(\hat{\theta}))$$

Using this distributional assumption, a sampling distribution of  $\hat{\theta}$  which we refer to as  $\hat{\theta}^*$  can be generated by replacing the population parameters with sample estimates, that is,

$$\hat{\theta}^* \sim \mathcal{N}(\hat{\theta}, \hat{\mathbb{V}}(\hat{\theta})).$$

### Value

Returns an object of class `ctmedmcphi` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("MCPhiSigma").

**output** A list of simulated draws. Each element contains a drift matrix and a process noise covariance matrix.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**See Also**

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

**Examples**

```
set.seed(42)
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
sigma <- matrix(
  data = c(
    0.24455556, 0.02201587, -0.05004762,
    0.02201587, 0.07067800, 0.01539456,
    -0.05004762, 0.01539456, 0.07553061
  ),
  nrow = 3
)
MCPhiSigma(
  phi = phi,
  sigma = sigma,
  vcov_theta = 0.1 * diag(15),
  R = 100L # use a large value for R in actual research
)
```

---

|                |                                                                                                                                |
|----------------|--------------------------------------------------------------------------------------------------------------------------------|
| MCTotalCentral | <i>Monte Carlo Sampling Distribution of Total Effect Centrality Over a Specific Time Interval or a Range of Time Intervals</i> |
|----------------|--------------------------------------------------------------------------------------------------------------------------------|

---

## Description

This function generates a Monte Carlo method sampling distribution of the total effect centrality at a particular time interval  $\Delta t$  using the first-order stochastic differential equation model drift matrix  $\Phi$ .

## Usage

```
MCTotalCentral(
  phi,
  vcov_phi_vec,
  delta_t,
  R,
  test_phi = TRUE,
  ncores = NULL,
  seed = NULL,
  tol = 0.001
)
```

## Arguments

|              |                                                                                                                                                                                                                                             |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phi          | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system.                                                                                                                |
| vcov_phi_vec | Numeric matrix. The sampling variance-covariance matrix of $\text{vec}(\Phi)$ .                                                                                                                                                             |
| delta_t      | Numeric. Time interval ( $\Delta t$ ).                                                                                                                                                                                                      |
| R            | Positive integer. Number of replications.                                                                                                                                                                                                   |
| test_phi     | Logical. If test_phi = TRUE, the function tests the stability of the generated drift matrix $\Phi$ . If the test returns FALSE, the function generates a new drift matrix $\Phi$ and runs the test recursively until the test returns TRUE. |
| ncores       | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of replications R is a large value.                                                                                |
| seed         | Random seed.                                                                                                                                                                                                                                |
| tol          | Numeric. Smallest possible time interval to allow.                                                                                                                                                                                          |

## Details

See [TotalCentral\(\)](#) for more details.

### Monte Carlo Method:

Let  $\theta$  be  $\text{vec}(\Phi)$ , that is, the elements of the  $\Phi$  matrix in vector form sorted column-wise. Let  $\hat{\theta}$  be  $\text{vec}(\hat{\Phi})$ . Based on the asymptotic properties of maximum likelihood estimators, we can assume that estimators are normally distributed around the population parameters.

$$\hat{\theta} \sim \mathcal{N}(\theta, \mathbb{V}(\hat{\theta}))$$

Using this distributional assumption, a sampling distribution of  $\hat{\theta}$  which we refer to as  $\hat{\theta}^*$  can be generated by replacing the population parameters with sample estimates, that is,

$$\hat{\theta}^* \sim \mathcal{N}(\hat{\theta}, \hat{\mathbb{V}}(\hat{\theta})).$$

Let  $g(\hat{\theta})$  be a parameter that is a function of the estimated parameters. A sampling distribution of  $g(\hat{\theta})$ , which we refer to as  $g(\hat{\theta}^*)$ , can be generated by using the simulated estimates to calculate  $g$ . The standard deviations of the simulated estimates are the standard errors. Percentiles corresponding to  $100(1 - \alpha)\%$  are the confidence intervals.

## Value

Returns an object of class `ctmedmc` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("MCTotalCentral").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**est** A vector of total effect centrality.

**thetahatstar** A matrix of Monte Carlo total effect centrality.

## Author(s)

Ivan Jacob Agaloos Pesigan

## References

- Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. [doi:10.2307/271028](https://doi.org/10.2307/271028)
- Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. [doi:10.1080/10705511.2014.973960](https://doi.org/10.1080/10705511.2014.973960)
- Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. [doi:10.1037/met0000779](https://doi.org/10.1037/met0000779)
- Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. [doi:10.1007/s11336021097670](https://doi.org/10.1007/s11336021097670)



**See Also**

Other Continuous-Time Mediation Functions: `BootBeta()`, `BootBetaStd()`, `BootDirectCentral()`, `BootDirectCentralStd()`, `BootIndirectCentral()`, `BootIndirectCentralStd()`, `BootMed()`, `BootMedStd()`, `BootTotalCentral()`, `BootTotalCentralStd()`, `DeltaBeta()`, `DeltaBetaStd()`, `DeltaDirectCentral()`, `DeltaDirectCentralStd()`, `DeltaIndirectCentral()`, `DeltaMed()`, `DeltaMedStd()`, `DeltaTotalCentral()`, `DeltaTotalCentralStd()`, `Direct()`, `DirectCentral()`, `DirectCentralStd()`, `DirectStd()`, `Indirect()`, `IndirectCentral()`, `IndirectCentralStd()`, `IndirectStd()`, `MCBeta()`, `MCBetaStd()`, `MCDirectCentral()`, `MCDirectCentralStd()`, `MCIndirectCentral()`, `MCIndirectCentralStd()`, `MCMed()`, `MCMedStd()`, `MCPPhi()`, `MCPPhiSigma()`, `MCTotalCentralStd()`, `Med()`, `MedStd()`, `PosteriorBeta()`, `PosteriorBetaStd()`, `PosteriorDirectCentral()`, `PosteriorDirectCentralStd()`, `PosteriorIndirectCentral()`, `PosteriorIndirectCentralStd()`, `PosteriorMed()`, `PosteriorMedStd()`, `PosteriorTotalCentral()`, `PosteriorTotalCentralStd()`, `Total()`, `TotalCentral()`, `TotalCentralStd()`, `TotalStd()`, `Trajectory()`

**Examples**

```
set.seed(42)
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
vcov_phi_vec <- matrix(
  data = c(
    0.00843, 0.00040, -0.00151,
    -0.00600, -0.00033, 0.00110,
    0.00324, 0.00020, -0.00061,
    0.00040, 0.00374, 0.00016,
    -0.00022, -0.00273, -0.00016,
    0.00009, 0.00150, 0.00012,
    -0.00151, 0.00016, 0.00389,
    0.00103, -0.00007, -0.00283,
    -0.00050, 0.00000, 0.00156,
    -0.00600, -0.00022, 0.00103,
    0.00644, 0.00031, -0.00119,
    -0.00374, -0.00021, 0.00070,
    -0.00033, -0.00273, -0.00007,
    0.00031, 0.00287, 0.00013,
    -0.00014, -0.00170, -0.00012,
    0.00110, -0.00016, -0.00283,
    -0.00119, 0.00013, 0.00297,
    0.00063, -0.00004, -0.00177,
    0.00324, 0.00009, -0.00050,
    -0.00374, -0.00014, 0.00063,
    0.00495, 0.00024, -0.00093,
    0.00020, 0.00150, 0.00000,
    -0.00021, -0.00170, -0.00004,
  )
)
```

```

    0.00024, 0.00214, 0.00012,
    -0.00061, 0.00012, 0.00156,
    0.00070, -0.00012, -0.00177,
    -0.00093, 0.00012, 0.00223
  ),
  nrow = 9
)

# Specific time interval -----
MCTotalCentral(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  delta_t = 1,
  R = 100L # use a large value for R in actual research
)

# Range of time intervals -----
mc <- MCTotalCentral(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  delta_t = 1:5,
  R = 100L # use a large value for R in actual research
)
plot(mc)

# Methods -----
# MCTotalCentral has a number of methods including
# print, summary, confint, and plot
print(mc)
summary(mc)
confint(mc, level = 0.95)
plot(mc)

```

---

MCTotalCentralStd

---

*Monte Carlo Sampling Distribution of Standardized Total Effect Centrality Over a Specific Time Interval or a Range of Time Intervals*


---

## Description

This function generates a Monte Carlo method sampling distribution of the standardized total effect centrality at a particular time interval  $\Delta t$  using the first-order stochastic differential equation model drift matrix  $\Phi$  and process noise covariance matrix  $\Sigma$ .

## Usage

```

MCTotalCentralStd(
  phi,
  sigma,
  vcov_theta,

```

```

    delta_t,
    R,
    test_phi = TRUE,
    sigma_diag = FALSE,
    ncores = NULL,
    seed = NULL,
    tol = 0.001
)

```

### Arguments

|            |                                                                                                                                                                                                                                             |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phi        | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system.                                                                                                                |
| sigma      | Numeric matrix. The process noise covariance matrix ( $\Sigma$ ).                                                                                                                                                                           |
| vcov_theta | Numeric matrix. The sampling variance-covariance matrix of vec ( $\Phi$ ) and vech ( $\Sigma$ )                                                                                                                                             |
| delta_t    | Numeric. Time interval ( $\Delta t$ ).                                                                                                                                                                                                      |
| R          | Positive integer. Number of replications.                                                                                                                                                                                                   |
| test_phi   | Logical. If test_phi = TRUE, the function tests the stability of the generated drift matrix $\Phi$ . If the test returns FALSE, the function generates a new drift matrix $\Phi$ and runs the test recursively until the test returns TRUE. |
| sigma_diag | Logical. If sigma_diag = TRUE, treat $\Sigma$ as a diagonal matrix.                                                                                                                                                                         |
| ncores     | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of replications R is a large value.                                                                                |
| seed       | Random seed.                                                                                                                                                                                                                                |
| tol        | Numeric. Smallest possible time interval to allow.                                                                                                                                                                                          |

### Details

See [TotalCentralStd\(\)](#) for more details.

#### Monte Carlo Method:

Let  $\theta$  be a vector that combines vec ( $\Phi$ ), that is, the elements of the  $\Phi$  matrix in vector form sorted column-wise and vech ( $\Sigma$ ), that is, the unique elements of the  $\Sigma$  matrix in vector form sorted column-wise. Let  $\hat{\theta}$  be a vector that combines vec ( $\hat{\Phi}$ ) and vech ( $\hat{\Sigma}$ ). Based on the asymptotic properties of maximum likelihood estimators, we can assume that estimators are normally distributed around the population parameters.

$$\hat{\theta} \sim \mathcal{N}(\theta, \mathbb{V}(\hat{\theta}))$$

Using this distributional assumption, a sampling distribution of  $\hat{\theta}$  which we refer to as  $\hat{\theta}^*$  can be generated by replacing the population parameters with sample estimates, that is,

$$\hat{\theta}^* \sim \mathcal{N}(\hat{\theta}, \hat{\mathbb{V}}(\hat{\theta})).$$

Let  $g(\hat{\theta})$  be a parameter that is a function of the estimated parameters. A sampling distribution of  $g(\hat{\theta})$ , which we refer to as  $g(\hat{\theta}^*)$ , can be generated by using the simulated estimates to

calculate **g**. The standard deviations of the simulated estimates are the standard errors. Percentiles corresponding to  $100(1 - \alpha)\%$  are the confidence intervals.

### Value

Returns an object of class `ctmedmc` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("MCTotalCentralStd").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**est** A vector of standardized total effect centrality.

**thetahatstar** A matrix of Monte Carlo standardized total effect centrality.

### Author(s)

Ivan Jacob Agaloos Pesigan

### References

Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. doi:10.2307/271028

Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. doi:10.1080/10705511.2014.973960

Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. doi:10.1037/met0000779

Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. doi:10.1007/s11336021097670

### See Also

Other Continuous-Time Mediation Functions: `BootBeta()`, `BootBetaStd()`, `BootDirectCentral()`, `BootDirectCentralStd()`, `BootIndirectCentral()`, `BootIndirectCentralStd()`, `BootMed()`, `BootMedStd()`, `BootTotalCentral()`, `BootTotalCentralStd()`, `DeltaBeta()`, `DeltaBetaStd()`, `DeltaDirectCentral()`, `DeltaDirectCentralStd()`, `DeltaIndirectCentral()`, `DeltaMed()`, `DeltaMedStd()`, `DeltaTotalCentral()`, `DeltaTotalCentralStd()`, `Direct()`, `DirectCentral()`, `DirectCentralStd()`, `DirectStd()`, `Indirect()`, `IndirectCentral()`, `IndirectCentralStd()`, `IndirectStd()`, `MCBeta()`, `MCBetaStd()`, `MCDirectCentral()`, `MCDirectCentralStd()`, `MCIndirectCentral()`, `MCIndirectCentralStd()`, `MCMed()`, `MCMedStd()`, `MCPHi()`, `MCPHiSigma()`, `MCTotalCentral()`, `Med()`, `MedStd()`, `PosteriorBeta()`, `PosteriorBetaStd()`, `PosteriorDirectCentral()`, `PosteriorDirectCentralStd()`, `PosteriorIndirectCentral()`, `PosteriorIndirectCentralStd()`, `PosteriorMed()`, `PosteriorMedStd()`, `PosteriorTotalCentral()`, `PosteriorTotalCentralStd()`, `Total()`, `TotalCentral()`, `TotalCentralStd()`, `TotalStd()`, `Trajectory()`

**Examples**

```

phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
sigma <- matrix(
  data = c(
    0.24455556, 0.02201587, -0.05004762,
    0.02201587, 0.07067800, 0.01539456,
    -0.05004762, 0.01539456, 0.07553061
  ),
  nrow = 3
)
vcov_theta <- matrix(
  data = c(
    0.00843, 0.00040, -0.00151, -0.00600, -0.00033,
    0.00110, 0.00324, 0.00020, -0.00061, -0.00115,
    0.00011, 0.00015, 0.00001, -0.00002, -0.00001,
    0.00040, 0.00374, 0.00016, -0.00022, -0.00273,
    -0.00016, 0.00009, 0.00150, 0.00012, -0.00010,
    -0.00026, 0.00002, 0.00012, 0.00004, -0.00001,
    -0.00151, 0.00016, 0.00389, 0.00103, -0.00007,
    -0.00283, -0.00050, 0.00000, 0.00156, 0.00021,
    -0.00005, -0.00031, 0.00001, 0.00007, 0.00006,
    -0.00600, -0.00022, 0.00103, 0.00644, 0.00031,
    -0.00119, -0.00374, -0.00021, 0.00070, 0.00064,
    -0.00015, -0.00005, 0.00000, 0.00003, -0.00001,
    -0.00033, -0.00273, -0.00007, 0.00031, 0.00287,
    0.00013, -0.00014, -0.00170, -0.00012, 0.00006,
    0.00014, -0.00001, -0.00015, 0.00000, 0.00001,
    0.00110, -0.00016, -0.00283, -0.00119, 0.00013,
    0.00297, 0.00063, -0.00004, -0.00177, -0.00013,
    0.00005, 0.00017, -0.00002, -0.00008, 0.00001,
    0.00324, 0.00009, -0.00050, -0.00374, -0.00014,
    0.00063, 0.00495, 0.00024, -0.00093, -0.00020,
    0.00006, -0.00010, 0.00000, -0.00001, 0.00004,
    0.00020, 0.00150, 0.00000, -0.00021, -0.00170,
    -0.00004, 0.00024, 0.00214, 0.00012, -0.00002,
    -0.00004, 0.00000, 0.00006, -0.00005, -0.00001,
    -0.00061, 0.00012, 0.00156, 0.00070, -0.00012,
    -0.00177, -0.00093, 0.00012, 0.00223, 0.00004,
    -0.00002, -0.00003, 0.00001, 0.00003, -0.00013,
    -0.00115, -0.00010, 0.00021, 0.00064, 0.00006,
    -0.00013, -0.00020, -0.00002, 0.00004, 0.00057,
    0.00001, -0.00009, 0.00000, 0.00000, 0.00001,
    0.00011, -0.00026, -0.00005, -0.00015, 0.00014,
    0.00005, 0.00006, -0.00004, -0.00002, 0.00001,

```

```

0.00012, 0.00001, 0.00000, -0.00002, 0.00000,
0.00015, 0.00002, -0.00031, -0.00005, -0.00001,
0.00017, -0.00010, 0.00000, -0.00003, -0.00009,
0.00001, 0.00014, 0.00000, 0.00000, -0.00005,
0.00001, 0.00012, 0.00001, 0.00000, -0.00015,
-0.00002, 0.00000, 0.00006, 0.00001, 0.00000,
0.00000, 0.00000, 0.00010, 0.00001, 0.00000,
-0.00002, 0.00004, 0.00007, 0.00003, 0.00000,
-0.00008, -0.00001, -0.00005, 0.00003, 0.00000,
-0.00002, 0.00000, 0.00001, 0.00005, 0.00001,
-0.00001, -0.00001, 0.00006, -0.00001, 0.00001,
0.00001, 0.00004, -0.00001, -0.00013, 0.00001,
0.00000, -0.00005, 0.00000, 0.00001, 0.00012
),
nrow = 15
)

# Specific time interval -----
MCTotalCentralStd(
  phi = phi,
  sigma = sigma,
  vcov_theta = vcov_theta,
  delta_t = 1,
  R = 100L # use a large value for R in actual research
)

# Range of time intervals -----
mc <- MCTotalCentralStd(
  phi = phi,
  sigma = sigma,
  vcov_theta = vcov_theta,
  delta_t = 1:5,
  R = 100L # use a large value for R in actual research
)
plot(mc)

# Methods -----
# MCTotalCentralStd has a number of methods including
# print, summary, confint, and plot
print(mc)
summary(mc)
confint(mc, level = 0.95)
plot(mc)

```

### Description

This function computes the total, direct, and indirect effects of the independent variable  $X$  on the dependent variable  $Y$  through mediator variables  $\mathbf{m}$  over a specific time interval  $\Delta t$  or a range of time intervals using the first-order stochastic differential equation model's drift matrix  $\Phi$ .

### Usage

```
Med(phi, delta_t, from, to, med, tol = 0.001)
```

### Arguments

|         |                                                                                                                              |
|---------|------------------------------------------------------------------------------------------------------------------------------|
| phi     | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system. |
| delta_t | Vector of positive numbers. Time interval ( $\Delta t$ ).                                                                    |
| from    | Character string. Name of the independent variable $X$ in phi.                                                               |
| to      | Character string. Name of the dependent variable $Y$ in phi.                                                                 |
| med     | Character vector. Name/s of the mediator variable/s in phi.                                                                  |
| tol     | Numeric. Smallest possible time interval to allow.                                                                           |

### Details

See [Total\(\)](#), [Direct\(\)](#), and [Indirect\(\)](#) for more details.

#### Linear Stochastic Differential Equation Model:

The measurement model is given by

$$\mathbf{y}_{i,t} = \boldsymbol{\nu} + \mathbf{\Lambda} \boldsymbol{\eta}_{i,t} + \boldsymbol{\varepsilon}_{i,t}, \quad \text{with} \quad \boldsymbol{\varepsilon}_{i,t} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Theta})$$

where  $\mathbf{y}_{i,t}$ ,  $\boldsymbol{\eta}_{i,t}$ , and  $\boldsymbol{\varepsilon}_{i,t}$  are random variables and  $\boldsymbol{\nu}$ ,  $\mathbf{\Lambda}$ , and  $\boldsymbol{\Theta}$  are model parameters.  $\mathbf{y}_{i,t}$  represents a vector of observed random variables,  $\boldsymbol{\eta}_{i,t}$  a vector of latent random variables, and  $\boldsymbol{\varepsilon}_{i,t}$  a vector of random measurement errors, at time  $t$  and individual  $i$ .  $\boldsymbol{\nu}$  denotes a vector of intercepts,  $\mathbf{\Lambda}$  a matrix of factor loadings, and  $\boldsymbol{\Theta}$  the covariance matrix of  $\boldsymbol{\varepsilon}$ .

An alternative representation of the measurement error is given by

$$\boldsymbol{\varepsilon}_{i,t} = \boldsymbol{\Theta}^{\frac{1}{2}} \mathbf{z}_{i,t}, \quad \text{with} \quad \mathbf{z}_{i,t} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

where  $\mathbf{z}_{i,t}$  is a vector of independent standard normal random variables and  $\left(\boldsymbol{\Theta}^{\frac{1}{2}}\right) \left(\boldsymbol{\Theta}^{\frac{1}{2}}\right)' = \boldsymbol{\Theta}$ .

The dynamic structure is given by

$$d\boldsymbol{\eta}_{i,t} = (\boldsymbol{\iota} + \boldsymbol{\Phi} \boldsymbol{\eta}_{i,t}) dt + \boldsymbol{\Sigma}^{\frac{1}{2}} d\mathbf{W}_{i,t}$$

where  $\boldsymbol{\iota}$  is a term which is unobserved and constant over time,  $\boldsymbol{\Phi}$  is the drift matrix which represents the rate of change of the solution in the absence of any random fluctuations,  $\boldsymbol{\Sigma}$  is the matrix of volatility or randomness in the process, and  $d\mathbf{W}$  is a Wiener process or Brownian motion, which represents random fluctuations.

**Value**

Returns an object of class `ctmedmed` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("Med").

**output** A matrix of total, direct, and indirect effects.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**References**

Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. doi:[10.2307/271028](https://doi.org/10.2307/271028)

Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. doi:[10.1080/10705511.2014.973960](https://doi.org/10.1080/10705511.2014.973960)

Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. doi:[10.1037/met0000779](https://doi.org/10.1037/met0000779)

Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. doi:[10.1007/s11336021097670](https://doi.org/10.1007/s11336021097670)

**See Also**

Other Continuous-Time Mediation Functions: `BootBeta()`, `BootBetaStd()`, `BootDirectCentral()`, `BootDirectCentralStd()`, `BootIndirectCentral()`, `BootIndirectCentralStd()`, `BootMed()`, `BootMedStd()`, `BootTotalCentral()`, `BootTotalCentralStd()`, `DeltaBeta()`, `DeltaBetaStd()`, `DeltaDirectCentral()`, `DeltaDirectCentralStd()`, `DeltaIndirectCentral()`, `DeltaMed()`, `DeltaMedStd()`, `DeltaTotalCentral()`, `DeltaTotalCentralStd()`, `Direct()`, `DirectCentral()`, `DirectCentralStd()`, `DirectStd()`, `Indirect()`, `IndirectCentral()`, `IndirectCentralStd()`, `IndirectStd()`, `MCBeta()`, `MCBetaStd()`, `MCDirectCentral()`, `MCDirectCentralStd()`, `MCIndirectCentral()`, `MCIndirectCentralStd()`, `MCMed()`, `MCMedStd()`, `MCPhi()`, `MCPhiSigma()`, `MCTotalCentral()`, `MCTotalCentralStd()`, `MedStd()`, `PosteriorBeta()`, `PosteriorBetaStd()`, `PosteriorDirectCentral()`, `PosteriorDirectCentralStd()`, `PosteriorIndirectCentral()`, `PosteriorIndirectCentralStd()`, `PosteriorMed()`, `PosteriorMedStd()`, `PosteriorTotalCentral()`, `PosteriorTotalCentralStd()`, `Total()`, `TotalCentral()`, `TotalCentralStd()`, `TotalStd()`, `Trajectory()`

**Examples**

```
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
```



```

    nrow = 3
  )
  colnames(phi) <- rownames(phi) <- c("x", "m", "y")

  # Specific time interval -----
  Med(
    phi = phi,
    delta_t = 1,
    from = "x",
    to = "y",
    med = "m"
  )

  # Range of time intervals -----
  med <- Med(
    phi = phi,
    delta_t = 1:30,
    from = "x",
    to = "y",
    med = "m"
  )
  plot(med)

  # Methods -----
  # Med has a number of methods including
  # print, summary, and plot
  med <- Med(
    phi = phi,
    delta_t = 1:5,
    from = "x",
    to = "y",
    med = "m"
  )
  print(med)
  summary(med)
  plot(med)

```

---

MedStd

*Standardized Total, Direct, and Indirect Effects of X on Y Through M  
Over a Specific Time Interval or a Range of Time Intervals*

---

## Description

This function computes the standardized total, direct, and indirect effects of the independent variable  $X$  on the dependent variable  $Y$  through mediator variables  $\mathbf{m}$  over a specific time interval  $\Delta t$  or a range of time intervals using the first-order stochastic differential equation model's drift matrix  $\Phi$  and process noise covariance matrix  $\Sigma$ .

**Usage**

```
MedStd(phi, sigma, delta_t, from, to, med, tol = 0.001)
```

**Arguments**

|         |                                                                                                                              |
|---------|------------------------------------------------------------------------------------------------------------------------------|
| phi     | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system. |
| sigma   | Numeric matrix. The process noise covariance matrix ( $\Sigma$ ).                                                            |
| delta_t | Numeric. Time interval ( $\Delta t$ ).                                                                                       |
| from    | Character string. Name of the independent variable $X$ in phi.                                                               |
| to      | Character string. Name of the dependent variable $Y$ in phi.                                                                 |
| med     | Character vector. Name/s of the mediator variable/s in phi.                                                                  |
| tol     | Numeric. Smallest possible time interval to allow.                                                                           |

**Details**

See [TotalStd\(\)](#), [DirectStd\(\)](#), and [IndirectStd\(\)](#) for more details.

**Value**

Returns an object of class `ctmedmed` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("MedStd").

**output** A standardized matrix of total, direct, and indirect effects.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**References**

- Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. [doi:10.2307/271028](#)
- Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. [doi:10.1080/10705511.2014.973960](#)
- Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. [doi:10.1037/met0000779](#)
- Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. [doi:10.1007/s11336021097670](#)

**See Also**

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

**Examples**

```
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
sigma <- matrix(
  data = c(
    0.24455556, 0.02201587, -0.05004762,
    0.02201587, 0.07067800, 0.01539456,
    -0.05004762, 0.01539456, 0.07553061
  ),
  nrow = 3
)

# Specific time interval -----
MedStd(
  phi = phi,
  sigma = sigma,
  delta_t = 1,
  from = "x",
  to = "y",
  med = "m"
)

# Range of time intervals -----
med <- MedStd(
  phi = phi,
  sigma = sigma,
  delta_t = 1:30,
  from = "x",
  to = "y",
```

```

    med = "m"
  )
  plot(med)

# Methods -----
# MedStd has a number of methods including
# print, summary, and plot
med <- MedStd(
  phi = phi,
  sigma = sigma,
  delta_t = 1:5,
  from = "x",
  to = "y",
  med = "m"
)
print(med)
summary(med)
plot(med)

```

---

plot.ctmedboot

---

*Plot Method for an Object of Class ctmedboot*


---

## Description

Plot Method for an Object of Class ctmedboot

## Usage

```

## S3 method for class 'ctmedboot'
plot(x, alpha = 0.05, col = NULL, type = "pc", ...)

```

## Arguments

|       |                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------|
| x     | Object of class ctmedboot.                                                                                       |
| alpha | Numeric. Significance level                                                                                      |
| col   | Character vector. Optional argument. Character vector of colors.                                                 |
| type  | Character string. Confidence interval type, that is, type = "pc" for percentile; type = "bc" for bias corrected. |
| ...   | Additional arguments.                                                                                            |

## Value

Displays plots of point estimates and confidence intervals.

## Author(s)

Ivan Jacob Agaloos Pesigan

**Examples**

```
## Not run:
library(bootStateSpace)
# prepare parameters
## number of individuals
n <- 50
## time points
time <- 100
delta_t <- 0.10
## dynamic structure
p <- 3
mu0 <- rep(x = 0, times = p)
sigma0 <- matrix(
  data = c(
    1.0,
    0.2,
    0.2,
    0.2,
    1.0,
    0.2,
    0.2,
    0.2,
    1.0
  ),
  nrow = p
)
sigma0_l <- t(chol(sigma0))
mu <- rep(x = 0, times = p)
phi <- matrix(
  data = c(
    -0.357,
    0.771,
    -0.450,
    0.0,
    -0.511,
    0.729,
    0,
    0,
    -0.693
  ),
  nrow = p
)
sigma <- matrix(
  data = c(
    0.24455556,
    0.02201587,
    -0.05004762,
    0.02201587,
    0.07067800,
    0.01539456,
    -0.05004762,
    0.01539456,

```

```

      0.07553061
    ),
    nrow = p
  )
  sigma_l <- t(chol(sigma))
  ## measurement model
  k <- 3
  nu <- rep(x = 0, times = k)
  lambda <- diag(k)
  theta <- 0.2 * diag(k)
  theta_l <- t(chol(theta))

  boot <- PBSSMOUFixed(
    R = 1000L,
    path = getwd(),
    prefix = "ou",
    n = n,
    time = time,
    delta_t = delta_t,
    mu0 = mu0,
    sigma0_l = sigma0_l,
    mu = mu,
    phi = phi,
    sigma_l = sigma_l,
    nu = nu,
    lambda = lambda,
    theta_l = theta_l,
    ncores = parallel::detectCores() - 1,
    seed = 42
  )
  phi_hat <- phi
  colnames(phi_hat) <- rownames(phi_hat) <- c("x", "m", "y")
  phi <- extract(object = boot, what = "phi")

  # Range of time intervals -----
  boot <- BootMed(
    phi = phi,
    phi_hat = phi_hat,
    delta_t = 1:5,
    from = "x",
    to = "y",
    med = "m"
  )
  confint(boot)
  confint(boot, type = "bc") # bias-corrected

  ## End(Not run)

```

**Description**

Plot Method for an Object of Class ctmeddelta

**Usage**

```
## S3 method for class 'ctmeddelta'
plot(x, alpha = 0.05, col = NULL, ...)
```

**Arguments**

|       |                                                                  |
|-------|------------------------------------------------------------------|
| x     | Object of class ctmeddelta.                                      |
| alpha | Numeric. Significance level                                      |
| col   | Character vector. Optional argument. Character vector of colors. |
| ...   | Additional arguments.                                            |

**Value**

Displays plots of point estimates and confidence intervals.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**Examples**

```
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
vcov_phi_vec <- matrix(
  data = c(
    0.00843, 0.00040, -0.00151,
    -0.00600, -0.00033, 0.00110,
    0.00324, 0.00020, -0.00061,
    0.00040, 0.00374, 0.00016,
    -0.00022, -0.00273, -0.00016,
    0.00009, 0.00150, 0.00012,
    -0.00151, 0.00016, 0.00389,
    0.00103, -0.00007, -0.00283,
    -0.00050, 0.00000, 0.00156,
    -0.00600, -0.00022, 0.00103,
    0.00644, 0.00031, -0.00119,
    -0.00374, -0.00021, 0.00070,
    -0.00033, -0.00273, -0.00007,
    0.00031, 0.00287, 0.00013,
```

```

-0.00014, -0.00170, -0.00012,
0.00110, -0.00016, -0.00283,
-0.00119, 0.00013, 0.00297,
0.00063, -0.00004, -0.00177,
0.00324, 0.00009, -0.00050,
-0.00374, -0.00014, 0.00063,
0.00495, 0.00024, -0.00093,
0.00020, 0.00150, 0.00000,
-0.00021, -0.00170, -0.00004,
0.00024, 0.00214, 0.00012,
-0.00061, 0.00012, 0.00156,
0.00070, -0.00012, -0.00177,
-0.00093, 0.00012, 0.00223
),
nrow = 9
)

# Range of time intervals -----
delta <- DeltaMed(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  delta_t = 1:5,
  from = "x",
  to = "y",
  med = "m"
)
plot(delta)

```

---

plot.ctmedmc

---

*Plot Method for an Object of Class ctmedmc*


---

## Description

Plot Method for an Object of Class ctmedmc

## Usage

```
## S3 method for class 'ctmedmc'
plot(x, alpha = 0.05, col = NULL, ...)
```

## Arguments

|       |                                                                  |
|-------|------------------------------------------------------------------|
| x     | Object of class ctmedmc.                                         |
| alpha | Numeric. Significance level                                      |
| col   | Character vector. Optional argument. Character vector of colors. |
| ...   | Additional arguments.                                            |



**Value**

Displays plots of point estimates and confidence intervals.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**Examples**

```
set.seed(42)
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
vcov_phi_vec <- matrix(
  data = c(
    0.00843, 0.00040, -0.00151,
    -0.00600, -0.00033, 0.00110,
    0.00324, 0.00020, -0.00061,
    0.00040, 0.00374, 0.00016,
    -0.00022, -0.00273, -0.00016,
    0.00009, 0.00150, 0.00012,
    -0.00151, 0.00016, 0.00389,
    0.00103, -0.00007, -0.00283,
    -0.00050, 0.00000, 0.00156,
    -0.00600, -0.00022, 0.00103,
    0.00644, 0.00031, -0.00119,
    -0.00374, -0.00021, 0.00070,
    -0.00033, -0.00273, -0.00007,
    0.00031, 0.00287, 0.00013,
    -0.00014, -0.00170, -0.00012,
    0.00110, -0.00016, -0.00283,
    -0.00119, 0.00013, 0.00297,
    0.00063, -0.00004, -0.00177,
    0.00324, 0.00009, -0.00050,
    -0.00374, -0.00014, 0.00063,
    0.00495, 0.00024, -0.00093,
    0.00020, 0.00150, 0.00000,
    -0.00021, -0.00170, -0.00004,
    0.00024, 0.00214, 0.00012,
    -0.00061, 0.00012, 0.00156,
    0.00070, -0.00012, -0.00177,
    -0.00093, 0.00012, 0.00223
  ),
  nrow = 9
)
```

```
# Range of time intervals -----
mc <- MCMed(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  delta_t = 1:5,
  from = "x",
  to = "y",
  med = "m",
  R = 100L # use a large value for R in actual research
)
plot(mc)
```

---

|               |                                                    |
|---------------|----------------------------------------------------|
| plot.ctmedmed | <i>Plot Method for an Object of Class ctmedmed</i> |
|---------------|----------------------------------------------------|

---

### Description

Plot Method for an Object of Class ctmedmed

### Usage

```
## S3 method for class 'ctmedmed'
plot(x, col = NULL, legend_pos = "topright", ...)
```

### Arguments

|            |                                                                  |
|------------|------------------------------------------------------------------|
| x          | Object of class ctmedmed.                                        |
| col        | Character vector. Optional argument. Character vector of colors. |
| legend_pos | Character vector. Optional argument. Legend position.            |
| ...        | Additional arguments.                                            |

### Value

Displays plots of point estimates and confidence intervals.

### Author(s)

Ivan Jacob Agaloos Pesigan

### Examples

```
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
```

```

      nrow = 3
    )
    colnames(phi) <- rownames(phi) <- c("x", "m", "y")

    # Range of time intervals -----
    med <- Med(
      phi = phi,
      delta_t = 1:5,
      from = "x",
      to = "y",
      med = "m"
    )
    plot(med)

```

---

|                |                                                     |
|----------------|-----------------------------------------------------|
| plot.ctmedtraj | <i>Plot Method for an Object of Class ctmedtraj</i> |
|----------------|-----------------------------------------------------|

---

## Description

Plot Method for an Object of Class ctmedtraj

## Usage

```

## S3 method for class 'ctmedtraj'
plot(x, legend_pos = "topright", total = TRUE, ...)

```

## Arguments

|            |                                                                                                                       |
|------------|-----------------------------------------------------------------------------------------------------------------------|
| x          | Object of class ctmedtraj.                                                                                            |
| legend_pos | Character vector. Optional argument. Legend position.                                                                 |
| total      | Logical. If total = TRUE, include the total effect trajectory. If total = FALSE, exclude the total effect trajectory. |
| ...        | Additional arguments.                                                                                                 |

## Value

Displays trajectory plots of the effects.

## Author(s)

Ivan Jacob Agaloos Pesigan

Examples

```
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")

traj <- Trajectory(
  mu0 = c(3, 3, -3),
  time = 150,
  phi = phi,
  med = "m"
)

plot(traj)
```

---

|               |                                                                                                                                                         |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| PosteriorBeta | <i>Posterior Sampling Distribution for the Elements of the Matrix of Lagged Coefficients Over a Specific Time Interval or a Range of Time Intervals</i> |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|

---

Description

This function generates a posterior sampling distribution for the elements of the matrix of lagged coefficients  $\beta$  over a specific time interval  $\Delta t$  or a range of time intervals using the first-order stochastic differential equation model drift matrix  $\Phi$ .

Usage

```
PosteriorBeta(phi, delta_t, ncores = NULL, tol = 0.001)
```

Arguments

|         |                                                                                                                                                                                                                        |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phi     | List of numeric matrices. Each element of the list is a sample from the posterior distribution of the drift matrix ( $\Phi$ ). Each matrix should have row and column names pertaining to the variables in the system. |
| delta_t | Numeric. Time interval ( $\Delta t$ ).                                                                                                                                                                                 |
| ncores  | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of replications R is a large value.                                                           |
| tol     | Numeric. Smallest possible time interval to allow.                                                                                                                                                                     |

Details

See [Total\(\)](#).

**Value**

Returns an object of class `ctmedmc` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("PosteriorBeta").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**est** A vector of total, direct, and indirect effects.

**thetahatstar** A matrix of Monte Carlo total, direct, and indirect effects.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**References**

Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. [doi:10.2307/271028](https://doi.org/10.2307/271028)

Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. [doi:10.1080/10705511.2014.973960](https://doi.org/10.1080/10705511.2014.973960)

Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. [doi:10.1037/met0000779](https://doi.org/10.1037/met0000779)

Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. [doi:10.1007/s11336021097670](https://doi.org/10.1007/s11336021097670)

**See Also**

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

**Examples**

```

phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
vcov_phi_vec <- matrix(
  data = c(
    0.00843, 0.00040, -0.00151,
    -0.00600, -0.00033, 0.00110,
    0.00324, 0.00020, -0.00061,
    0.00040, 0.00374, 0.00016,
    -0.00022, -0.00273, -0.00016,
    0.00009, 0.00150, 0.00012,
    -0.00151, 0.00016, 0.00389,
    0.00103, -0.00007, -0.00283,
    -0.00050, 0.00000, 0.00156,
    -0.00600, -0.00022, 0.00103,
    0.00644, 0.00031, -0.00119,
    -0.00374, -0.00021, 0.00070,
    -0.00033, -0.00273, -0.00007,
    0.00031, 0.00287, 0.00013,
    -0.00014, -0.00170, -0.00012,
    0.00110, -0.00016, -0.00283,
    -0.00119, 0.00013, 0.00297,
    0.00063, -0.00004, -0.00177,
    0.00324, 0.00009, -0.00050,
    -0.00374, -0.00014, 0.00063,
    0.00495, 0.00024, -0.00093,
    0.00020, 0.00150, 0.00000,
    -0.00021, -0.00170, -0.00004,
    0.00024, 0.00214, 0.00012,
    -0.00061, 0.00012, 0.00156,
    0.00070, -0.00012, -0.00177,
    -0.00093, 0.00012, 0.00223
  ),
  nrow = 9
)

phi <- MCPHi(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  R = 1000L
)$output

# Specific time interval -----
PosteriorBeta(
  phi = phi,

```

```

    delta_t = 1
  )

  # Range of time intervals -----
  posterior <- PosteriorBeta(
    phi = phi,
    delta_t = 1:5
  )
  plot(posterior)

  # Methods -----
  # PosteriorBeta has a number of methods including
  # print, summary, confint, and plot
  print(posterior)
  summary(posterior)
  confint(posterior, level = 0.95)
  plot(posterior)

```

---

|                  |                                                                                                                                                             |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PosteriorBetaStd | <i>Posterior Distribution for the Elements of the Standardized Matrix of Lagged Coefficients Over a Specific Time Interval or a Range of Time Intervals</i> |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|

---

## Description

This function generates a posterior distribution for the elements of the standardized matrix of lagged coefficients  $\beta$  over a specific time interval  $\Delta t$  or a range of time intervals using the posterior distribution of the first-order stochastic differential equation model drift matrix  $\Phi$  and process noise covariance matrix  $\Sigma$ .

## Usage

```
PosteriorBetaStd(phi, sigma, delta_t, ncores = NULL, tol = 0.001)
```

## Arguments

|         |                                                                                                                                                                                                                                             |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phi     | List of numeric matrices. Each element of the list is a sample from the posterior distribution of the drift matrix ( $\Phi$ ). Each matrix should have row and column names pertaining to the variables in the system.                      |
| sigma   | List of numeric matrices. Each element of the list is a sample from the posterior distribution of the process noise covariance matrix ( $\Sigma$ ). Each matrix should have row and column names pertaining to the variables in the system. |
| delta_t | Numeric. Time interval ( $\Delta t$ ).                                                                                                                                                                                                      |
| ncores  | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of replications R is a large value.                                                                                |
| tol     | Numeric. Smallest possible time interval to allow.                                                                                                                                                                                          |

## Details

See [TotalStd\(\)](#) for more details.

## Value

Returns an object of class `ctmedmc` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("PosteriorBetaStd").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**est** Mean of the posterior distribution of the elements of the standardized matrix of lagged coefficients.

**thetahatstar** Posterior distribution of the elements of the standardized matrix of lagged coefficients.

## Author(s)

Ivan Jacob Agaloos Pesigan

## References

Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. [doi:10.2307/271028](#)

Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. [doi:10.1080/10705511.2014.973960](#)

Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. [doi:10.1037/met0000779](#)

Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. [doi:10.1007/s11336021097670](#)

## See Also

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#),



PosteriorTotalCentral(),PosteriorTotalCentralStd(),Total(),TotalCentral(),TotalCentralStd(),  
TotalStd(),Trajectory()

## Examples

```
set.seed(42)
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.000, -0.511, 0.729,
    0.000, 0.000, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
sigma <- matrix(
  data = c(
    0.24455556, 0.02201587, -0.05004762,
    0.02201587, 0.07067800, 0.01539456,
    -0.05004762, 0.01539456, 0.07553061
  ),
  nrow = 3
)
colnames(sigma) <- rownames(sigma) <- c("x", "m", "y")
input <- MCPHiSigma(
  phi = phi,
  sigma = sigma,
  vcov_theta = 0.001 * diag(15),
  R = 100L,
  seed = 42
)$output
phi <- lapply(
  X = input,
  FUN = function(x) {
    x[[1]]
  }
)
sigma <- lapply(
  X = input,
  FUN = function(x) {
    x[[2]]
  }
)

# Specific time interval -----
PosteriorBetaStd(
  phi = phi,
  sigma = sigma,
  delta_t = 1
)

# Range of time intervals -----
posterior <- PosteriorBetaStd(
```

```

    phi = phi,
    sigma = sigma,
    delta_t = 1:5
  )
  plot(posterior)

# Methods -----
# PosteriorBetaStd has a number of methods including
# print, summary, confint, and plot
print(posterior)
summary(posterior)
confint(posterior, level = 0.95)
plot(posterior)

```

---

PosteriorDirectCentral

*Posterior Distribution of the Direct Effect Centrality Over a Specific Time Interval or a Range of Time Intervals*

---

## Description

This function generates a posterior distribution of the direct effect centrality over a specific time interval  $\Delta t$  or a range of time intervals using the posterior distribution of the first-order stochastic differential equation model drift matrix  $\Phi$ .

## Usage

```
PosteriorDirectCentral(phi, delta_t, ncores = NULL, tol = 0.001)
```

## Arguments

|         |                                                                                                                                                                                                                        |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phi     | List of numeric matrices. Each element of the list is a sample from the posterior distribution of the drift matrix ( $\Phi$ ). Each matrix should have row and column names pertaining to the variables in the system. |
| delta_t | Numeric. Time interval ( $\Delta t$ ).                                                                                                                                                                                 |
| ncores  | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of replications R is a large value.                                                           |
| tol     | Numeric. Smallest possible time interval to allow.                                                                                                                                                                     |

## Details

See [DirectCentral\(\)](#) for more details.

**Value**

Returns an object of class `ctmedmc` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("PosteriorDirectCentral").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**est** Mean of the posterior distribution of the direct effect centrality.

**thetahatstar** Posterior distribution of the direct effect centrality measure.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**References**

Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. [doi:10.2307/271028](https://doi.org/10.2307/271028)

Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. [doi:10.1080/10705511.2014.973960](https://doi.org/10.1080/10705511.2014.973960)

Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. [doi:10.1037/met0000779](https://doi.org/10.1037/met0000779)

Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. [doi:10.1007/s11336021097670](https://doi.org/10.1007/s11336021097670)

**See Also**

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

**Examples**

```

phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
vcov_phi_vec <- matrix(
  data = c(
    0.00843, 0.00040, -0.00151,
    -0.00600, -0.00033, 0.00110,
    0.00324, 0.00020, -0.00061,
    0.00040, 0.00374, 0.00016,
    -0.00022, -0.00273, -0.00016,
    0.00009, 0.00150, 0.00012,
    -0.00151, 0.00016, 0.00389,
    0.00103, -0.00007, -0.00283,
    -0.00050, 0.00000, 0.00156,
    -0.00600, -0.00022, 0.00103,
    0.00644, 0.00031, -0.00119,
    -0.00374, -0.00021, 0.00070,
    -0.00033, -0.00273, -0.00007,
    0.00031, 0.00287, 0.00013,
    -0.00014, -0.00170, -0.00012,
    0.00110, -0.00016, -0.00283,
    -0.00119, 0.00013, 0.00297,
    0.00063, -0.00004, -0.00177,
    0.00324, 0.00009, -0.00050,
    -0.00374, -0.00014, 0.00063,
    0.00495, 0.00024, -0.00093,
    0.00020, 0.00150, 0.00000,
    -0.00021, -0.00170, -0.00004,
    0.00024, 0.00214, 0.00012,
    -0.00061, 0.00012, 0.00156,
    0.00070, -0.00012, -0.00177,
    -0.00093, 0.00012, 0.00223
  ),
  nrow = 9
)

phi <- MCPHi(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  R = 1000L
)$output

# Specific time interval -----
PosteriorDirectCentral(
  phi = phi,

```

```

    delta_t = 1
  )

  # Range of time intervals -----
  posterior <- PosteriorDirectCentral(
    phi = phi,
    delta_t = 1:5
  )

  # Methods -----
  # PosteriorDirectCentral has a number of methods including
  # print, summary, confint, and plot
  print(posterior)
  summary(posterior)
  confint(posterior, level = 0.95)
  plot(posterior)

```

---

PosteriorDirectCentralStd

*Posterior Distribution of the Standardized Direct Effect Centrality  
Over a Specific Time Interval or a Range of Time Intervals*

---

## Description

This function generates a posterior distribution of the standardized direct effect centrality over a specific time interval  $\Delta t$  or a range of time intervals using the posterior distribution of the first-order stochastic differential equation model drift matrix  $\Phi$  and process noise covariance matrix  $\Sigma$ .

## Usage

```
PosteriorDirectCentralStd(phi, sigma, delta_t, ncores = NULL, tol = 0.001)
```

## Arguments

|         |                                                                                                                                                                                                                        |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phi     | List of numeric matrices. Each element of the list is a sample from the posterior distribution of the drift matrix ( $\Phi$ ). Each matrix should have row and column names pertaining to the variables in the system. |
| sigma   | List of numeric matrices. Each element is a posterior draw of the diffusion covariance matrix.                                                                                                                         |
| delta_t | Numeric. Time interval ( $\Delta t$ ).                                                                                                                                                                                 |
| ncores  | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of replications R is a large value.                                                           |
| tol     | Numeric. Smallest possible time interval to allow.                                                                                                                                                                     |

## Details

See [DirectCentralStd\(\)](#) for more details.

## Value

Returns an object of class `ctmedmc` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("PosteriorDirectCentralStd").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**est** Mean of the posterior distribution of the standardized direct effect centrality.

**thetahatstar** Posterior distribution of the standardized direct effect centrality measure.

## Author(s)

Ivan Jacob Agaloos Pesigan

## References

Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. [doi:10.2307/271028](#)

Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. [doi:10.1080/10705511.2014.973960](#)

Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. [doi:10.1037/met0000779](#)

Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. [doi:10.1007/s11336021097670](#)

## See Also

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

**Examples**

```

set.seed(42)
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.000, -0.511, 0.729,
    0.000, 0.000, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")

sigma <- matrix(
  data = c(
    0.24455556, 0.02201587, -0.05004762,
    0.02201587, 0.07067800, 0.01539456,
    -0.05004762, 0.01539456, 0.07553061
  ),
  nrow = 3
)
colnames(sigma) <- rownames(sigma) <- c("x", "m", "y")

input <- MCPhiSigma(
  phi = phi,
  sigma = sigma,
  vcov_theta = 0.001 * diag(15),
  R = 100L,
  seed = 42
)$output

phi <- lapply(
  X = input,
  FUN = function(x) {
    x[[1]]
  }
)

sigma <- lapply(
  X = input,
  FUN = function(x) {
    x[[2]]
  }
)

# Specific time interval -----
PosteriorDirectCentralStd(
  phi = phi,
  sigma = sigma,
  delta_t = 1
)

# Range of time intervals -----

```

```

posterior <- PosteriorDirectCentralStd(
  phi = phi,
  sigma = sigma,
  delta_t = 1:5
)

# Methods -----
# PosteriorDirectCentralStd has a number of methods including
# print, summary, confint, and plot
print(posterior)
summary(posterior)
confint(posterior, level = 0.95)
plot(posterior)

```

---

## PosteriorIndirectCentral

*Posterior Distribution of the Indirect Effect Centrality Over a Specific Time Interval or a Range of Time Intervals*

---

### Description

This function generates a posterior distribution of the indirect effect centrality over a specific time interval  $\Delta t$  or a range of time intervals using the posterior distribution of the first-order stochastic differential equation model drift matrix  $\Phi$ .

### Usage

```
PosteriorIndirectCentral(phi, delta_t, ncores = NULL, tol = 0.001)
```

### Arguments

|         |                                                                                                                                                                                                                        |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phi     | List of numeric matrices. Each element of the list is a sample from the posterior distribution of the drift matrix ( $\Phi$ ). Each matrix should have row and column names pertaining to the variables in the system. |
| delta_t | Numeric. Time interval ( $\Delta t$ ).                                                                                                                                                                                 |
| ncores  | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of replications R is a large value.                                                           |
| tol     | Numeric. Smallest possible time interval to allow.                                                                                                                                                                     |

### Details

See [IndirectCentral\(\)](#) for more details.



**Value**

Returns an object of class `ctmedmc` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("PosteriorIndirectCentral").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**est** Mean of the posterior distribution of the indirect effect centrality.

**thetahatstar** Posterior distribution of the indirect effect centrality measure.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**References**

Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. [doi:10.2307/271028](https://doi.org/10.2307/271028)

Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. [doi:10.1080/10705511.2014.973960](https://doi.org/10.1080/10705511.2014.973960)

Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. [doi:10.1037/met0000779](https://doi.org/10.1037/met0000779)

Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. [doi:10.1007/s11336021097670](https://doi.org/10.1007/s11336021097670)

**See Also**

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

**Examples**

```

phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
vcov_phi_vec <- matrix(
  data = c(
    0.00843, 0.00040, -0.00151,
    -0.00600, -0.00033, 0.00110,
    0.00324, 0.00020, -0.00061,
    0.00040, 0.00374, 0.00016,
    -0.00022, -0.00273, -0.00016,
    0.00009, 0.00150, 0.00012,
    -0.00151, 0.00016, 0.00389,
    0.00103, -0.00007, -0.00283,
    -0.00050, 0.00000, 0.00156,
    -0.00600, -0.00022, 0.00103,
    0.00644, 0.00031, -0.00119,
    -0.00374, -0.00021, 0.00070,
    -0.00033, -0.00273, -0.00007,
    0.00031, 0.00287, 0.00013,
    -0.00014, -0.00170, -0.00012,
    0.00110, -0.00016, -0.00283,
    -0.00119, 0.00013, 0.00297,
    0.00063, -0.00004, -0.00177,
    0.00324, 0.00009, -0.00050,
    -0.00374, -0.00014, 0.00063,
    0.00495, 0.00024, -0.00093,
    0.00020, 0.00150, 0.00000,
    -0.00021, -0.00170, -0.00004,
    0.00024, 0.00214, 0.00012,
    -0.00061, 0.00012, 0.00156,
    0.00070, -0.00012, -0.00177,
    -0.00093, 0.00012, 0.00223
  ),
  nrow = 9
)

phi <- MCPHi(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  R = 1000L
)$output

# Specific time interval -----
PosteriorIndirectCentral(
  phi = phi,

```

```

    delta_t = 1
  )

  # Range of time intervals -----
  posterior <- PosteriorIndirectCentral(
    phi = phi,
    delta_t = 1:5
  )

  # Methods -----
  # PosteriorIndirectCentral has a number of methods including
  # print, summary, confint, and plot
  print(posterior)
  summary(posterior)
  confint(posterior, level = 0.95)
  plot(posterior)

```

---

PosteriorIndirectCentralStd

*Posterior Distribution of the Standardized Indirect Effect Centrality  
Over a Specific Time Interval or a Range of Time Intervals*

---

## Description

This function generates a posterior distribution of the standardized indirect effect centrality over a specific time interval  $\Delta t$  or a range of time intervals using the posterior distribution of the first-order stochastic differential equation model drift matrix  $\Phi$  and process noise covariance matrix  $\Sigma$ .

## Usage

```
PosteriorIndirectCentralStd(phi, sigma, delta_t, ncores = NULL, tol = 0.001)
```

## Arguments

|         |                                                                                                                                                                                                                        |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phi     | List of numeric matrices. Each element of the list is a sample from the posterior distribution of the drift matrix ( $\Phi$ ). Each matrix should have row and column names pertaining to the variables in the system. |
| sigma   | List of numeric matrices. Each element is a posterior draw of the diffusion covariance matrix.                                                                                                                         |
| delta_t | Numeric. Time interval ( $\Delta t$ ).                                                                                                                                                                                 |
| ncores  | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of replications R is a large value.                                                           |
| tol     | Numeric. Smallest possible time interval to allow.                                                                                                                                                                     |

## Details

See [IndirectCentralStd\(\)](#) for more details.

**Value**

Returns an object of class `ctmedmc` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("PosteriorIndirectCentralStd").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**est** Mean of the posterior distribution of the standardized indirect effect centrality.

**thetahatstar** Posterior distribution of the standardized indirect effect centrality measure.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**References**

Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. [doi:10.2307/271028](https://doi.org/10.2307/271028)

Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. [doi:10.1080/10705511.2014.973960](https://doi.org/10.1080/10705511.2014.973960)

Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. [doi:10.1037/met0000779](https://doi.org/10.1037/met0000779)

Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. [doi:10.1007/s11336021097670](https://doi.org/10.1007/s11336021097670)

**See Also**

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

**Examples**

```

set.seed(42)
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.000, -0.511, 0.729,
    0.000, 0.000, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")

sigma <- matrix(
  data = c(
    0.24455556, 0.02201587, -0.05004762,
    0.02201587, 0.07067800, 0.01539456,
    -0.05004762, 0.01539456, 0.07553061
  ),
  nrow = 3
)
colnames(sigma) <- rownames(sigma) <- c("x", "m", "y")

input <- MCPHiSigma(
  phi = phi,
  sigma = sigma,
  vcov_theta = 0.001 * diag(15),
  R = 100L,
  seed = 42
)$output

phi <- lapply(
  X = input,
  FUN = function(x) {
    x[[1]]
  }
)

sigma <- lapply(
  X = input,
  FUN = function(x) {
    x[[2]]
  }
)

# Specific time interval -----
PosteriorIndirectCentralStd(
  phi = phi,
  sigma = sigma,
  delta_t = 1
)

# Range of time intervals -----

```

```

posterior <- PosteriorIndirectCentralStd(
  phi = phi,
  sigma = sigma,
  delta_t = 1:5
)

# Methods -----
# PosteriorIndirectCentralStd has a number of methods including
# print, summary, confint, and plot
print(posterior)
summary(posterior)
confint(posterior, level = 0.95)
plot(posterior)

```

---

PosteriorMed

---

*Posterior Distribution of Total, Direct, and Indirect Effects of  $X$  on  $Y$  Through  $M$  Over a Specific Time Interval or a Range of Time Intervals*


---

## Description

This function generates a posterior distribution of the total, direct and indirect effects of the independent variable  $X$  on the dependent variable  $Y$  through mediator variables  $\mathbf{m}$  over a specific time interval  $\Delta t$  or a range of time intervals using the posterior distribution of the first-order stochastic differential equation model drift matrix  $\Phi$ .

## Usage

```
PosteriorMed(phi, delta_t, from, to, med, ncores = NULL, tol = 0.001)
```

## Arguments

|         |                                                                                                                                                                                                                        |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phi     | List of numeric matrices. Each element of the list is a sample from the posterior distribution of the drift matrix ( $\Phi$ ). Each matrix should have row and column names pertaining to the variables in the system. |
| delta_t | Numeric. Time interval ( $\Delta t$ ).                                                                                                                                                                                 |
| from    | Character string. Name of the independent variable $X$ in phi.                                                                                                                                                         |
| to      | Character string. Name of the dependent variable $Y$ in phi.                                                                                                                                                           |
| med     | Character vector. Name/s of the mediator variable/s in phi.                                                                                                                                                            |
| ncores  | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of replications $R$ is a large value.                                                         |
| tol     | Numeric. Smallest possible time interval to allow.                                                                                                                                                                     |

## Details

See [Total\(\)](#), [Direct\(\)](#), and [Indirect\(\)](#) for more details.

**Value**

Returns an object of class `ctmedmc` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("PosteriorMed").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**est** Mean of the posterior distribution of the total, direct, and indirect effects.

**thetahatstar** Posterior distribution of the total, direct, and indirect effects.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**References**

Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. [doi:10.2307/271028](https://doi.org/10.2307/271028)

Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. [doi:10.1080/10705511.2014.973960](https://doi.org/10.1080/10705511.2014.973960)

Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. [doi:10.1037/met0000779](https://doi.org/10.1037/met0000779)

Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. [doi:10.1007/s11336021097670](https://doi.org/10.1007/s11336021097670)

**See Also**

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

**Examples**

```

phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
vcov_phi_vec <- matrix(
  data = c(
    0.00843, 0.00040, -0.00151,
    -0.00600, -0.00033, 0.00110,
    0.00324, 0.00020, -0.00061,
    0.00040, 0.00374, 0.00016,
    -0.00022, -0.00273, -0.00016,
    0.00009, 0.00150, 0.00012,
    -0.00151, 0.00016, 0.00389,
    0.00103, -0.00007, -0.00283,
    -0.00050, 0.00000, 0.00156,
    -0.00600, -0.00022, 0.00103,
    0.00644, 0.00031, -0.00119,
    -0.00374, -0.00021, 0.00070,
    -0.00033, -0.00273, -0.00007,
    0.00031, 0.00287, 0.00013,
    -0.00014, -0.00170, -0.00012,
    0.00110, -0.00016, -0.00283,
    -0.00119, 0.00013, 0.00297,
    0.00063, -0.00004, -0.00177,
    0.00324, 0.00009, -0.00050,
    -0.00374, -0.00014, 0.00063,
    0.00495, 0.00024, -0.00093,
    0.00020, 0.00150, 0.00000,
    -0.00021, -0.00170, -0.00004,
    0.00024, 0.00214, 0.00012,
    -0.00061, 0.00012, 0.00156,
    0.00070, -0.00012, -0.00177,
    -0.00093, 0.00012, 0.00223
  ),
  nrow = 9
)

phi <- MCPHi(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  R = 1000L
)$output

# Specific time interval -----
PosteriorMed(
  phi = phi,

```



```

    delta_t = 1,
    from = "x",
    to = "y",
    med = "m"
)

# Range of time intervals -----
posterior <- PosteriorMed(
  phi = phi,
  delta_t = 1:5,
  from = "x",
  to = "y",
  med = "m"
)

# Methods -----
# PosteriorMed has a number of methods including
# print, summary, confint, and plot
print(posterior)
summary(posterior)
confint(posterior, level = 0.95)
plot(posterior)
```

---

|                 |                                                                                                                                                                  |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PosteriorMedStd | <i>Posterior Distribution of Standardized Total, Direct, and Indirect Effects of X on Y Through M Over a Specific Time Interval or a Range of Time Intervals</i> |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|

---

**Description**

This function generates a posterior distribution of the standardized total, direct, and indirect effects of the independent variable  $X$  on the dependent variable  $Y$  through mediator variables  $\mathbf{m}$  over a specific time interval  $\Delta t$  or a range of time intervals using the posterior distribution of the first-order stochastic differential equation model drift matrix  $\Phi$  and process noise covariance matrix  $\Sigma$ .

**Usage**

```
PosteriorMedStd(phi, sigma, delta_t, from, to, med, ncores = NULL, tol = 0.001)
```

**Arguments**

|         |                                                                                                                                                                                                                                             |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phi     | List of numeric matrices. Each element of the list is a sample from the posterior distribution of the drift matrix ( $\Phi$ ). Each matrix should have row and column names pertaining to the variables in the system.                      |
| sigma   | List of numeric matrices. Each element of the list is a sample from the posterior distribution of the process noise covariance matrix ( $\Sigma$ ). Each matrix should have row and column names pertaining to the variables in the system. |
| delta_t | Numeric. Time interval ( $\Delta t$ ).                                                                                                                                                                                                      |

|        |                                                                                                                                                                |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| from   | Character string. Name of the independent variable $X$ in $\phi$ .                                                                                             |
| to     | Character string. Name of the dependent variable $Y$ in $\phi$ .                                                                                               |
| med    | Character vector. Name/s of the mediator variable/s in $\phi$ .                                                                                                |
| ncores | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of replications $R$ is a large value. |
| tol    | Numeric. Smallest possible time interval to allow.                                                                                                             |

## Details

See [TotalStd\(\)](#), [DirectStd\(\)](#), and [IndirectStd\(\)](#) for more details.

## Value

Returns an object of class `ctmedmc` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("PosteriorMedStd").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**est** Mean of the posterior distribution of the standardized total, direct, and indirect effects.

**thetahatstar** Posterior distribution of the standardized total, direct, and indirect effects.

## Author(s)

Ivan Jacob Agaloos Pesigan

## References

- Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. [doi:10.2307/271028](#)
- Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. [doi:10.1080/10705511.2014.973960](#)
- Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. [doi:10.1037/met0000779](#)
- Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. [doi:10.1007/s11336021097670](#)

**See Also**

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

**Examples**

```
set.seed(42)
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.000, -0.511, 0.729,
    0.000, 0.000, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
sigma <- matrix(
  data = c(
    0.24455556, 0.02201587, -0.05004762,
    0.02201587, 0.07067800, 0.01539456,
    -0.05004762, 0.01539456, 0.07553061
  ),
  nrow = 3
)
colnames(sigma) <- rownames(sigma) <- c("x", "m", "y")
input <- MCPhiSigma(
  phi = phi,
  sigma = sigma,
  vcov_theta = 0.001 * diag(15),
  R = 100L,
  seed = 42
)$output
phi <- lapply(
  X = input,
  FUN = function(x) {
    x[[1]]
  }
)
sigma <- lapply(
  X = input,
  FUN = function(x) {
```

```

        x[[2]]
    }
)

# Specific time interval -----
PosteriorMedStd(
  phi = phi,
  sigma = sigma,
  delta_t = 1,
  from = "x",
  to = "y",
  med = "m"
)

# Range of time intervals -----
posterior <- PosteriorMedStd(
  phi = phi,
  sigma = sigma,
  delta_t = 1:5,
  from = "x",
  to = "y",
  med = "m"
)
plot(posterior)

# Methods -----
# PosteriorMedStd has a number of methods including
# print, summary, confint, and plot
print(posterior)
summary(posterior)
confint(posterior, level = 0.95)
plot(posterior)

```

---

PosteriorTotalCentral *Posterior Distribution of the Total Effect Centrality Over a Specific Time Interval or a Range of Time Intervals*

---

## Description

This function generates a posterior distribution of the total effect centrality over a specific time interval  $\Delta t$  or a range of time intervals using the posterior distribution of the first-order stochastic differential equation model drift matrix  $\Phi$ .

## Usage

```
PosteriorTotalCentral(phi, delta_t, ncores = NULL, tol = 0.001)
```

**Arguments**

|                      |                                                                                                                                                                                                                        |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>phi</code>     | List of numeric matrices. Each element of the list is a sample from the posterior distribution of the drift matrix ( $\Phi$ ). Each matrix should have row and column names pertaining to the variables in the system. |
| <code>delta_t</code> | Numeric. Time interval ( $\Delta t$ ).                                                                                                                                                                                 |
| <code>ncores</code>  | Positive integer. Number of cores to use. If <code>ncores = NULL</code> , use a single core. Consider using multiple cores when number of replications <code>R</code> is a large value.                                |
| <code>tol</code>     | Numeric. Smallest possible time interval to allow.                                                                                                                                                                     |

**Details**

See [TotalCentral\(\)](#) for more details.

**Value**

Returns an object of class `ctmedmc` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("PosteriorTotalCentral").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**est** Mean of the posterior distribution of the total, direct, and indirect effects.

**thetahatstar** Posterior distribution of the total effect centrality measure.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**References**

- Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. [doi:10.2307/271028](#)
- Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. [doi:10.1080/10705511.2014.973960](#)
- Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. [doi:10.1037/met0000779](#)
- Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. [doi:10.1007/s11336021097670](#)

**See Also**

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

**Examples**

```
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
vcov_phi_vec <- matrix(
  data = c(
    0.00843, 0.00040, -0.00151,
    -0.00600, -0.00033, 0.00110,
    0.00324, 0.00020, -0.00061,
    0.00040, 0.00374, 0.00016,
    -0.00022, -0.00273, -0.00016,
    0.00009, 0.00150, 0.00012,
    -0.00151, 0.00016, 0.00389,
    0.00103, -0.00007, -0.00283,
    -0.00050, 0.00000, 0.00156,
    -0.00600, -0.00022, 0.00103,
    0.00644, 0.00031, -0.00119,
    -0.00374, -0.00021, 0.00070,
    -0.00033, -0.00273, -0.00007,
    0.00031, 0.00287, 0.00013,
    -0.00014, -0.00170, -0.00012,
    0.00110, -0.00016, -0.00283,
    -0.00119, 0.00013, 0.00297,
    0.00063, -0.00004, -0.00177,
    0.00324, 0.00009, -0.00050,
    -0.00374, -0.00014, 0.00063,
    0.00495, 0.00024, -0.00093,
    0.00020, 0.00150, 0.00000,
    -0.00021, -0.00170, -0.00004,
    0.00024, 0.00214, 0.00012,
  )
)
```

```

      -0.00061, 0.00012, 0.00156,
      0.00070, -0.00012, -0.00177,
      -0.00093, 0.00012, 0.00223
    ),
    nrow = 9
  )

  phi <- MCPHi(
    phi = phi,
    vcov_phi_vec = vcov_phi_vec,
    R = 1000L
  )$output

  # Specific time interval -----
  PosteriorTotalCentral(
    phi = phi,
    delta_t = 1
  )

  # Range of time intervals -----
  posterior <- PosteriorTotalCentral(
    phi = phi,
    delta_t = 1:5
  )

  # Methods -----
  # PosteriorTotalCentral has a number of methods including
  # print, summary, confint, and plot
  print(posterior)
  summary(posterior)
  confint(posterior, level = 0.95)
  plot(posterior)

```

---

## PosteriorTotalCentralStd

*Posterior Distribution of the Standardized Total Effect Centrality Over a Specific Time Interval or a Range of Time Intervals*

---

### Description

This function generates a posterior distribution of the standardized total effect centrality over a specific time interval  $\Delta t$  or a range of time intervals using the posterior distribution of the first-order stochastic differential equation model drift matrix  $\Phi$  and process noise covariance matrix  $\Sigma$ .

### Usage

```
PosteriorTotalCentralStd(phi, sigma, delta_t, ncores = NULL, tol = 0.001)
```

**Arguments**

|                      |                                                                                                                                                                                                                        |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>phi</code>     | List of numeric matrices. Each element of the list is a sample from the posterior distribution of the drift matrix ( $\Phi$ ). Each matrix should have row and column names pertaining to the variables in the system. |
| <code>sigma</code>   | List of numeric matrices. Each element is a posterior draw of the diffusion covariance matrix.                                                                                                                         |
| <code>delta_t</code> | Numeric. Time interval ( $\Delta t$ ).                                                                                                                                                                                 |
| <code>ncores</code>  | Positive integer. Number of cores to use. If <code>ncores = NULL</code> , use a single core. Consider using multiple cores when number of replications <code>R</code> is a large value.                                |
| <code>tol</code>     | Numeric. Smallest possible time interval to allow.                                                                                                                                                                     |

**Details**

See [TotalCentralStd\(\)](#) for more details.

**Value**

Returns an object of class `ctmedmc` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("PosteriorTotalCentralStd").

**output** A list of length `length(delta_t)`.

Each element in the output list has the following elements:

**est** Mean of the posterior distribution of the standardized total effect centrality.

**thetahatstar** Posterior distribution of the standardized total effect centrality measure.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**References**

- Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. [doi:10.2307/271028](#)
- Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. [doi:10.1080/10705511.2014.973960](#)
- Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. [doi:10.1037/met0000779](#)
- Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. [doi:10.1007/s11336021097670](#)



**See Also**

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

**Examples**

```
set.seed(42)
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.000, -0.511, 0.729,
    0.000, 0.000, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")

sigma <- matrix(
  data = c(
    0.24455556, 0.02201587, -0.05004762,
    0.02201587, 0.07067800, 0.01539456,
    -0.05004762, 0.01539456, 0.07553061
  ),
  nrow = 3
)
colnames(sigma) <- rownames(sigma) <- c("x", "m", "y")

input <- MCPhiSigma(
  phi = phi,
  sigma = sigma,
  vcov_theta = 0.001 * diag(15),
  R = 100L,
  seed = 42
)$output

phi <- lapply(
  X = input,
  FUN = function(x) {
    x[[1]]
  }
)
```

```

sigma <- lapply(
  X = input,
  FUN = function(x) {
    x[[2]]
  }
)

# Specific time interval -----
PosteriorTotalCentralStd(
  phi = phi,
  sigma = sigma,
  delta_t = 1
)

# Range of time intervals -----
posterior <- PosteriorTotalCentralStd(
  phi = phi,
  sigma = sigma,
  delta_t = 1:5
)

# Methods -----
# PosteriorTotalCentralStd has a number of methods including
# print, summary, confint, and plot
print(posterior)
summary(posterior)
confint(posterior, level = 0.95)
plot(posterior)

```

---

|                 |                                                   |
|-----------------|---------------------------------------------------|
| print.ctmedboot | <i>Print Method for Object of Class ctmedboot</i> |
|-----------------|---------------------------------------------------|

---

## Description

Print Method for Object of Class ctmedboot

## Usage

```

## S3 method for class 'ctmedboot'
print(x, alpha = 0.05, digits = 4, type = "pc", ...)

```

## Arguments

|        |                                                             |
|--------|-------------------------------------------------------------|
| x      | an object of class ctmedboot.                               |
| alpha  | Numeric vector. Significance level $\alpha$ .               |
| digits | Integer indicating the number of decimal places to display. |

|      |                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------|
| type | Character string. Confidence interval type, that is, type = "pc" for percentile; type = "bc" for bias corrected. |
| ...  | further arguments.                                                                                               |

**Value**

Prints a list of matrices of time intervals, estimates, standard errors, number of bootstrap replications, and confidence intervals.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**Examples**

```
## Not run:
library(bootStateSpace)
# prepare parameters
## number of individuals
n <- 50
## time points
time <- 100
delta_t <- 0.10
## dynamic structure
p <- 3
mu0 <- rep(x = 0, times = p)
sigma0 <- matrix(
  data = c(
    1.0,
    0.2,
    0.2,
    0.2,
    1.0,
    0.2,
    0.2,
    0.2,
    1.0
  ),
  nrow = p
)
sigma0_l <- t(chol(sigma0))
mu <- rep(x = 0, times = p)
phi <- matrix(
  data = c(
    -0.357,
    0.771,
    -0.450,
    0.0,
    -0.511,
    0.729,
    0,
    0,
    0
  )
)
```

```

      -0.693
    ),
    nrow = p
  )
  sigma <- matrix(
    data = c(
      0.24455556,
      0.02201587,
      -0.05004762,
      0.02201587,
      0.07067800,
      0.01539456,
      -0.05004762,
      0.01539456,
      0.07553061
    ),
    nrow = p
  )
  sigma_l <- t(chol(sigma))
  ## measurement model
  k <- 3
  nu <- rep(x = 0, times = k)
  lambda <- diag(k)
  theta <- 0.2 * diag(k)
  theta_l <- t(chol(theta))

  boot <- PBSSMOUFixed(
    R = 1000L,
    path = getwd(),
    prefix = "ou",
    n = n,
    time = time,
    delta_t = delta_t,
    mu0 = mu0,
    sigma0_l = sigma0_l,
    mu = mu,
    phi = phi,
    sigma_l = sigma_l,
    nu = nu,
    lambda = lambda,
    theta_l = theta_l,
    ncores = parallel::detectCores() - 1,
    seed = 42
  )
  phi_hat <- phi
  colnames(phi_hat) <- rownames(phi_hat) <- c("x", "m", "y")
  phi <- extract(object = boot, what = "phi")

  # Specific time interval -----
  boot <- BootMed(
    phi = phi,
    phi_hat = phi_hat,
    delta_t = 1,

```

```

    from = "x",
    to = "y",
    med = "m"
  )
  print(boot)
  print(boot, type = "bc") # bias-corrected

# Range of time intervals -----
boot <- BootMed(
  phi = phi,
  phi_hat = phi_hat,
  delta_t = 1:5,
  from = "x",
  to = "y",
  med = "m"
)
print(boot)
print(boot, type = "bc") # bias-corrected

## End(Not run)

```

---

|                  |                                                    |
|------------------|----------------------------------------------------|
| print.ctmeddelta | <i>Print Method for Object of Class ctmeddelta</i> |
|------------------|----------------------------------------------------|

---

## Description

Print Method for Object of Class ctmeddelta

## Usage

```

## S3 method for class 'ctmeddelta'
print(x, alpha = 0.05, digits = 4, ...)

```

## Arguments

|        |                                                             |
|--------|-------------------------------------------------------------|
| x      | an object of class ctmeddelta.                              |
| alpha  | Numeric vector. Significance level $\alpha$ .               |
| digits | Integer indicating the number of decimal places to display. |
| ...    | further arguments.                                          |

## Value

Prints a list of matrices of time intervals, estimates, standard errors, test statistics, p-values, and confidence intervals.

## Author(s)

Ivan Jacob Agaloos Pesigan

**Examples**

```

phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
vcov_phi_vec <- matrix(
  data = c(
    0.00843, 0.00040, -0.00151,
    -0.00600, -0.00033, 0.00110,
    0.00324, 0.00020, -0.00061,
    0.00040, 0.00374, 0.00016,
    -0.00022, -0.00273, -0.00016,
    0.00009, 0.00150, 0.00012,
    -0.00151, 0.00016, 0.00389,
    0.00103, -0.00007, -0.00283,
    -0.00050, 0.00000, 0.00156,
    -0.00600, -0.00022, 0.00103,
    0.00644, 0.00031, -0.00119,
    -0.00374, -0.00021, 0.00070,
    -0.00033, -0.00273, -0.00007,
    0.00031, 0.00287, 0.00013,
    -0.00014, -0.00170, -0.00012,
    0.00110, -0.00016, -0.00283,
    -0.00119, 0.00013, 0.00297,
    0.00063, -0.00004, -0.00177,
    0.00324, 0.00009, -0.00050,
    -0.00374, -0.00014, 0.00063,
    0.00495, 0.00024, -0.00093,
    0.00020, 0.00150, 0.00000,
    -0.00021, -0.00170, -0.00004,
    0.00024, 0.00214, 0.00012,
    -0.00061, 0.00012, 0.00156,
    0.00070, -0.00012, -0.00177,
    -0.00093, 0.00012, 0.00223
  ),
  nrow = 9
)

# Specific time interval -----
delta <- DeltaMed(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  delta_t = 1,
  from = "x",
  to = "y",
  med = "m"
)

```

```

print(delta)

# Range of time intervals -----
delta <- DeltaMed(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  delta_t = 1:5,
  from = "x",
  to = "y",
  med = "m"
)
print(delta)

```

---

|                   |                                                     |
|-------------------|-----------------------------------------------------|
| print.ctmedeffect | <i>Print Method for Object of Class ctmedeffect</i> |
|-------------------|-----------------------------------------------------|

---

## Description

Print Method for Object of Class ctmedeffect

## Usage

```

## S3 method for class 'ctmedeffect'
print(x, digits = 4, ...)

```

## Arguments

|        |                                                             |
|--------|-------------------------------------------------------------|
| x      | an object of class ctmedeffect.                             |
| digits | Integer indicating the number of decimal places to display. |
| ...    | further arguments.                                          |

## Value

Prints the effects.

## Author(s)

Ivan Jacob Agaloos Pesigan

## Examples

```

phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)

```

```

)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
delta_t <- 1

# Time Interval of One -----

## Total Effect -----
total_dt <- Total(
  phi = phi,
  delta_t = delta_t
)
print(total_dt)

## Direct Effect -----
direct_dt <- Direct(
  phi = phi,
  delta_t = delta_t,
  from = "x",
  to = "y",
  med = "m"
)
print(direct_dt)

## Indirect Effect -----
indirect_dt <- Indirect(
  phi = phi,
  delta_t = delta_t,
  from = "x",
  to = "y",
  med = "m"
)
print(indirect_dt)

```

---

print.ctmedmc

---

*Print Method for Object of Class ctmedmc*


---

## Description

Print Method for Object of Class ctmedmc

## Usage

```

## S3 method for class 'ctmedmc'
print(x, alpha = 0.05, digits = 4, ...)

```

## Arguments

|       |                                               |
|-------|-----------------------------------------------|
| x     | an object of class ctmedmc.                   |
| alpha | Numeric vector. Significance level $\alpha$ . |



digits            Integer indicating the number of decimal places to display.  
 ...              further arguments.

### Value

Prints a list of matrices of time intervals, estimates, standard errors, number of Monte Carlo replications, and confidence intervals.

### Author(s)

Ivan Jacob Agaloos Pesigan

### Examples

```
set.seed(42)
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
vcov_phi_vec <- matrix(
  data = c(
    0.00843, 0.00040, -0.00151,
    -0.00600, -0.00033, 0.00110,
    0.00324, 0.00020, -0.00061,
    0.00040, 0.00374, 0.00016,
    -0.00022, -0.00273, -0.00016,
    0.00009, 0.00150, 0.00012,
    -0.00151, 0.00016, 0.00389,
    0.00103, -0.00007, -0.00283,
    -0.00050, 0.00000, 0.00156,
    -0.00600, -0.00022, 0.00103,
    0.00644, 0.00031, -0.00119,
    -0.00374, -0.00021, 0.00070,
    -0.00033, -0.00273, -0.00007,
    0.00031, 0.00287, 0.00013,
    -0.00014, -0.00170, -0.00012,
    0.00110, -0.00016, -0.00283,
    -0.00119, 0.00013, 0.00297,
    0.00063, -0.00004, -0.00177,
    0.00324, 0.00009, -0.00050,
    -0.00374, -0.00014, 0.00063,
    0.00495, 0.00024, -0.00093,
    0.00020, 0.00150, 0.00000,
    -0.00021, -0.00170, -0.00004,
    0.00024, 0.00214, 0.00012,
    -0.00061, 0.00012, 0.00156,
    0.00070, -0.00012, -0.00177,
```

```

      -0.00093, 0.00012, 0.00223
    ),
    nrow = 9
  )

# Specific time interval -----
mc <- MCMed(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  delta_t = 1,
  from = "x",
  to = "y",
  med = "m",
  R = 100L # use a large value for R in actual research
)
print(mc)

# Range of time intervals -----
mc <- MCMed(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  delta_t = 1:5,
  from = "x",
  to = "y",
  med = "m",
  R = 100L # use a large value for R in actual research
)
print(mc)

```

---

|                  |                                                    |
|------------------|----------------------------------------------------|
| print.ctmedmcphi | <i>Print Method for Object of Class ctmedmcphi</i> |
|------------------|----------------------------------------------------|

---

## Description

Print Method for Object of Class ctmedmcphi

## Usage

```
## S3 method for class 'ctmedmcphi'
print(x, digits = 4, ...)
```

## Arguments

|        |                                                             |
|--------|-------------------------------------------------------------|
| x      | an object of class ctmedmcphi.                              |
| digits | Integer indicating the number of decimal places to display. |
| ...    | further arguments.                                          |

**Value**

Prints a list of drift matrices.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**Examples**

```
set.seed(42)
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
mc <- MCPHi(
  phi = phi,
  vcov_phi_vec = 0.1 * diag(9),
  R = 100L # use a large value for R in actual research
)
print(mc)
```

---

print.ctmedmed

---

*Print Method for Object of Class ctmedmed*


---

**Description**

Print Method for Object of Class ctmedmed

**Usage**

```
## S3 method for class 'ctmedmed'
print(x, digits = 4, ...)
```

**Arguments**

|        |                                                             |
|--------|-------------------------------------------------------------|
| x      | an object of class ctmedmed.                                |
| digits | Integer indicating the number of decimal places to display. |
| ...    | further arguments.                                          |

**Value**

Prints a matrix of effects.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**Examples**

```
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")

# Specific time interval -----
med <- Med(
  phi = phi,
  delta_t = 1,
  from = "x",
  to = "y",
  med = "m"
)
print(med)

# Range of time intervals -----
med <- Med(
  phi = phi,
  delta_t = 1:5,
  from = "x",
  to = "y",
  med = "m"
)
print(med)
```

---

|                 |                                                   |
|-----------------|---------------------------------------------------|
| print.ctmedtraj | <i>Print Method for Object of Class ctmedtraj</i> |
|-----------------|---------------------------------------------------|

---

**Description**

Print Method for Object of Class ctmedtraj

**Usage**

```
## S3 method for class 'ctmedtraj'
print(x, digits = 4, ...)
```

**Arguments**

`x` an object of class `ctmedtraj`.  
`digits` Integer indicating the number of decimal places to display.  
`...` further arguments.

**Value**

Prints a data frame of simulated data.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**Examples**

```
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")

traj <- Trajectory(
  mu0 = c(3, 3, -3),
  time = 150,
  phi = phi,
  med = "m"
)

print(traj)
```

---

|                   |                                                        |
|-------------------|--------------------------------------------------------|
| summary.ctmedboot | <i>Summary Method for an Object of Class ctmedboot</i> |
|-------------------|--------------------------------------------------------|

---

**Description**

Summary Method for an Object of Class `ctmedboot`

**Usage**

```
## S3 method for class 'ctmedboot'
summary(object, alpha = 0.05, type = "pc", digits = 4, ...)
```

**Arguments**

|        |                                                                                                                  |
|--------|------------------------------------------------------------------------------------------------------------------|
| object | Object of class ctmedboot.                                                                                       |
| alpha  | Numeric vector. Significance level $\alpha$ .                                                                    |
| type   | Character string. Confidence interval type, that is, type = "pc" for percentile; type = "bc" for bias corrected. |
| digits | Integer indicating the number of decimal places to display.                                                      |
| ...    | additional arguments.                                                                                            |

**Value**

Returns a data frame of effects, time intervals, estimates, standard errors, number of bootstrap replications, and confidence intervals.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**Examples**

```
## Not run:
library(bootStateSpace)
# prepare parameters
## number of individuals
n <- 50
## time points
time <- 100
delta_t <- 0.10
## dynamic structure
p <- 3
mu0 <- rep(x = 0, times = p)
sigma0 <- matrix(
  data = c(
    1.0,
    0.2,
    0.2,
    0.2,
    1.0,
    0.2,
    0.2,
    0.2,
    1.0
  ),
  nrow = p
)
sigma0_l <- t(chol(sigma0))
mu <- rep(x = 0, times = p)
phi <- matrix(
  data = c(
    -0.357,
    0.771,
```

```

      -0.450,
      0.0,
      -0.511,
      0.729,
      0,
      0,
      -0.693
    ),
    nrow = p
  )
  sigma <- matrix(
    data = c(
      0.24455556,
      0.02201587,
      -0.05004762,
      0.02201587,
      0.07067800,
      0.01539456,
      -0.05004762,
      0.01539456,
      0.07553061
    ),
    nrow = p
  )
  sigma_l <- t(chol(sigma))
  ## measurement model
  k <- 3
  nu <- rep(x = 0, times = k)
  lambda <- diag(k)
  theta <- 0.2 * diag(k)
  theta_l <- t(chol(theta))

  boot <- PBSSMOUFixed(
    R = 1000L,
    path = getwd(),
    prefix = "ou",
    n = n,
    time = time,
    delta_t = delta_t,
    mu0 = mu0,
    sigma0_l = sigma0_l,
    mu = mu,
    phi = phi,
    sigma_l = sigma_l,
    nu = nu,
    lambda = lambda,
    theta_l = theta_l,
    ncores = parallel::detectCores() - 1,
    seed = 42
  )
  phi_hat <- phi
  colnames(phi_hat) <- rownames(phi_hat) <- c("x", "m", "y")
  phi <- extract(object = boot, what = "phi")

```

```

# Specific time interval -----
boot <- BootMed(
  phi = phi,
  phi_hat = phi_hat,
  delta_t = 1,
  from = "x",
  to = "y",
  med = "m"
)
summary(boot)
summary(boot, type = "bc") # bias-corrected

# Range of time intervals -----
boot <- BootMed(
  phi = phi,
  phi_hat = phi_hat,
  delta_t = 1:5,
  from = "x",
  to = "y",
  med = "m"
)
summary(boot)
summary(boot, type = "bc") # bias-corrected

## End(Not run)

```

---

|                    |                                                         |
|--------------------|---------------------------------------------------------|
| summary.ctmeddelta | <i>Summary Method for an Object of Class ctmeddelta</i> |
|--------------------|---------------------------------------------------------|

---

## Description

Summary Method for an Object of Class ctmeddelta

## Usage

```

## S3 method for class 'ctmeddelta'
summary(object, alpha = 0.05, digits = 4, ...)

```

## Arguments

|        |                                                             |
|--------|-------------------------------------------------------------|
| object | Object of class ctmeddelta.                                 |
| alpha  | Numeric vector. Significance level $\alpha$ .               |
| digits | Integer indicating the number of decimal places to display. |
| ...    | additional arguments.                                       |



**Value**

Returns a data frame of effects, time intervals, estimates, standard errors, test statistics, p-values, and confidence intervals.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**Examples**

```
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
vcov_phi_vec <- matrix(
  data = c(
    0.00843, 0.00040, -0.00151,
    -0.00600, -0.00033, 0.00110,
    0.00324, 0.00020, -0.00061,
    0.00040, 0.00374, 0.00016,
    -0.00022, -0.00273, -0.00016,
    0.00009, 0.00150, 0.00012,
    -0.00151, 0.00016, 0.00389,
    0.00103, -0.00007, -0.00283,
    -0.00050, 0.00000, 0.00156,
    -0.00600, -0.00022, 0.00103,
    0.00644, 0.00031, -0.00119,
    -0.00374, -0.00021, 0.00070,
    -0.00033, -0.00273, -0.00007,
    0.00031, 0.00287, 0.00013,
    -0.00014, -0.00170, -0.00012,
    0.00110, -0.00016, -0.00283,
    -0.00119, 0.00013, 0.00297,
    0.00063, -0.00004, -0.00177,
    0.00324, 0.00009, -0.00050,
    -0.00374, -0.00014, 0.00063,
    0.00495, 0.00024, -0.00093,
    0.00020, 0.00150, 0.00000,
    -0.00021, -0.00170, -0.00004,
    0.00024, 0.00214, 0.00012,
    -0.00061, 0.00012, 0.00156,
    0.00070, -0.00012, -0.00177,
    -0.00093, 0.00012, 0.00223
  ),
  nrow = 9
)
```

```

# Specific time interval -----
delta <- DeltaMed(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  delta_t = 1,
  from = "x",
  to = "y",
  med = "m"
)
summary(delta)

# Range of time intervals -----
delta <- DeltaMed(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  delta_t = 1:5,
  from = "x",
  to = "y",
  med = "m"
)
summary(delta)

```

summary.ctmedmc

*Summary Method for an Object of Class ctmedmc***Description**

Summary Method for an Object of Class ctmedmc

**Usage**

```

## S3 method for class 'ctmedmc'
summary(object, alpha = 0.05, digits = 4, ...)

```

**Arguments**

|        |                                                             |
|--------|-------------------------------------------------------------|
| object | Object of class ctmedmc.                                    |
| alpha  | Numeric vector. Significance level $\alpha$ .               |
| digits | Integer indicating the number of decimal places to display. |
| ...    | additional arguments.                                       |

**Value**

Returns a data frame of effects, time intervals, estimates, standard errors, number of Monte Carlo replications, and confidence intervals.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**Examples**

```

set.seed(42)
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
vcov_phi_vec <- matrix(
  data = c(
    0.00843, 0.00040, -0.00151,
    -0.00600, -0.00033, 0.00110,
    0.00324, 0.00020, -0.00061,
    0.00040, 0.00374, 0.00016,
    -0.00022, -0.00273, -0.00016,
    0.00009, 0.00150, 0.00012,
    -0.00151, 0.00016, 0.00389,
    0.00103, -0.00007, -0.00283,
    -0.00050, 0.00000, 0.00156,
    -0.00600, -0.00022, 0.00103,
    0.00644, 0.00031, -0.00119,
    -0.00374, -0.00021, 0.00070,
    -0.00033, -0.00273, -0.00007,
    0.00031, 0.00287, 0.00013,
    -0.00014, -0.00170, -0.00012,
    0.00110, -0.00016, -0.00283,
    -0.00119, 0.00013, 0.00297,
    0.00063, -0.00004, -0.00177,
    0.00324, 0.00009, -0.00050,
    -0.00374, -0.00014, 0.00063,
    0.00495, 0.00024, -0.00093,
    0.00020, 0.00150, 0.00000,
    -0.00021, -0.00170, -0.00004,
    0.00024, 0.00214, 0.00012,
    -0.00061, 0.00012, 0.00156,
    0.00070, -0.00012, -0.00177,
    -0.00093, 0.00012, 0.00223
  ),
  nrow = 9
)

# Specific time interval -----
mc <- MCMed(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,

```

```

    delta_t = 1,
    from = "x",
    to = "y",
    med = "m",
    R = 100L # use a large value for R in actual research
  )
summary(mc)

# Range of time intervals -----
mc <- MCMed(
  phi = phi,
  vcov_phi_vec = vcov_phi_vec,
  delta_t = 1:5,
  from = "x",
  to = "y",
  med = "m",
  R = 100L # use a large value for R in actual research
)
summary(mc)

```

---

summary.ctmedmed

*Summary Method for an Object of Class ctmedmed*


---

## Description

Summary Method for an Object of Class ctmedmed

## Usage

```
## S3 method for class 'ctmedmed'
summary(object, digits = 4, ...)
```

## Arguments

|        |                                                             |
|--------|-------------------------------------------------------------|
| object | an object of class ctmedmed.                                |
| digits | Integer indicating the number of decimal places to display. |
| ...    | further arguments.                                          |

## Value

Returns a matrix of effects.

## Author(s)

Ivan Jacob Agaloos Pesigan

**Examples**

```

phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")

# Specific time interval -----
med <- Med(
  phi = phi,
  delta_t = 1,
  from = "x",
  to = "y",
  med = "m"
)
summary(med)

# Range of time intervals -----
med <- Med(
  phi = phi,
  delta_t = 1:5,
  from = "x",
  to = "y",
  med = "m"
)
summary(med)

```

---

summary.ctmedposteriorphi

*Summary Method for Object of Class ctmedposteriorphi*


---

**Description**

Summary Method for Object of Class ctmedposteriorphi

**Usage**

```

## S3 method for class 'ctmedposteriorphi'
summary(object, ...)

```

**Arguments**

|        |                                       |
|--------|---------------------------------------|
| object | an object of class ctmedposteriorphi. |
| ...    | further arguments.                    |

**Value**

Returns a list of the posterior means (in matrix form) and covariance matrix.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|                   |                                                        |
|-------------------|--------------------------------------------------------|
| summary.ctmedtraj | <i>Summary Method for an Object of Class ctmedtraj</i> |
|-------------------|--------------------------------------------------------|

---

**Description**

Summary Method for an Object of Class ctmedtraj

**Usage**

```
## S3 method for class 'ctmedtraj'
summary(object, digits = 4, ...)
```

**Arguments**

|        |                                                             |
|--------|-------------------------------------------------------------|
| object | an object of class ctmedtraj.                               |
| digits | Integer indicating the number of decimal places to display. |
| ...    | further arguments.                                          |

**Value**

Returns a data frame of simulated data.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**Examples**

```
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")

traj <- Trajectory(
  mu0 = c(3, 3, -3),
  time = 150,
```

```

    phi = phi,
    med = "m"
)

summary(traj)

```

Total

*Total Effect Matrix Over a Specific Time Interval***Description**

This function computes the total effects matrix over a specific time interval  $\Delta t$  using the first-order stochastic differential equation model's drift matrix  $\Phi$ .

**Usage**

```
Total(phi, delta_t)
```

**Arguments**

**phi** Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system.

**delta\_t** Numeric. Time interval ( $\Delta t$ ).

**Details**

The total effect matrix over a specific time interval  $\Delta t$  is given by

$$\text{Total}_{\Delta t} = \exp(\Delta t \Phi)$$

where  $\Phi$  denotes the drift matrix, and  $\Delta t$  the time interval.

**Linear Stochastic Differential Equation Model:**

The measurement model is given by

$$\mathbf{y}_{i,t} = \boldsymbol{\nu} + \mathbf{\Lambda} \boldsymbol{\eta}_{i,t} + \boldsymbol{\varepsilon}_{i,t}, \quad \text{with} \quad \boldsymbol{\varepsilon}_{i,t} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Theta})$$

where  $\mathbf{y}_{i,t}$ ,  $\boldsymbol{\eta}_{i,t}$ , and  $\boldsymbol{\varepsilon}_{i,t}$  are random variables and  $\boldsymbol{\nu}$ ,  $\mathbf{\Lambda}$ , and  $\boldsymbol{\Theta}$  are model parameters.  $\mathbf{y}_{i,t}$  represents a vector of observed random variables,  $\boldsymbol{\eta}_{i,t}$  a vector of latent random variables, and  $\boldsymbol{\varepsilon}_{i,t}$  a vector of random measurement errors, at time  $t$  and individual  $i$ .  $\boldsymbol{\nu}$  denotes a vector of intercepts,  $\mathbf{\Lambda}$  a matrix of factor loadings, and  $\boldsymbol{\Theta}$  the covariance matrix of  $\boldsymbol{\varepsilon}$ .

An alternative representation of the measurement error is given by

$$\boldsymbol{\varepsilon}_{i,t} = \boldsymbol{\Theta}^{\frac{1}{2}} \mathbf{z}_{i,t}, \quad \text{with} \quad \mathbf{z}_{i,t} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

where  $\mathbf{z}_{i,t}$  is a vector of independent standard normal random variables and  $\left(\boldsymbol{\Theta}^{\frac{1}{2}}\right) \left(\boldsymbol{\Theta}^{\frac{1}{2}}\right)' = \boldsymbol{\Theta}$ .

The dynamic structure is given by

$$d\boldsymbol{\eta}_{i,t} = (\boldsymbol{\iota} + \boldsymbol{\Phi}\boldsymbol{\eta}_{i,t}) dt + \boldsymbol{\Sigma}^{\frac{1}{2}} d\mathbf{W}_{i,t}$$

where  $\boldsymbol{\iota}$  is a term which is unobserved and constant over time,  $\boldsymbol{\Phi}$  is the drift matrix which represents the rate of change of the solution in the absence of any random fluctuations,  $\boldsymbol{\Sigma}$  is the matrix of volatility or randomness in the process, and  $d\mathbf{W}$  is a Wiener process or Brownian motion, which represents random fluctuations.

## Value

Returns an object of class `ctmedeffect` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("Total").

**output** The matrix of total effects.

## Author(s)

Ivan Jacob Agaloos Pesigan

## References

Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. [doi:10.2307/271028](https://doi.org/10.2307/271028)

Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. [doi:10.1080/10705511.2014.973960](https://doi.org/10.1080/10705511.2014.973960)

Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. [doi:10.1037/met0000779](https://doi.org/10.1037/met0000779)

Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. [doi:10.1007/s11336021097670](https://doi.org/10.1007/s11336021097670)

## See Also

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)



**Examples**

```

phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
delta_t <- 1
Total(
  phi = phi,
  delta_t = delta_t
)
phi <- matrix(
  data = c(
    -6, 5.5, 0, 0,
    1.25, -2.5, 5.9, -7.3,
    0, 0, -6, 2.5,
    5, 0, 0, -6
  ),
  nrow = 4
)
colnames(phi) <- rownames(phi) <- paste0("y", 1:4)
Total(
  phi = phi,
  delta_t = delta_t
)

```

---

|              |                                |
|--------------|--------------------------------|
| TotalCentral | <i>Total Effect Centrality</i> |
|--------------|--------------------------------|

---

**Description**

Total Effect Centrality

**Usage**

```
TotalCentral(phi, delta_t, tol = 0.001)
```

**Arguments**

|         |                                                                                                                              |
|---------|------------------------------------------------------------------------------------------------------------------------------|
| phi     | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system. |
| delta_t | Vector of positive numbers. Time interval ( $\Delta t$ ).                                                                    |
| tol     | Numeric. Smallest possible time interval to allow.                                                                           |

## Details

The total effect centrality of a variable is the sum of the total effects of a variable on all other variables at a particular time interval.

## Value

Returns an object of class `ctmedmed` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("TotalCentral").

**output** A matrix of total effect centrality.

## Author(s)

Ivan Jacob Agaloos Pesigan

## References

Bollen, K. A. (1987). Total, direct, and indirect effects in structural equation models. *Sociological Methodology*, 17, 37. [doi:10.2307/271028](https://doi.org/10.2307/271028)

Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, 23 (1), 61-75. [doi:10.1080/10705511.2014.973960](https://doi.org/10.1080/10705511.2014.973960)

Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. [doi:10.1037/met0000779](https://doi.org/10.1037/met0000779)

Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87 (1), 214-252. [doi:10.1007/s11336021097670](https://doi.org/10.1007/s11336021097670)

## See Also

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

**Examples**

```

phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")

# Specific time interval -----
TotalCentral(
  phi = phi,
  delta_t = 1
)

# Range of time intervals -----
total_central <- TotalCentral(
  phi = phi,
  delta_t = 1:30
)
plot(total_central)

# Methods -----
# TotalCentral has a number of methods including
# print, summary, and plot
total_central <- TotalCentral(
  phi = phi,
  delta_t = 1:5
)
print(total_central)
summary(total_central)
plot(total_central)

```

---

TotalCentralStd

*Standardized Total Effect Centrality*


---

**Description**

Standardized Total Effect Centrality

**Usage**

```
TotalCentralStd(phi, sigma, delta_t, tol = 0.001)
```

**Arguments**

|         |                                                                                                                              |
|---------|------------------------------------------------------------------------------------------------------------------------------|
| phi     | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system. |
| sigma   | Numeric matrix. The process noise covariance matrix ( $\Sigma$ ).                                                            |
| delta_t | Vector of positive numbers. Time interval ( $\Delta t$ ).                                                                    |
| tol     | Numeric. Smallest possible time interval to allow.                                                                           |

**Details**

The standardized total effect centrality of a variable is the sum of the standardized total effects of a variable on all other variables at a particular time interval.

**Value**

Returns an object of class `ctmedmed` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("TotalCentralStd").

**output** A matrix of standardized total effect centrality.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**See Also**

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalStd\(\)](#), [Trajectory\(\)](#)

**Examples**

```
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
```

```

    nrow = 3
  )
  colnames(phi) <- rownames(phi) <- c("x", "m", "y")
  sigma <- matrix(
    data = c(
      0.24455556, 0.02201587, -0.05004762,
      0.02201587, 0.07067800, 0.01539456,
      -0.05004762, 0.01539456, 0.07553061
    ),
    nrow = 3
  )

# Specific time interval -----
TotalCentralStd(
  phi = phi,
  sigma = sigma,
  delta_t = 1
)

# Range of time intervals -----
total_central_std <- TotalCentralStd(
  phi = phi,
  sigma = sigma,
  delta_t = 1:30
)
plot(total_central_std)

# Methods -----
# TotalCentralStd has a number of methods including
# print, summary, and plot
total_central_std <- TotalCentralStd(
  phi = phi,
  sigma = sigma,
  delta_t = 1:5
)
print(total_central_std)
summary(total_central_std)
plot(total_central_std)

```

---

TotalStd

---

Standardized Total Effect Matrix Over a Specific Time Interval

---

### Description

This function computes the standardized total effects matrix over a specific time interval  $\Delta t$  using the first-order stochastic differential equation model's drift matrix  $\Phi$  and process noise covariance matrix  $\Sigma$ .

**Usage**

```
TotalStd(phi, sigma, delta_t)
```

**Arguments**

|         |                                                                                                                              |
|---------|------------------------------------------------------------------------------------------------------------------------------|
| phi     | Numeric matrix. The drift matrix ( $\Phi$ ). phi should have row and column names pertaining to the variables in the system. |
| sigma   | Numeric matrix. The process noise covariance matrix ( $\Sigma$ ).                                                            |
| delta_t | Numeric. Time interval ( $\Delta t$ ).                                                                                       |

**Details**

The standardized total effect matrix over a specific time interval  $\Delta t$  is given by

$$\text{Total}_{\Delta t, i, j}^* = \text{Total}_{\Delta t, i, j} \left( \frac{\sigma_{x_j}}{\sigma_{y_i}} \right)$$

where  $\Phi$  denotes the drift matrix,  $\sigma_{x_j}$  and  $\sigma_{y_i}$  are the steady-state model-implied standard deviations of the state independent and dependent variables, respectively, and  $\Delta t$  the time interval.

**Value**

Returns an object of class `ctmedeffect` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("TotalStd").

**output** The standardized matrix of total effects.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**References**

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**See Also**

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [Trajectory\(\)](#)

**Examples**

```
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")
sigma <- matrix(
  data = c(
    0.24455556, 0.02201587, -0.05004762,
    0.02201587, 0.07067800, 0.01539456,
    -0.05004762, 0.01539456, 0.07553061
  ),
  nrow = 3
)
delta_t <- 1
TotalStd(
  phi = phi,
  sigma = sigma,
  delta_t = delta_t
)
```

**Description**

This function simulates trajectories of variables without measurement error or process noise. `total` corresponds to the total effect and `direct` corresponds to the portion of the total effect where the indirect effect is removed.

**Usage**

```
Trajectory(mu0, time, phi, med)
```

**Arguments**

|                   |                                                                                                                                           |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <code>mu0</code>  | Numeric vector. Initial values of the variables.                                                                                          |
| <code>time</code> | Positive integer. Number of time points.                                                                                                  |
| <code>phi</code>  | Numeric matrix. The drift matrix ( $\Phi$ ). <code>phi</code> should have row and column names pertaining to the variables in the system. |
| <code>med</code>  | Character vector. Name/s of the mediator variable/s in <code>phi</code> .                                                                 |

**Value**

Returns an object of class `ctmedtraj` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**fun** Function used ("Trajectory").

**output** A list containing simulated trajectories for total, direct, and indirect effects.

**See Also**

Other Continuous-Time Mediation Functions: [BootBeta\(\)](#), [BootBetaStd\(\)](#), [BootDirectCentral\(\)](#), [BootDirectCentralStd\(\)](#), [BootIndirectCentral\(\)](#), [BootIndirectCentralStd\(\)](#), [BootMed\(\)](#), [BootMedStd\(\)](#), [BootTotalCentral\(\)](#), [BootTotalCentralStd\(\)](#), [DeltaBeta\(\)](#), [DeltaBetaStd\(\)](#), [DeltaDirectCentral\(\)](#), [DeltaDirectCentralStd\(\)](#), [DeltaIndirectCentral\(\)](#), [DeltaMed\(\)](#), [DeltaMedStd\(\)](#), [DeltaTotalCentral\(\)](#), [DeltaTotalCentralStd\(\)](#), [Direct\(\)](#), [DirectCentral\(\)](#), [DirectCentralStd\(\)](#), [DirectStd\(\)](#), [Indirect\(\)](#), [IndirectCentral\(\)](#), [IndirectCentralStd\(\)](#), [IndirectStd\(\)](#), [MCBeta\(\)](#), [MCBetaStd\(\)](#), [MCDirectCentral\(\)](#), [MCDirectCentralStd\(\)](#), [MCIndirectCentral\(\)](#), [MCIndirectCentralStd\(\)](#), [MCMed\(\)](#), [MCMedStd\(\)](#), [MCPhi\(\)](#), [MCPhiSigma\(\)](#), [MCTotalCentral\(\)](#), [MCTotalCentralStd\(\)](#), [Med\(\)](#), [MedStd\(\)](#), [PosteriorBeta\(\)](#), [PosteriorBetaStd\(\)](#), [PosteriorDirectCentral\(\)](#), [PosteriorDirectCentralStd\(\)](#), [PosteriorIndirectCentral\(\)](#), [PosteriorIndirectCentralStd\(\)](#), [PosteriorMed\(\)](#), [PosteriorMedStd\(\)](#), [PosteriorTotalCentral\(\)](#), [PosteriorTotalCentralStd\(\)](#), [Total\(\)](#), [TotalCentral\(\)](#), [TotalCentralStd\(\)](#), [TotalStd\(\)](#)

**Examples**

```
phi <- matrix(
  data = c(
    -0.357, 0.771, -0.450,
    0.0, -0.511, 0.729,
    0, 0, -0.693
  ),
  nrow = 3
)
colnames(phi) <- rownames(phi) <- c("x", "m", "y")

traj <- Trajectory(
```



```
    mu0 = c(3, 3, -3),
    time = 150,
    phi = phi,
    med = "m"
)
plot(traj)

# Methods -----
# Trajectory has a number of methods including
# print, summary, and plot

traj <- Trajectory(
  mu0 = c(3, 3, -3),
  time = 25,
  phi = phi,
  med = "m"
)
print(traj)
summary(traj)
plot(traj)
```

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